

Supplemental Materials - Model MCMC Diagnostic Plots

For each model, we show histograms of the autocorrelation (AC.####), potential scale reduction factor (PSRF), and effective sample size (ESS) values for all parameters of interest in the model. Following these are the posterior predictive check distribution (PPD) expressed as the distribution of the deviation of the observed values from the PPD values. Finally, for the parameters of interest for each model, we show the posterior trace, histogram, expected cumulative distribution (ECDF), and autocorrelation distribution.

Model I

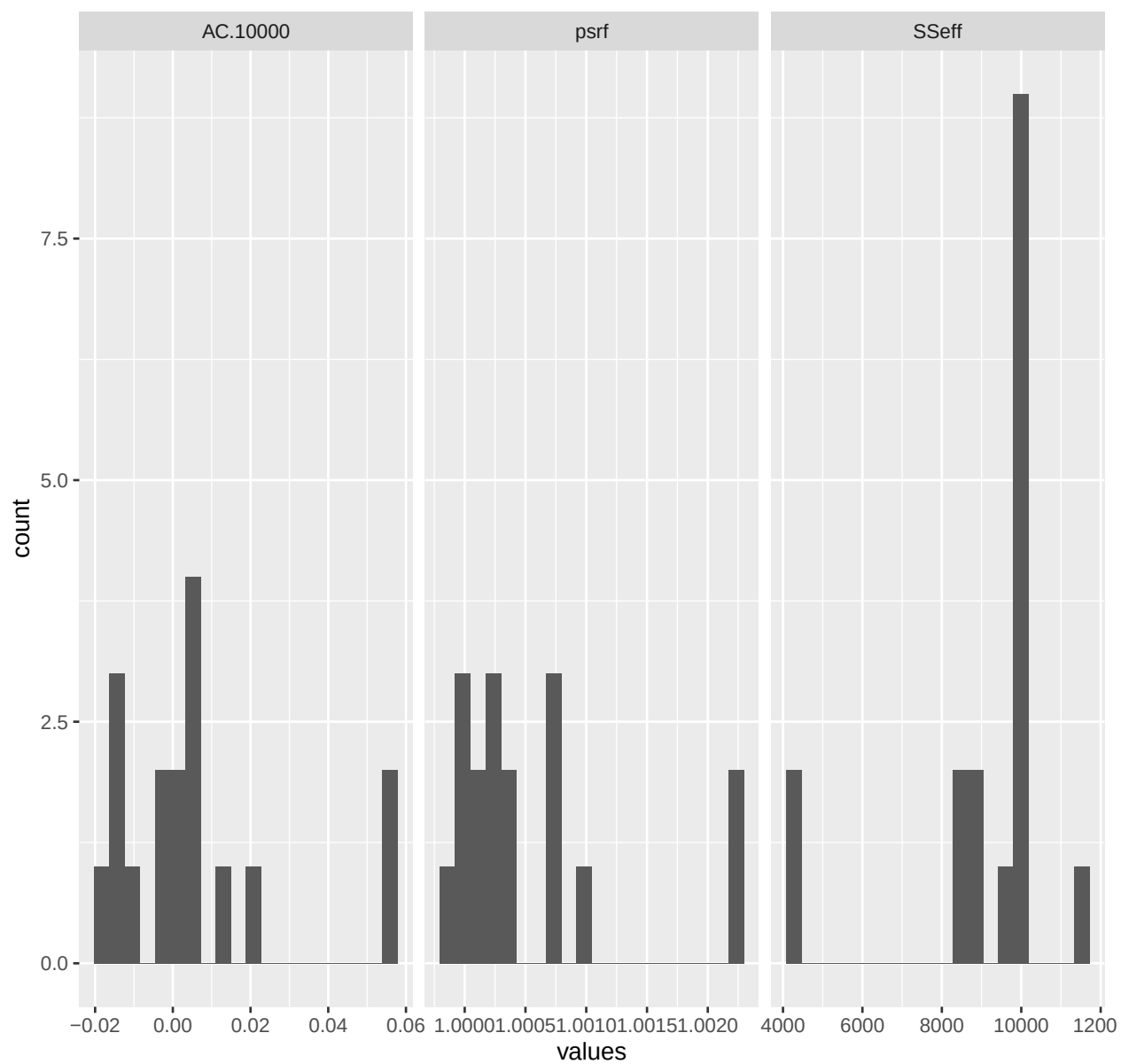
Number of chains: 10

Burnin: 100100

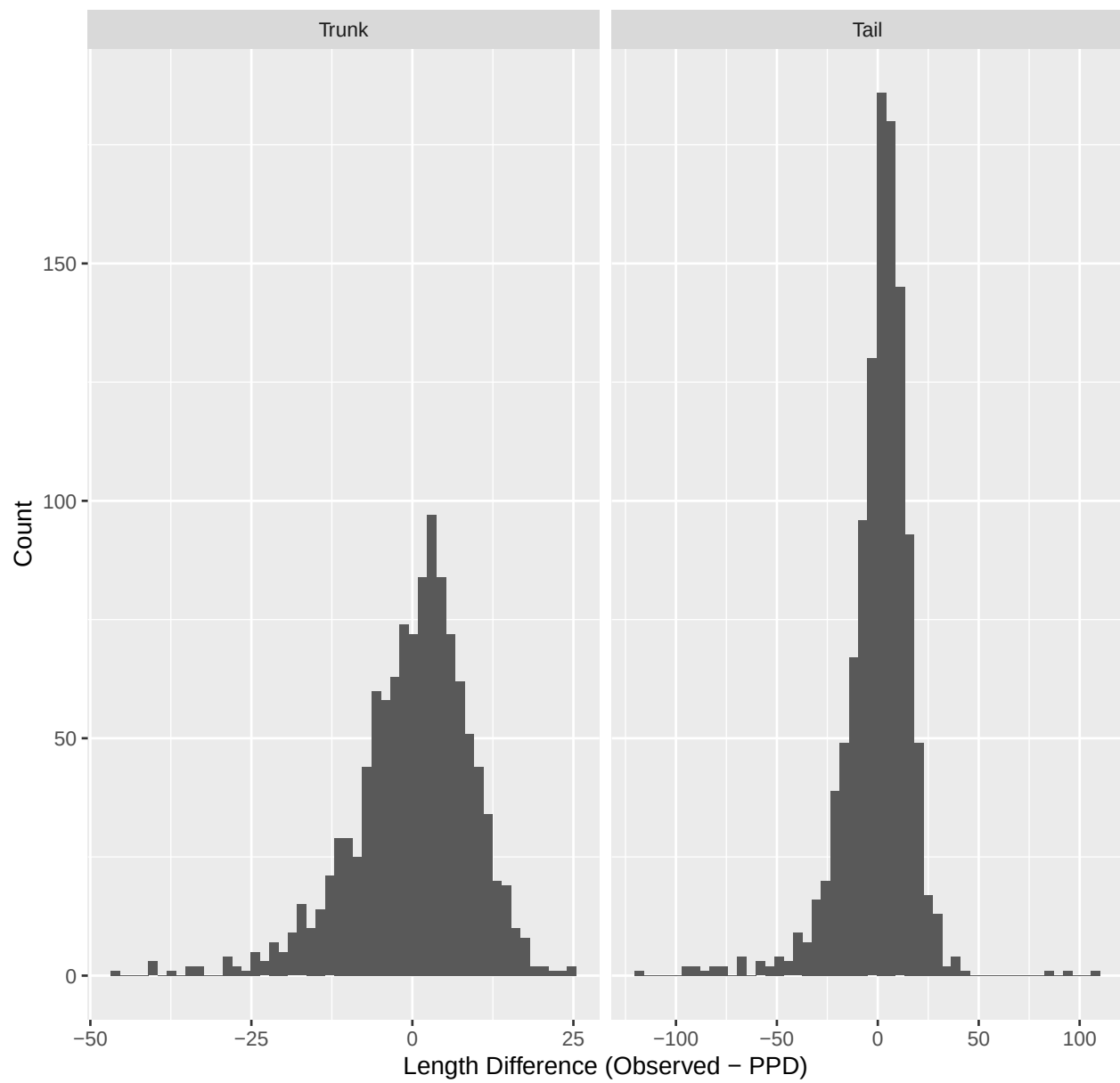
Samples per chain: 1000

Total samples: 10000

A) Posterior distribution diagnostics



B) Posterior predictive check distribution



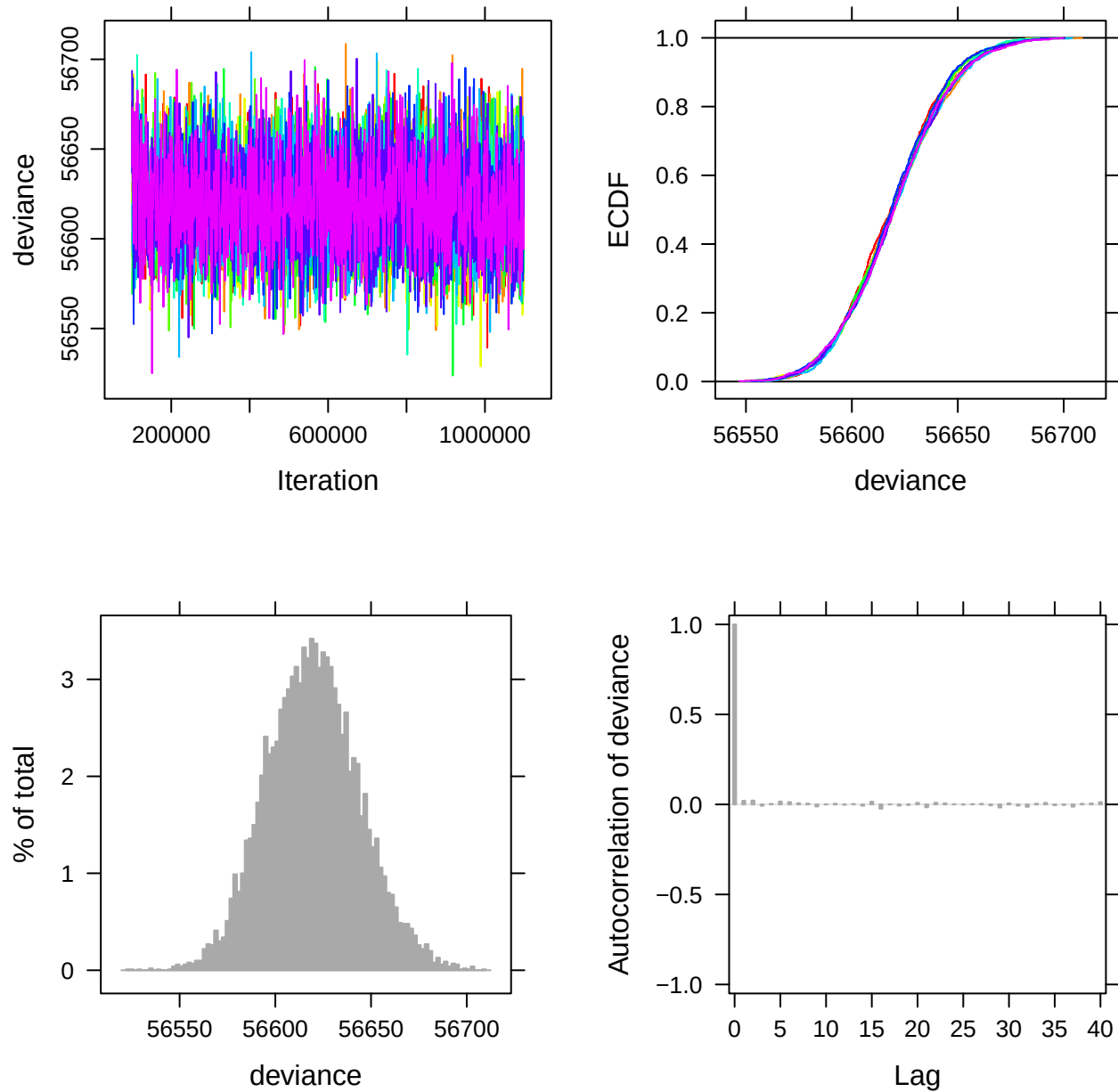
C) By-parameter posterior summaries

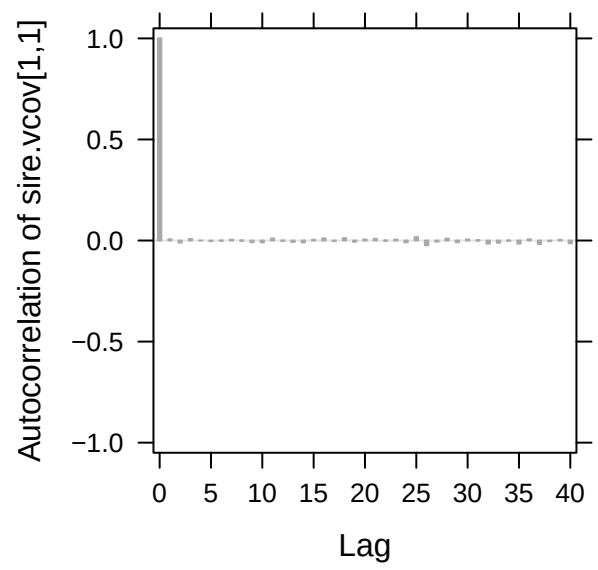
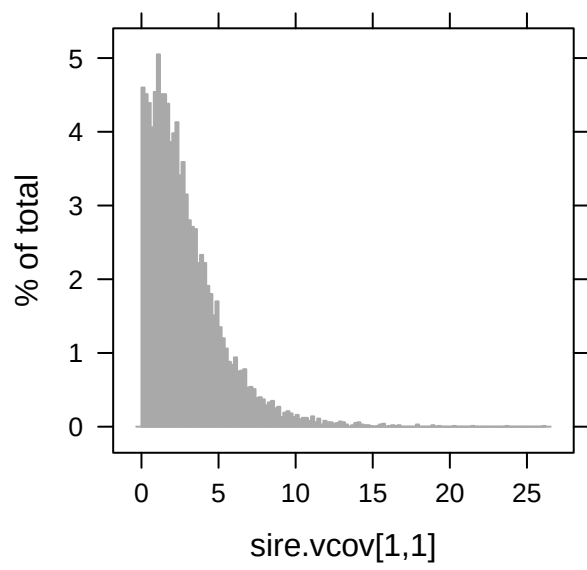
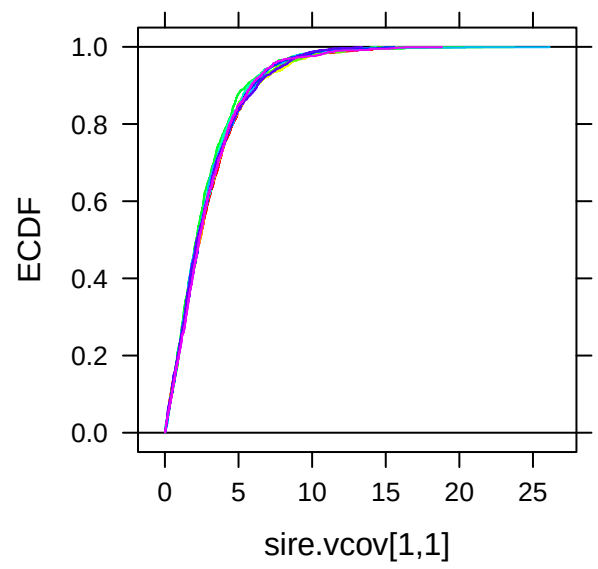
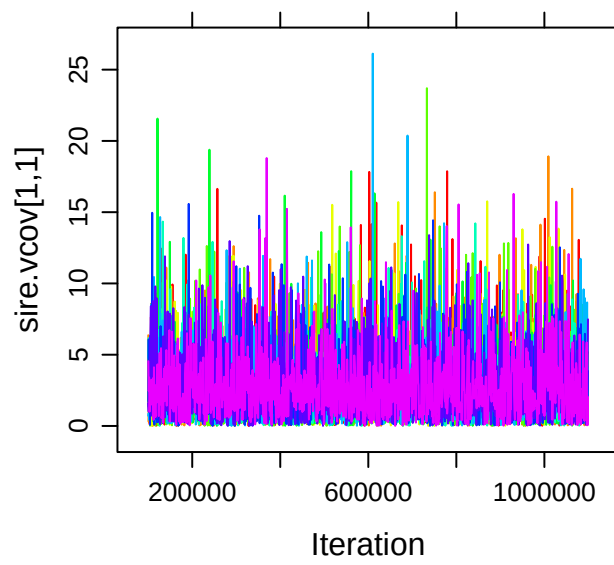
Generating summary statistics and plots (these will NOT be saved for reuse)...

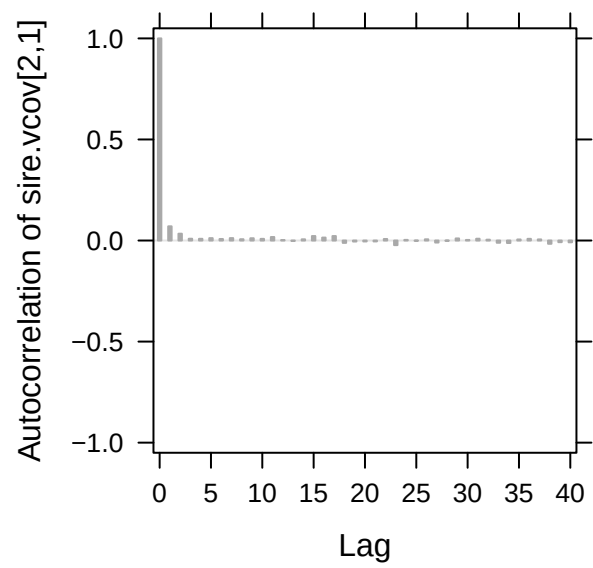
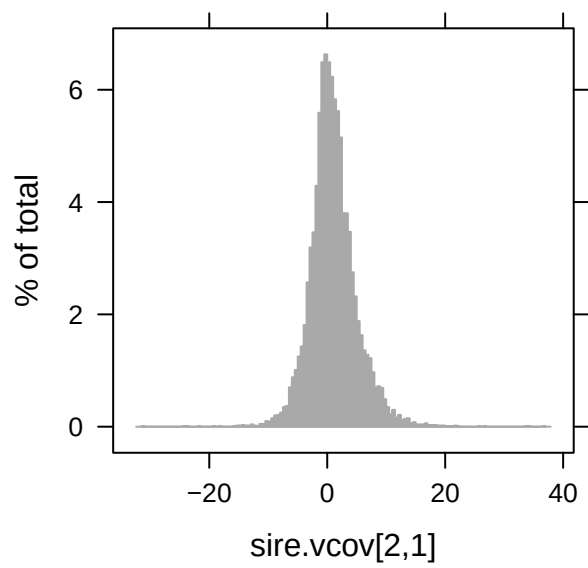
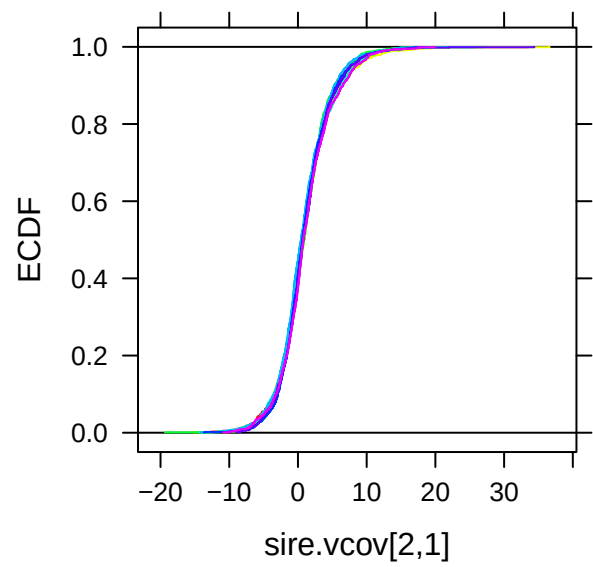
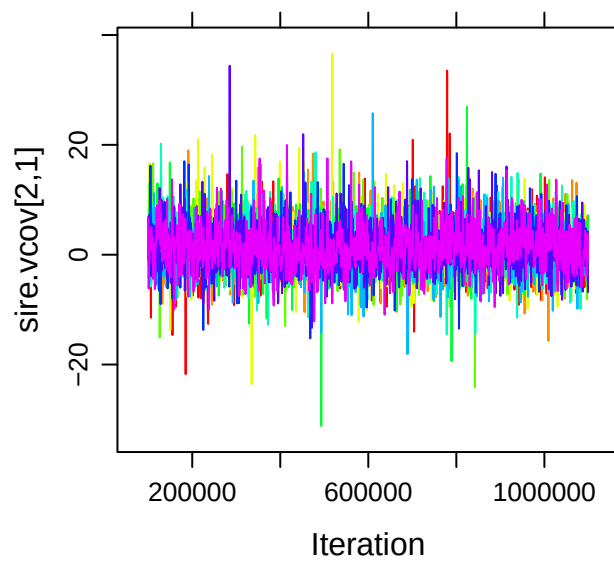
Calculating summary statistics...

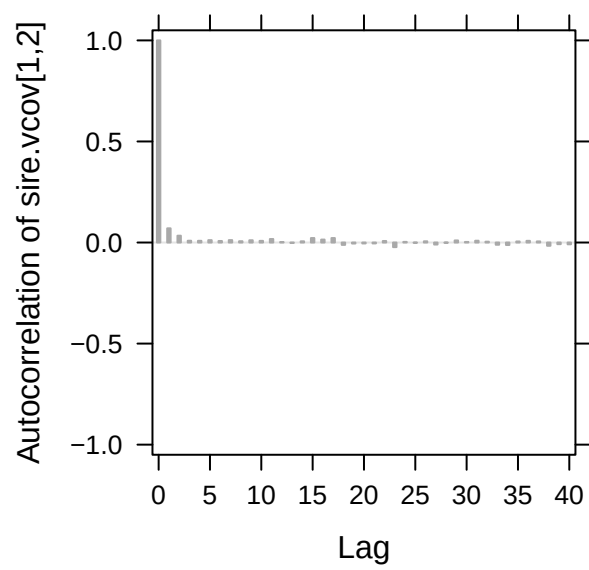
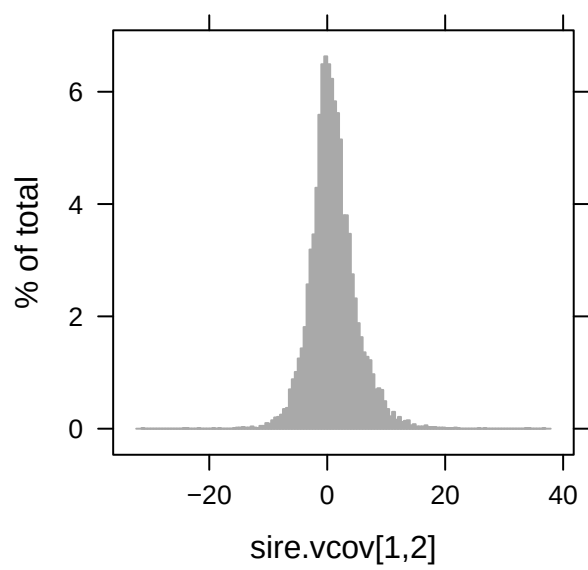
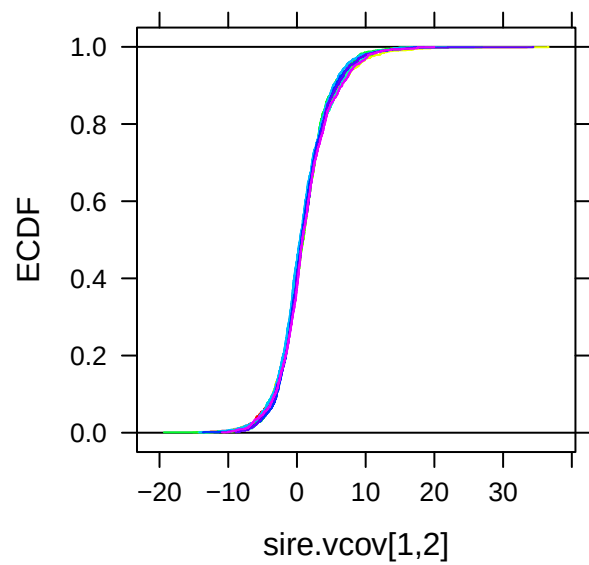
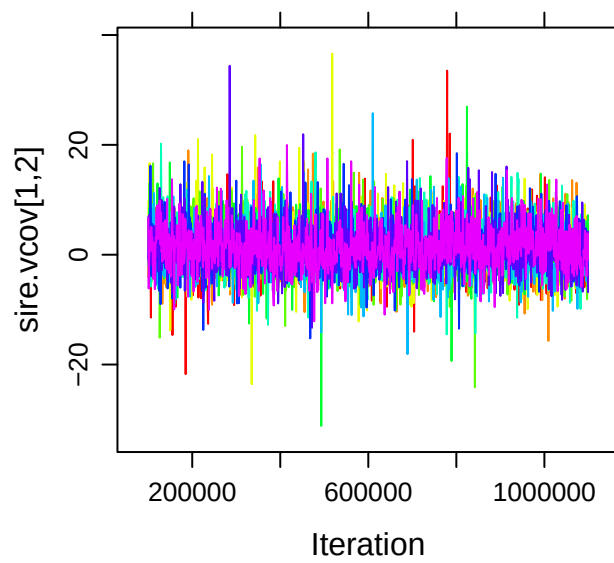
Calculating the Gelman-Rubin statistic for 17 variables....

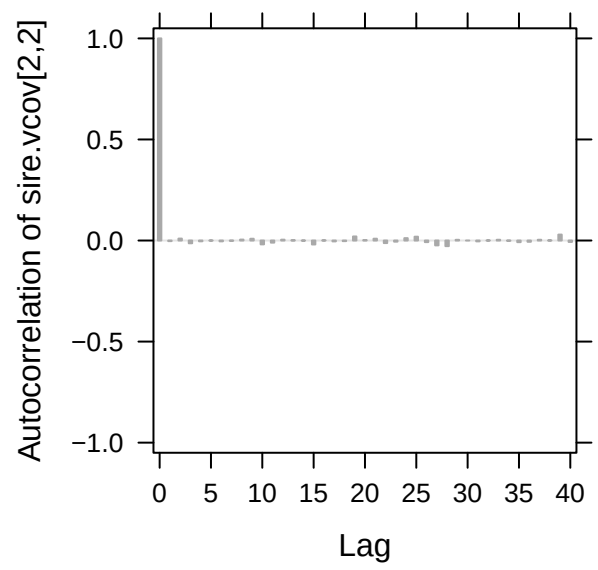
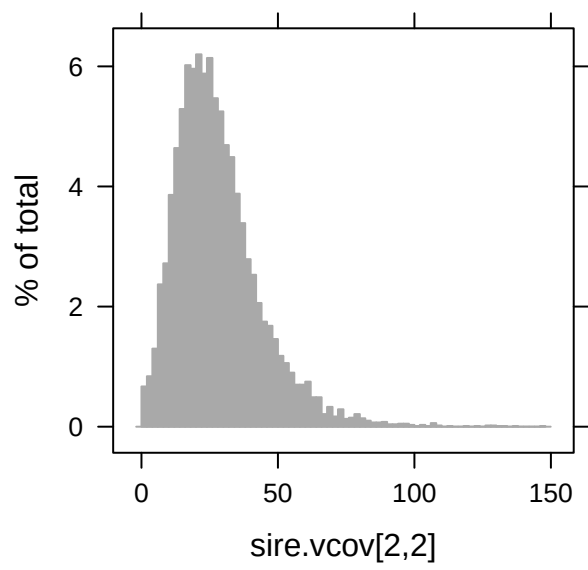
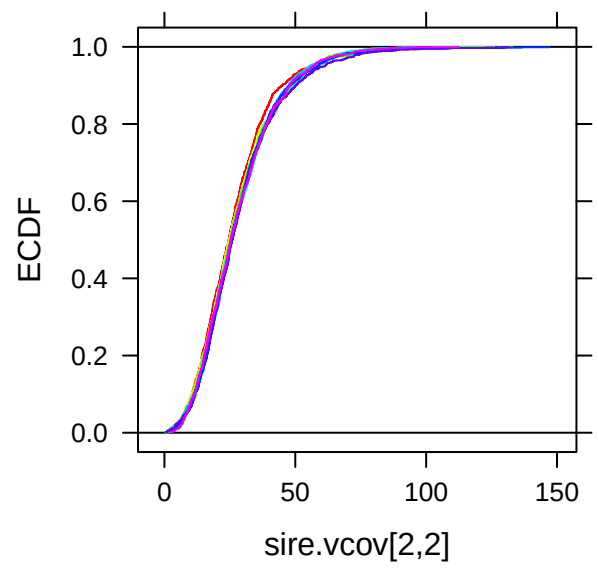
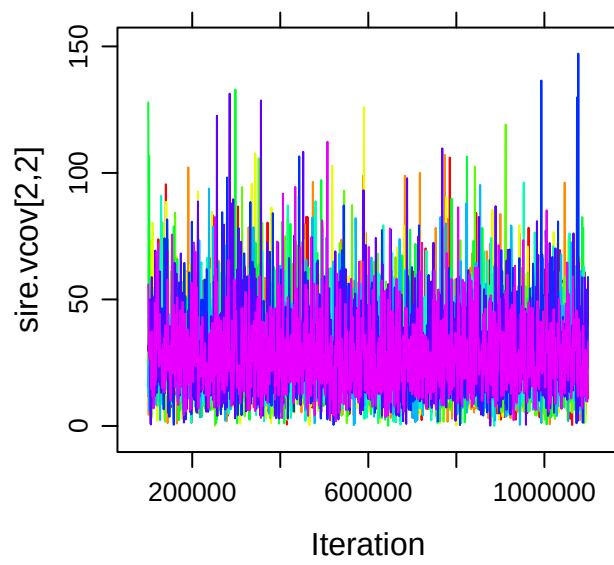
Note: Unable to calculate the multivariate psrf

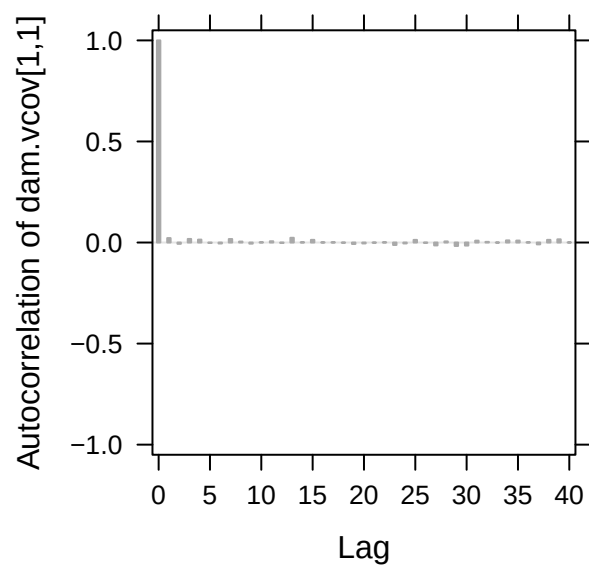
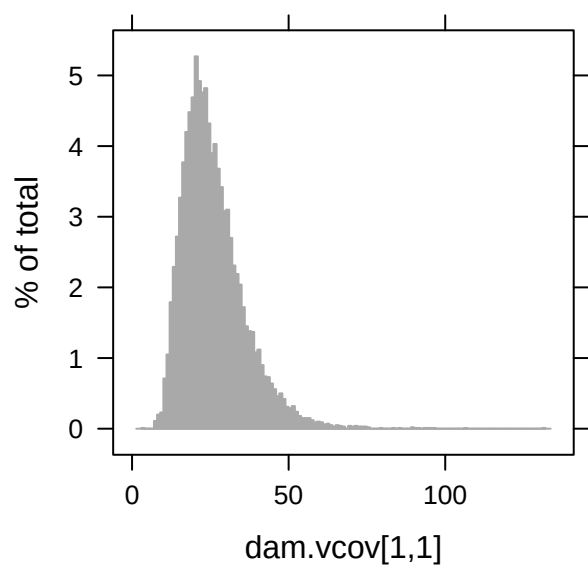
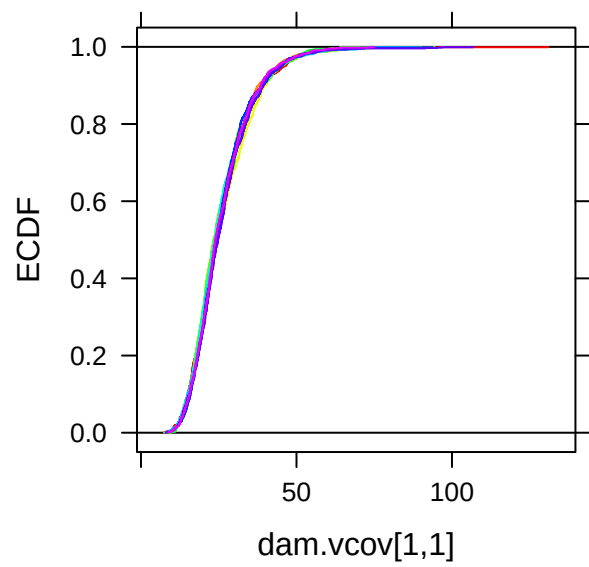
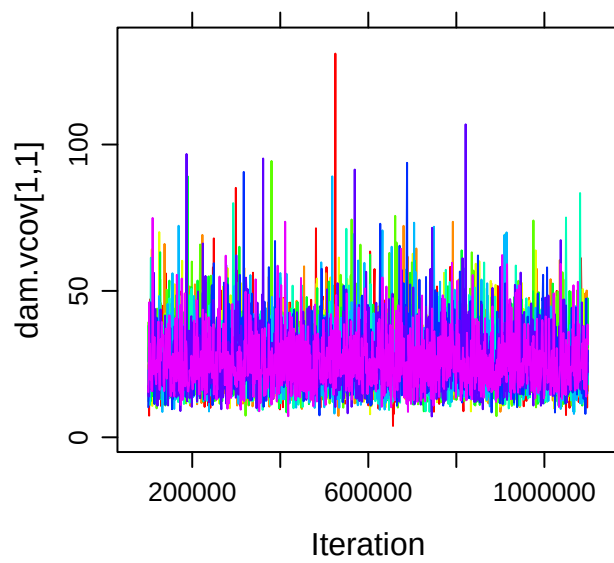


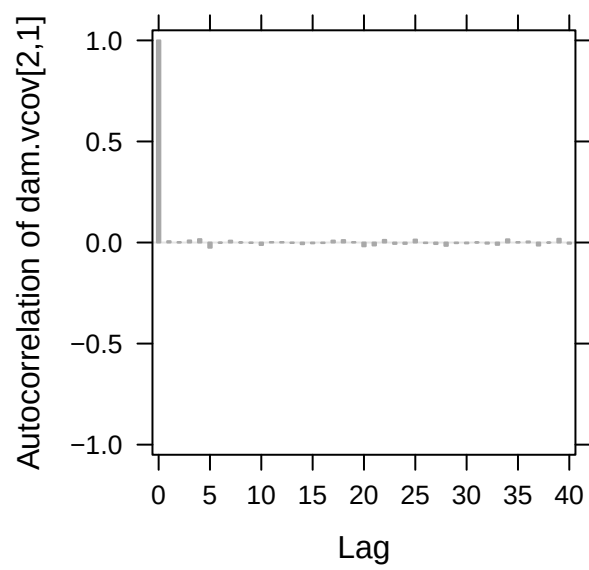
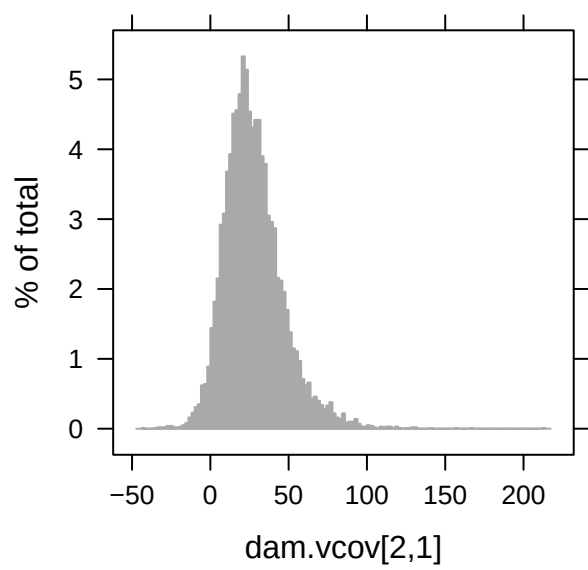
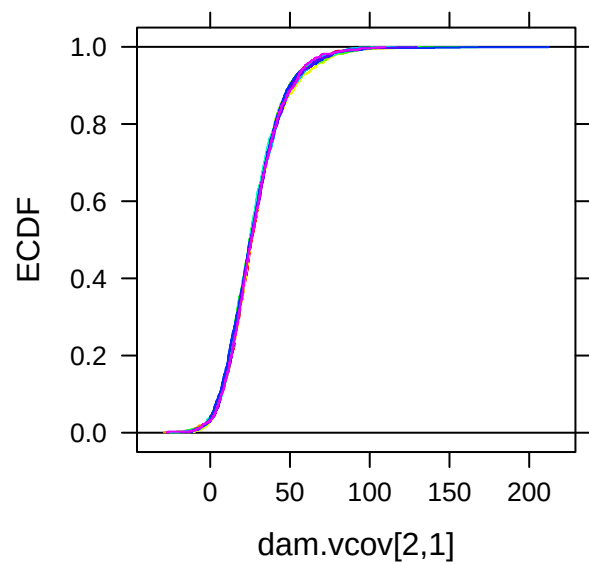
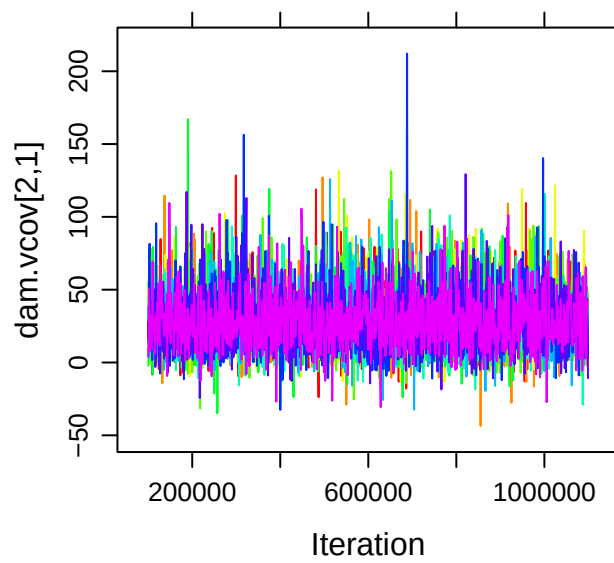


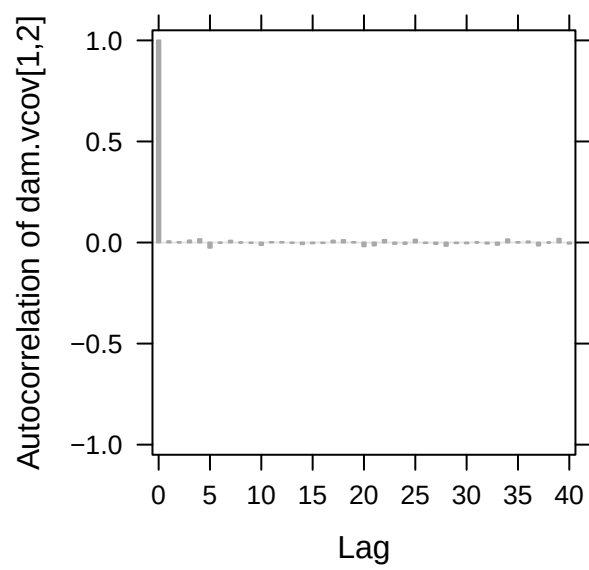
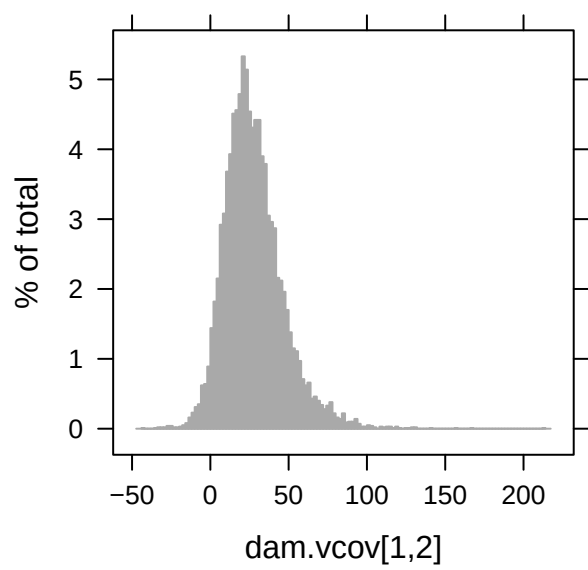
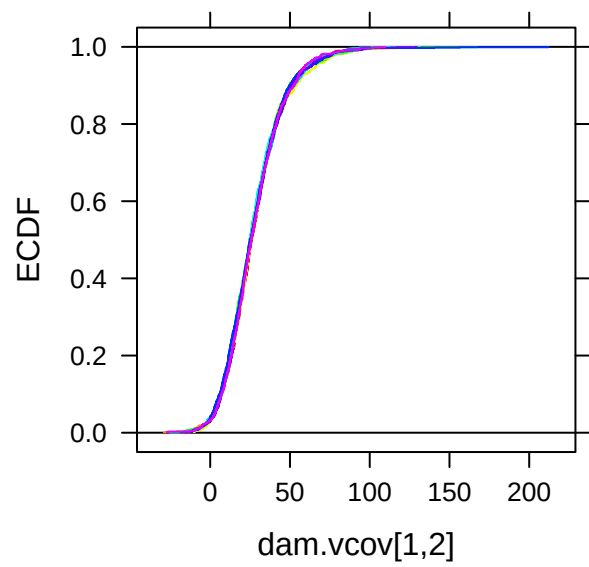
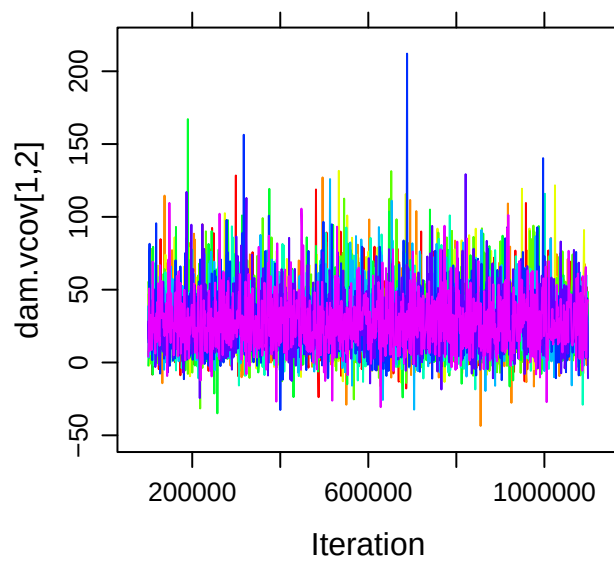


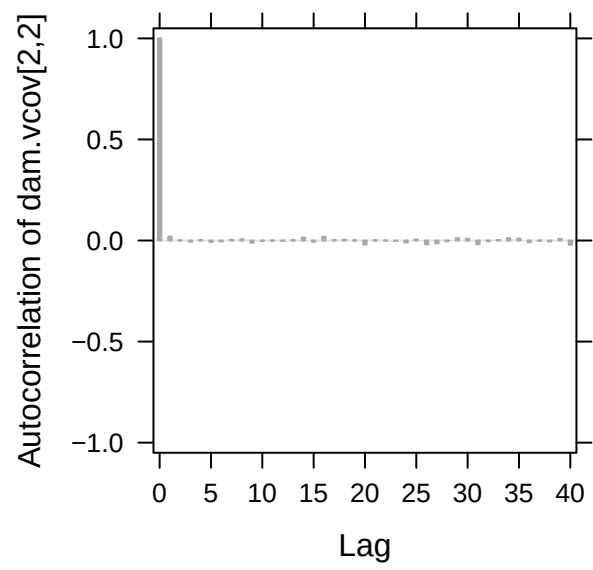
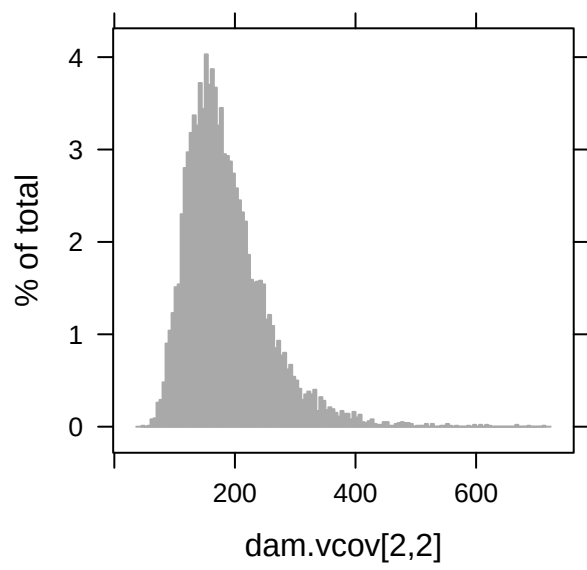
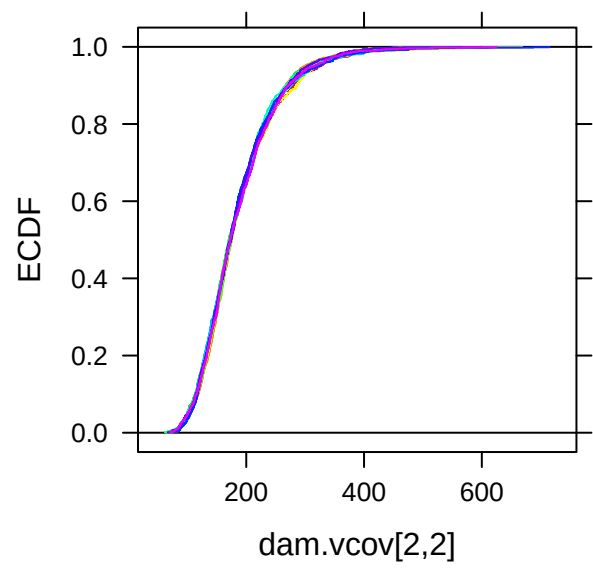
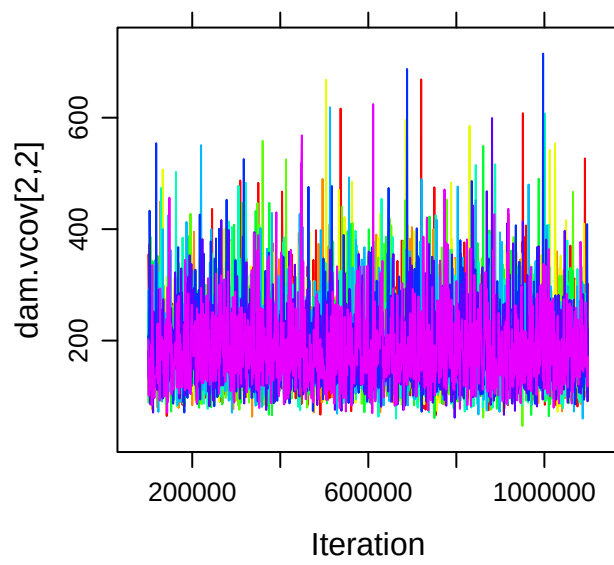


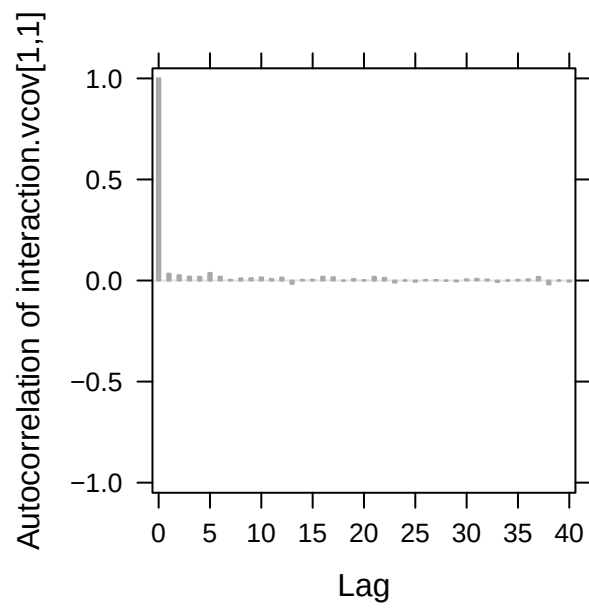
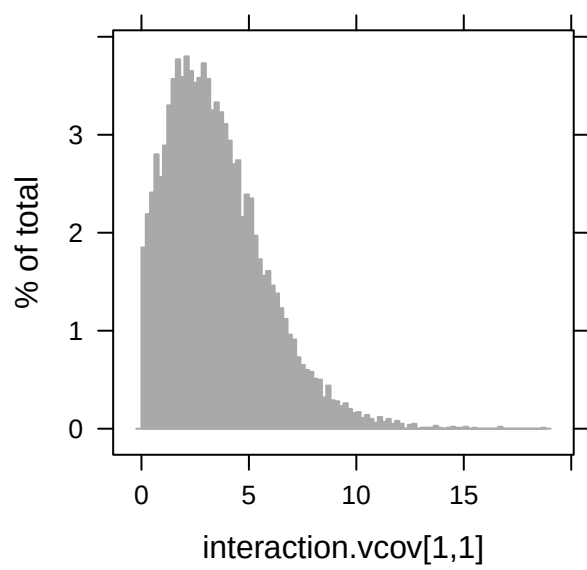
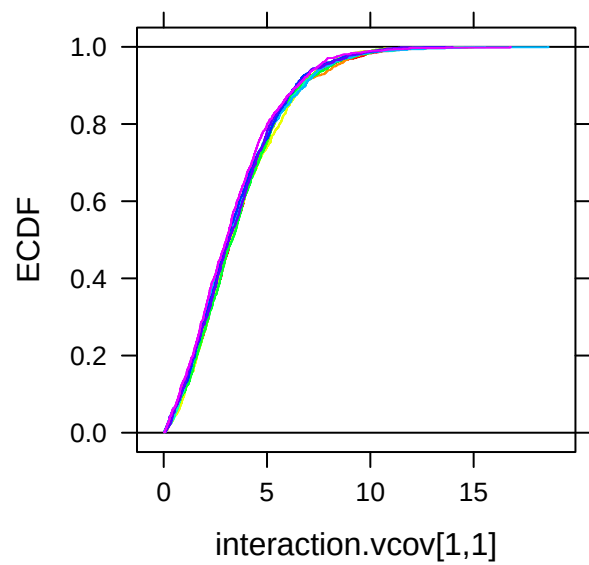
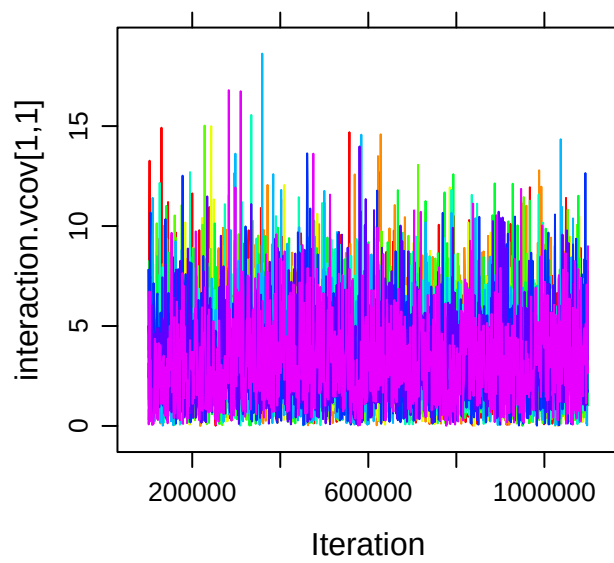


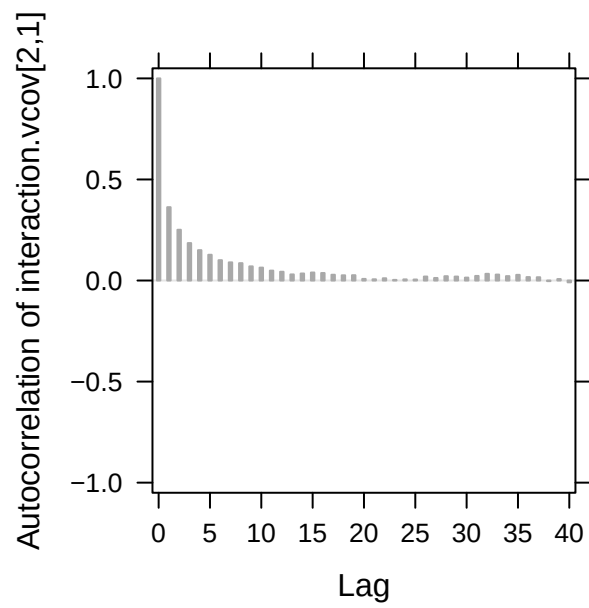
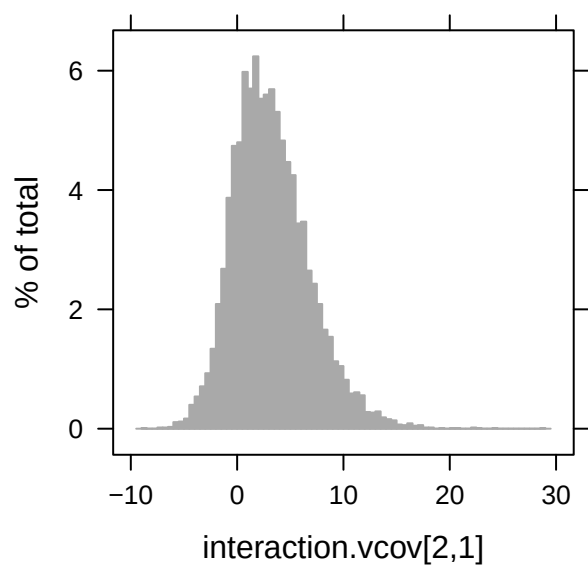
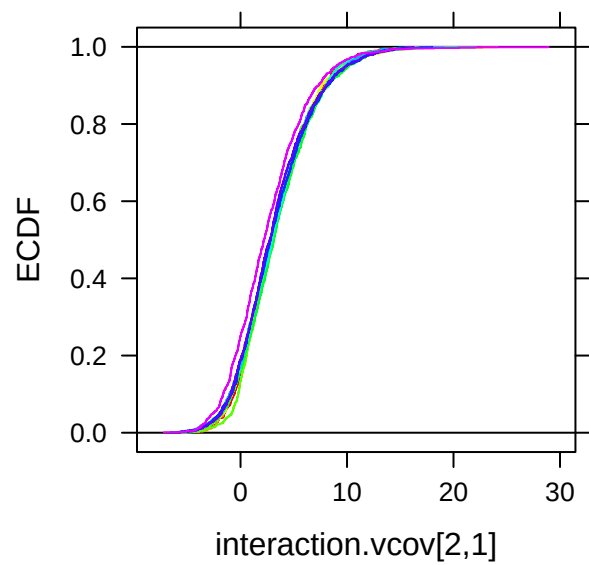
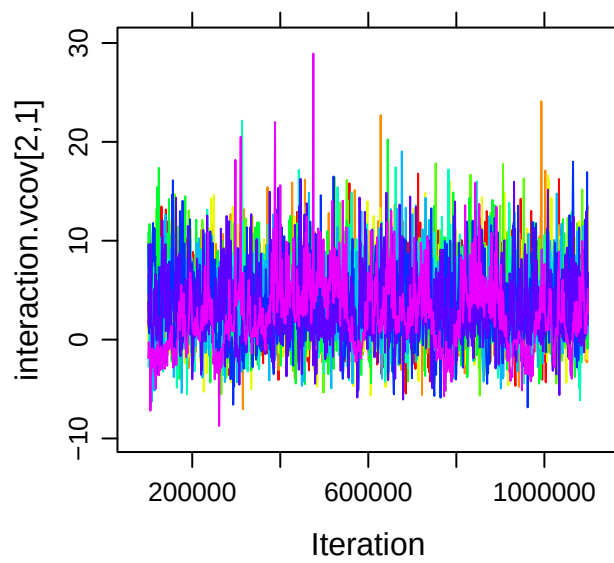


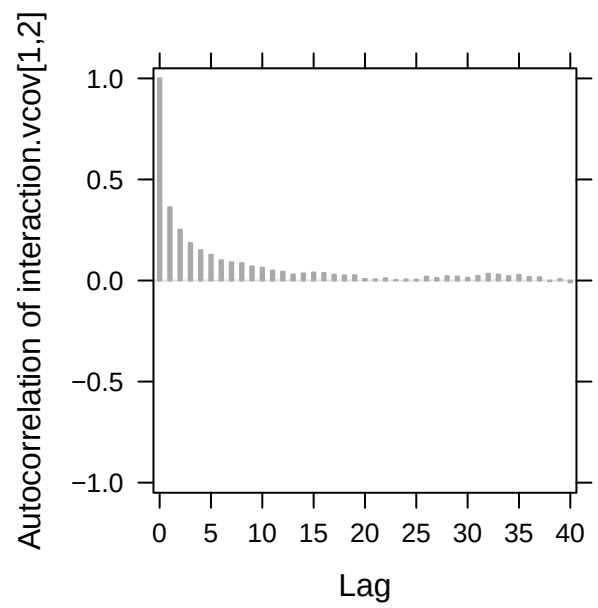
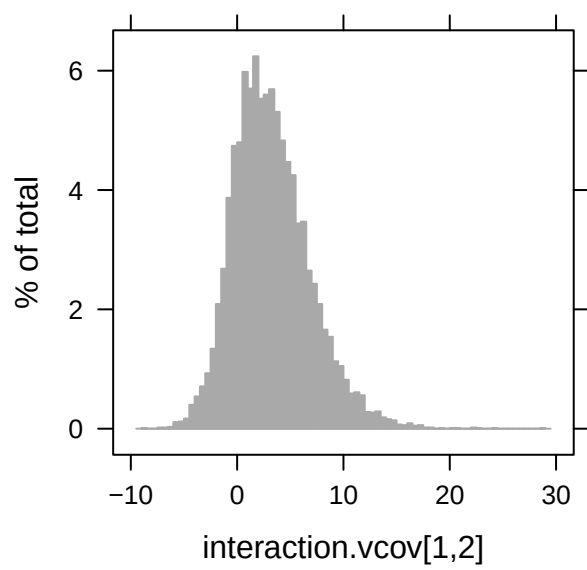
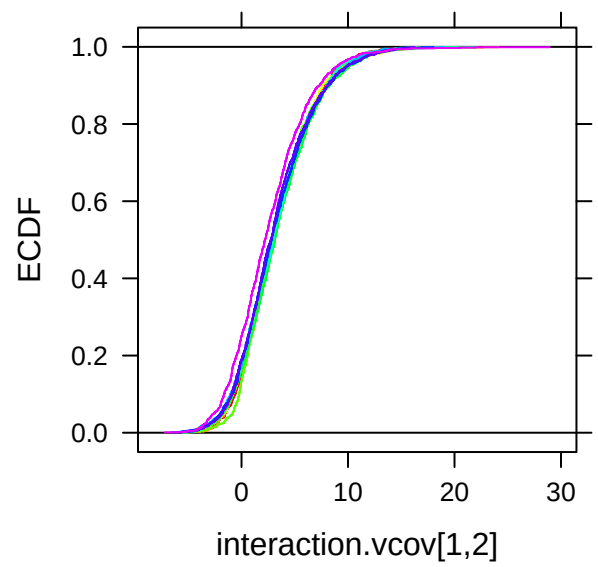
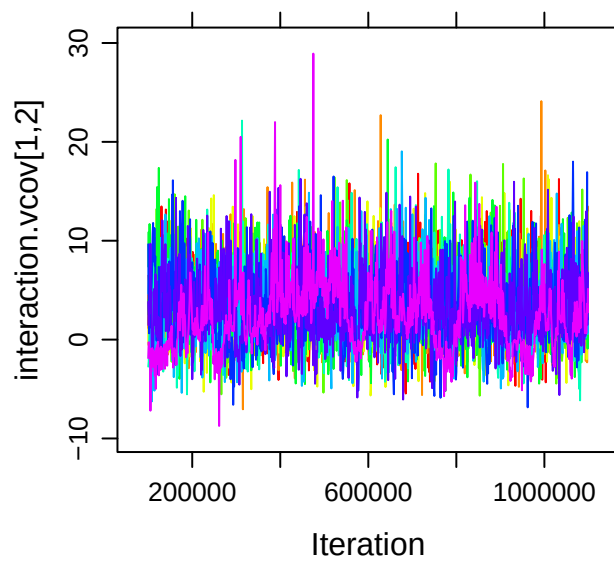


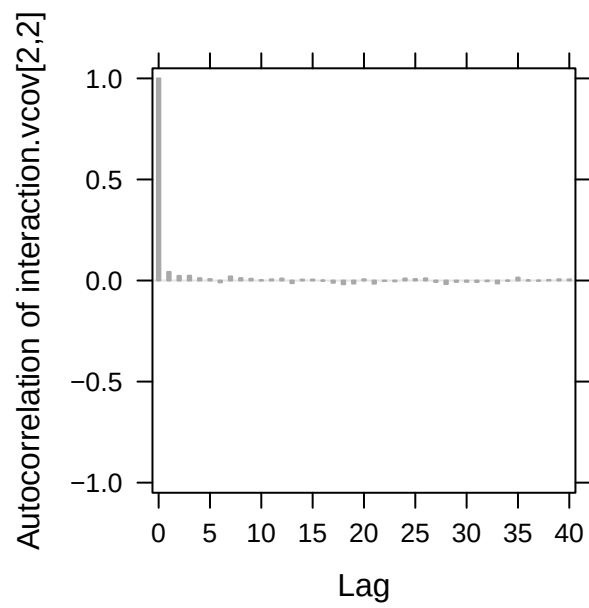
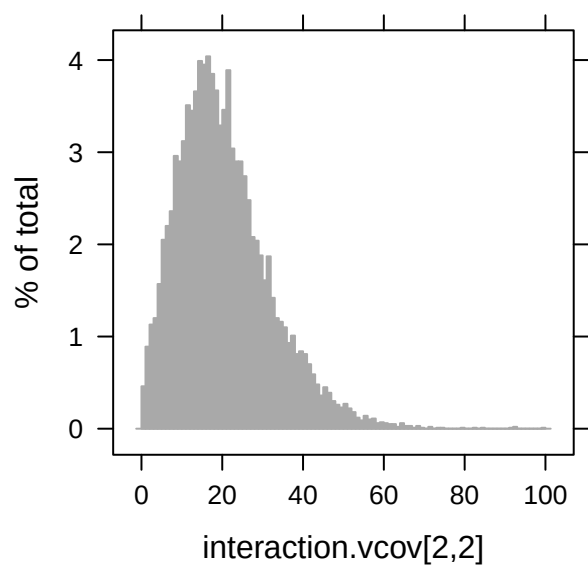
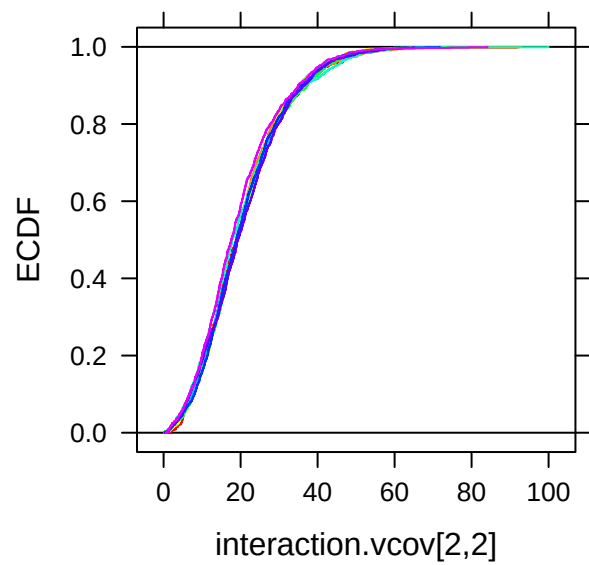
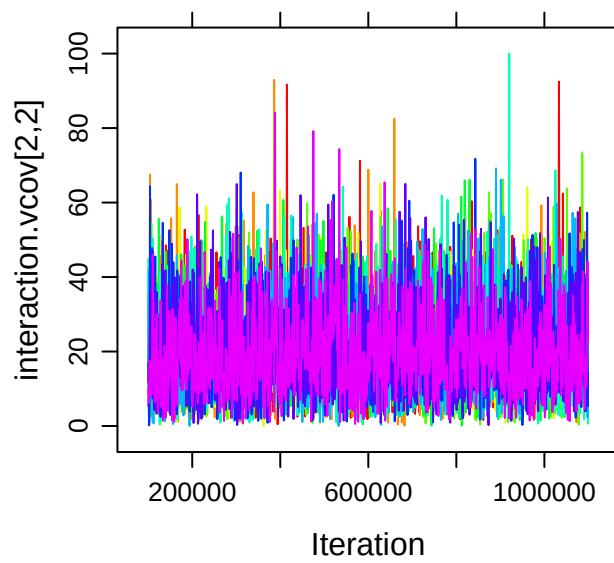


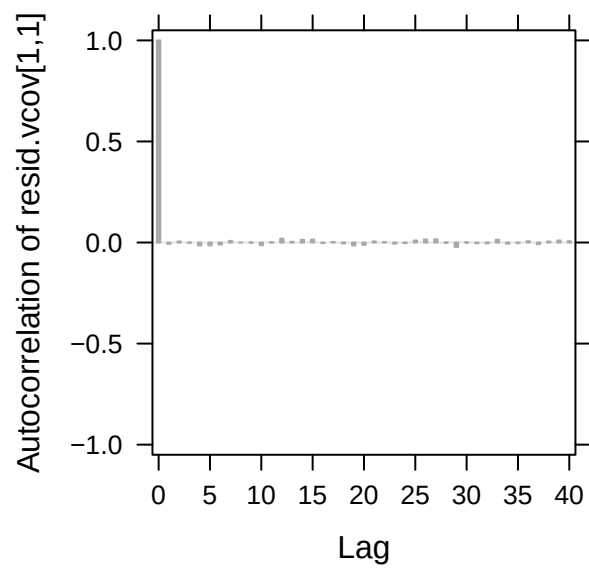
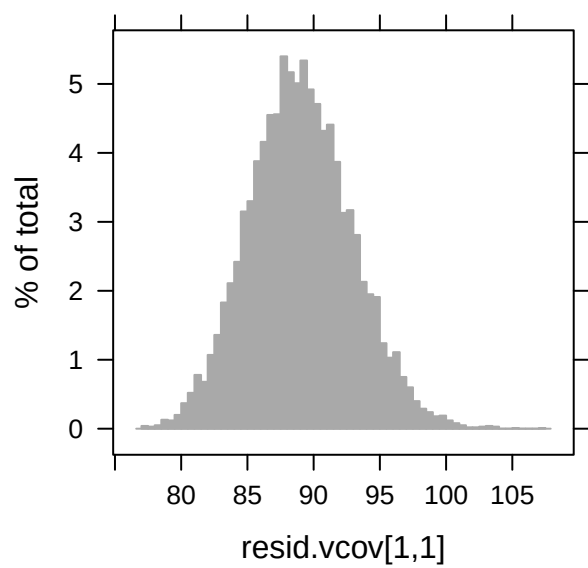
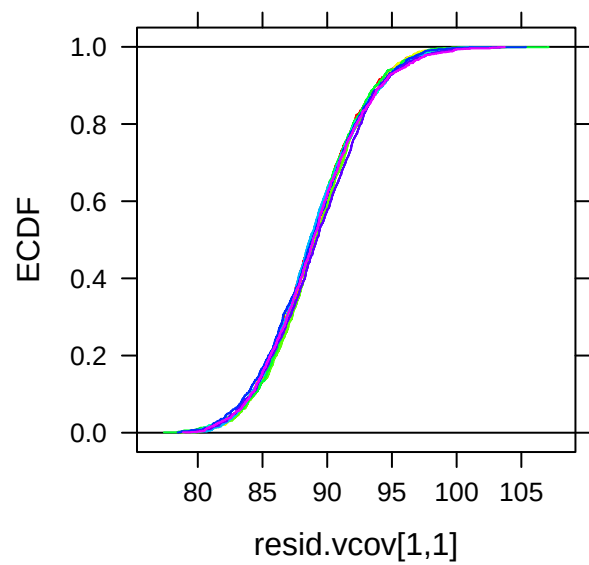
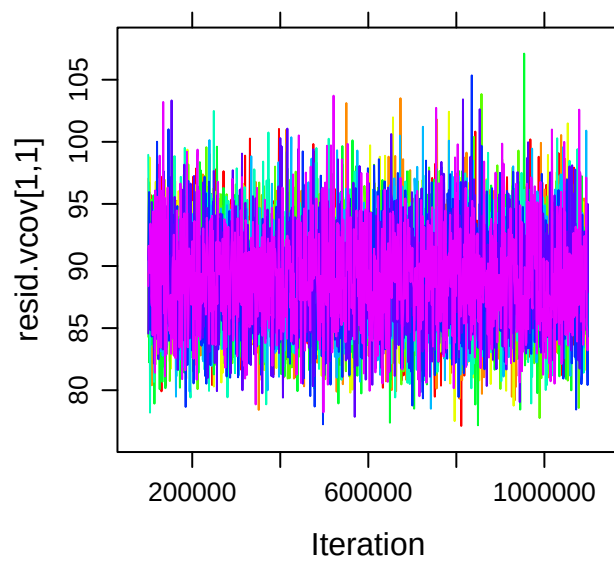


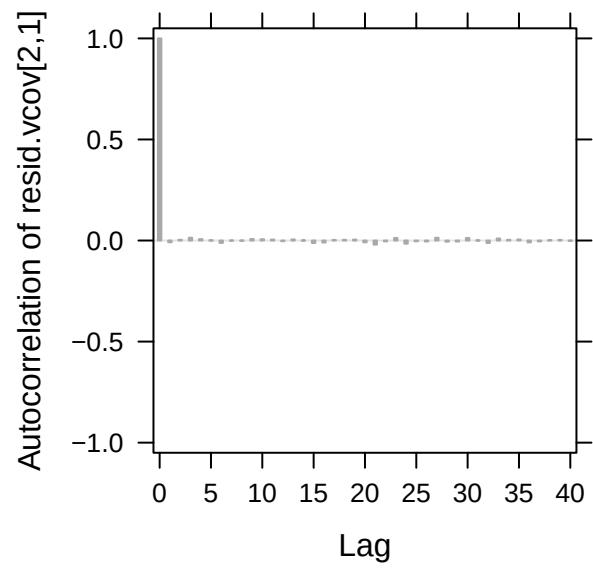
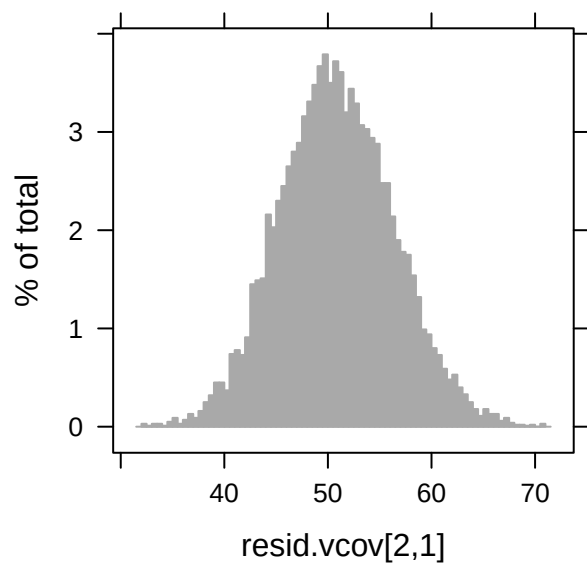
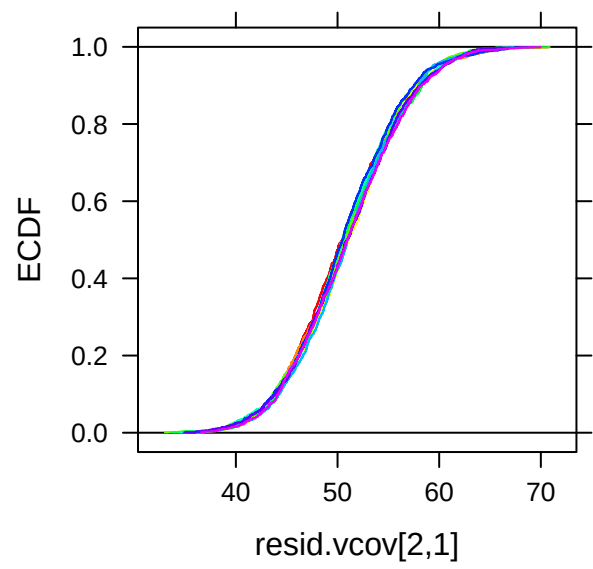
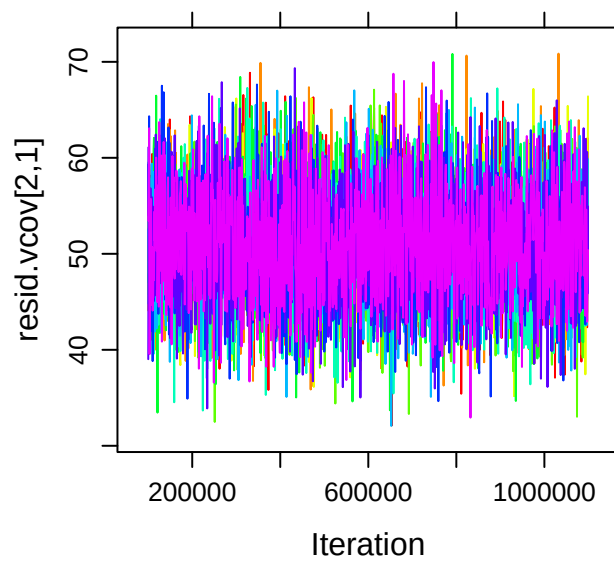


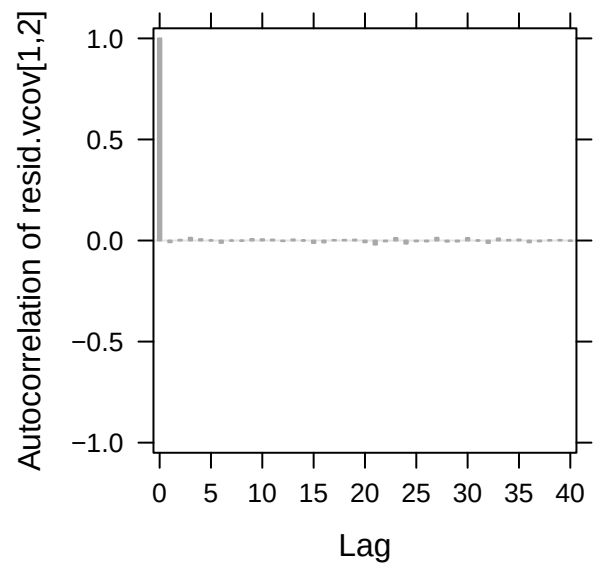
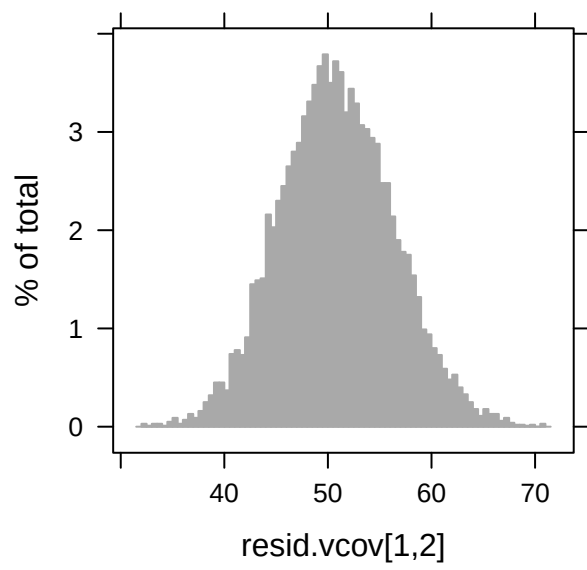
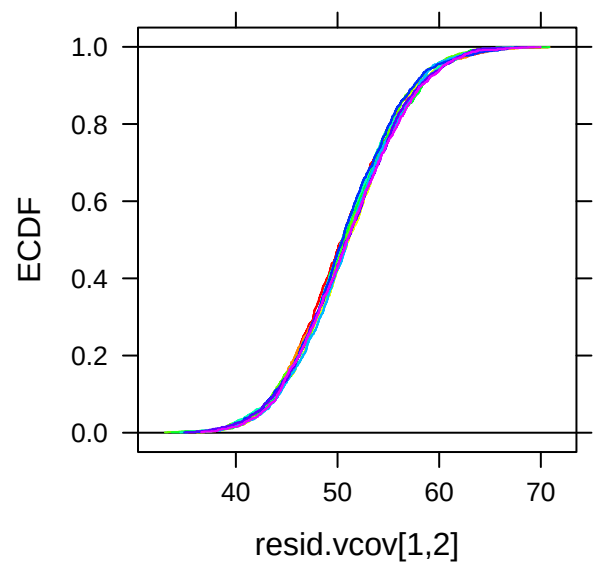
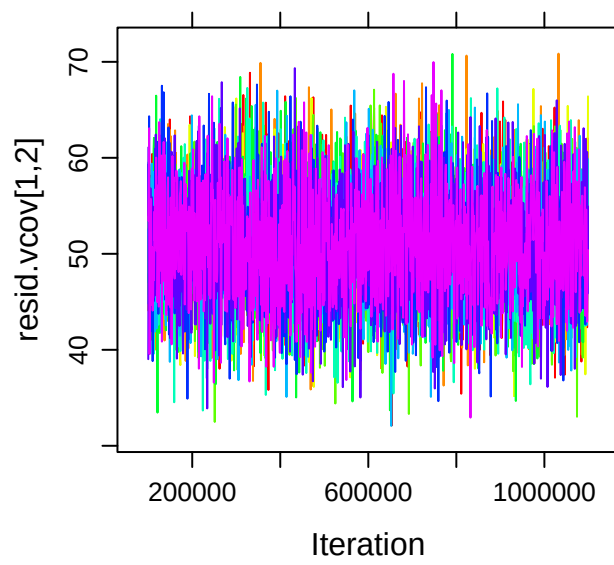


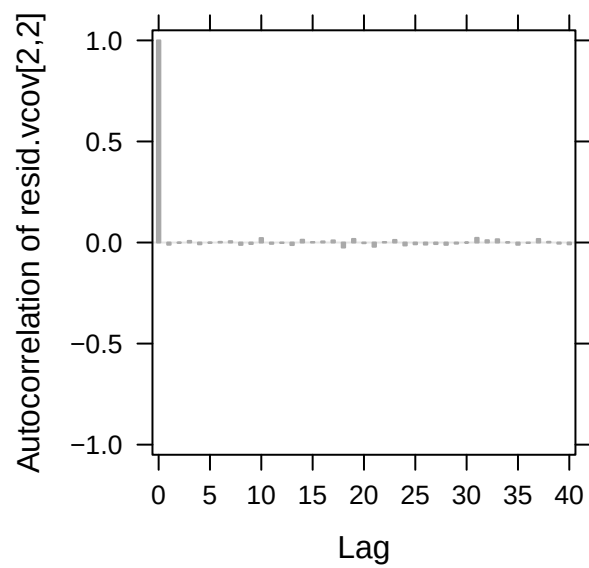
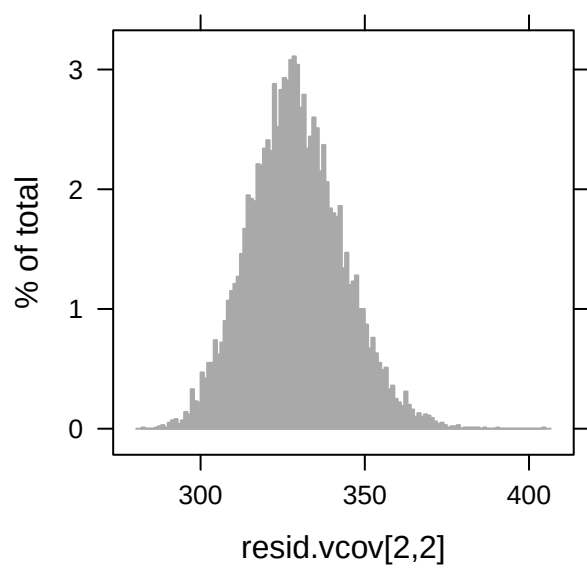
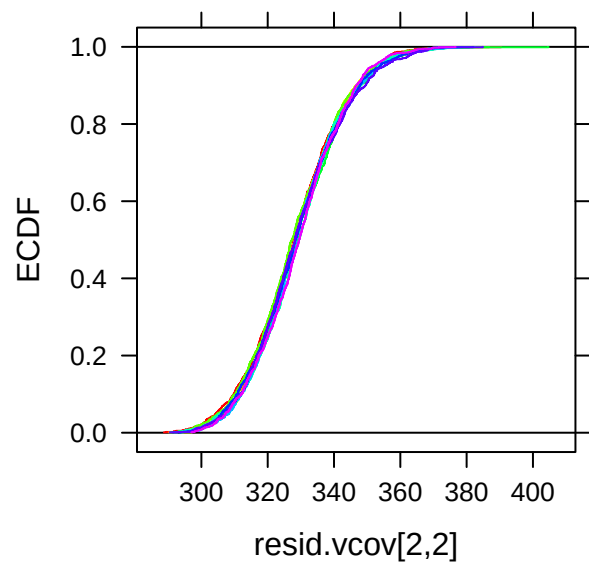
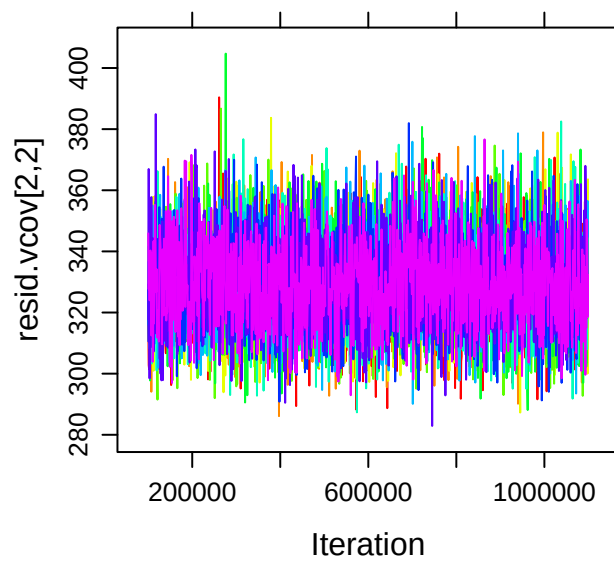


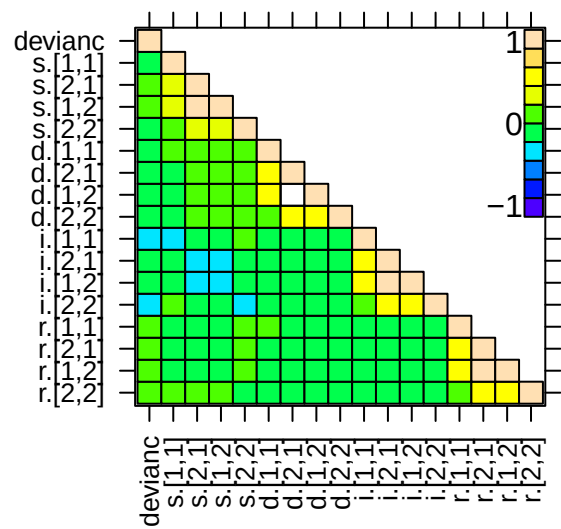












Model II

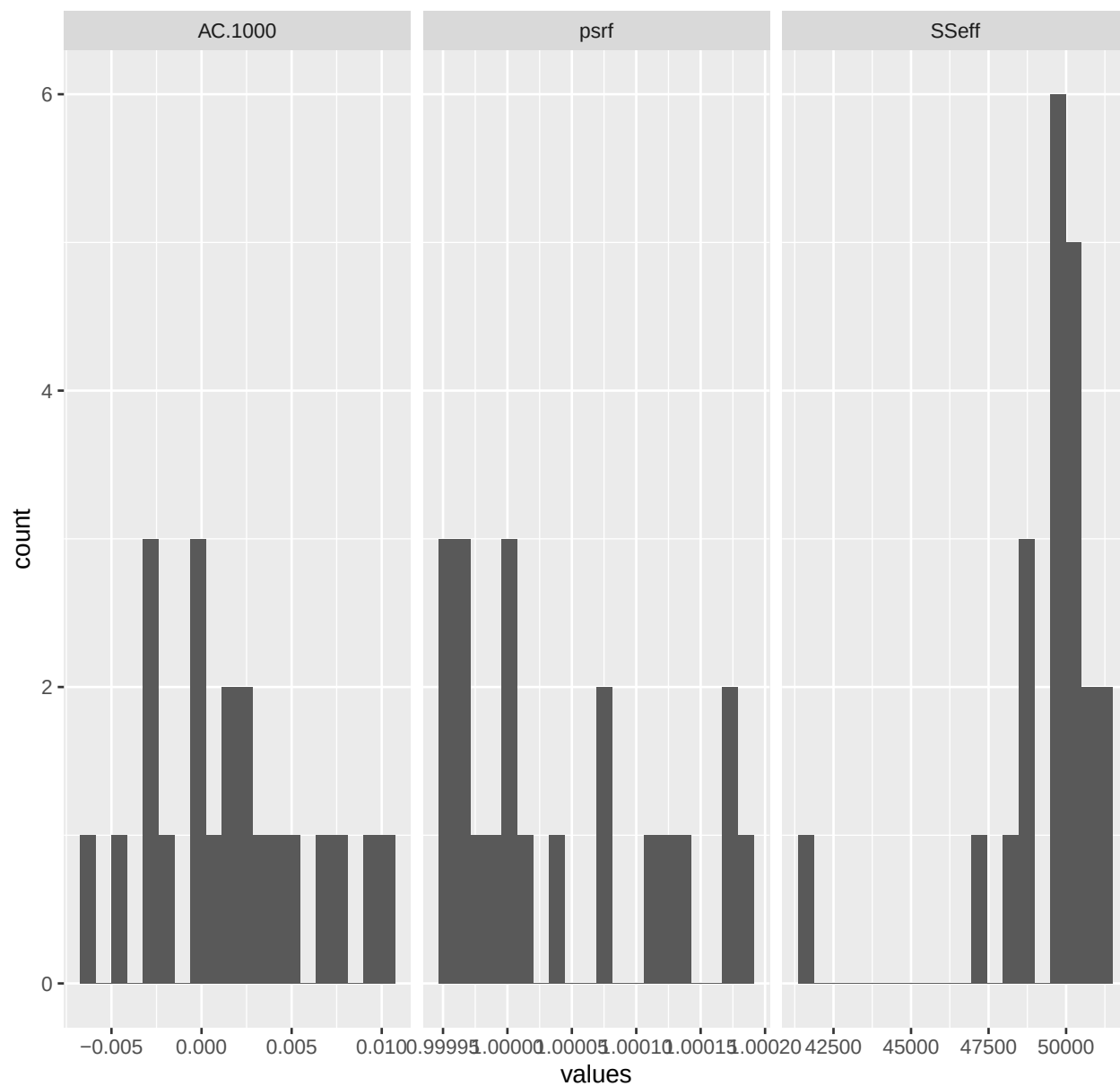
Number of chains: 10

Burnin: 50100

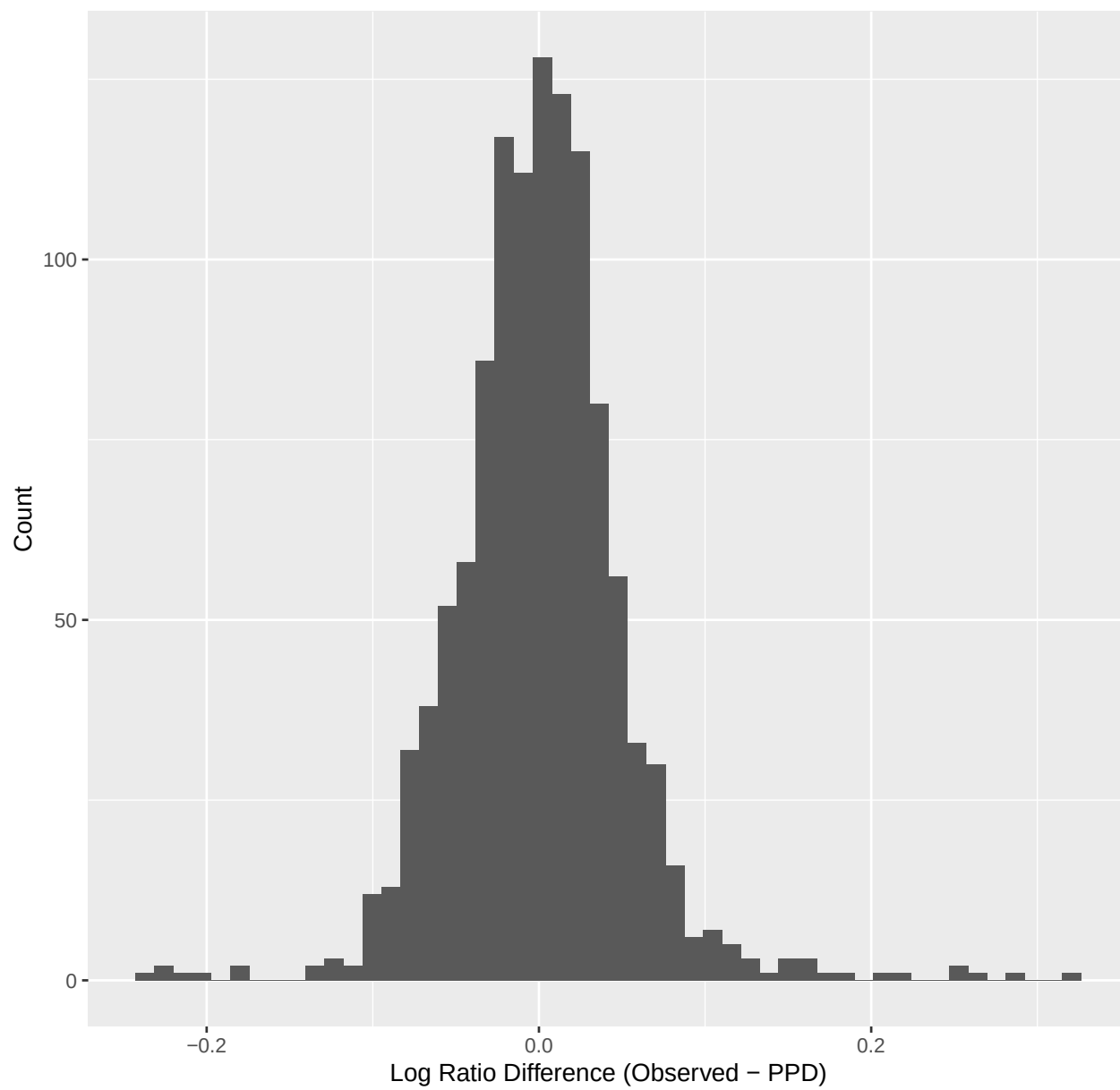
Samples per chain: 5000

Total samples: 50000

A) Posterior distribution diagnostics



B) Posterior predictive check distribution

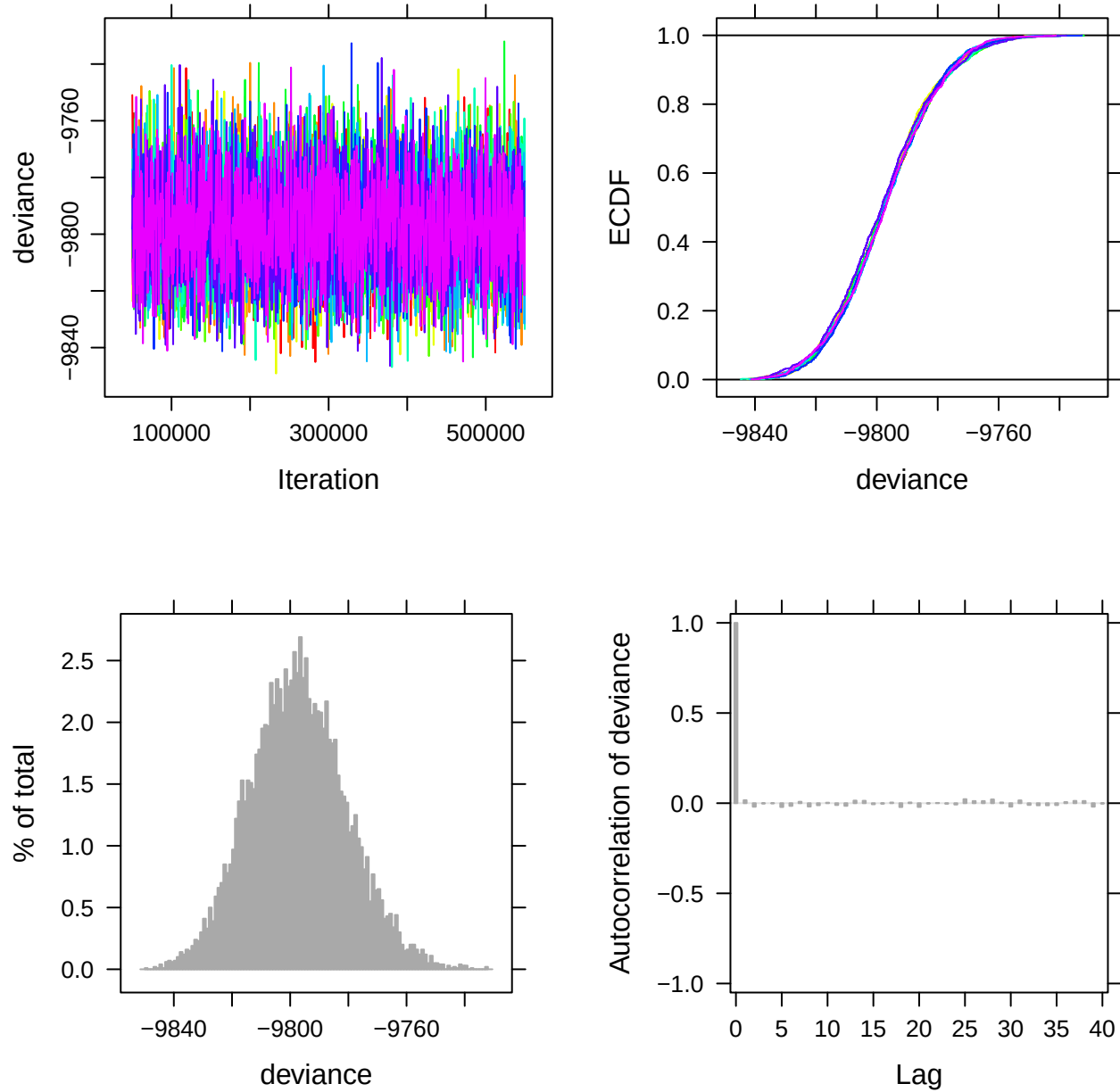


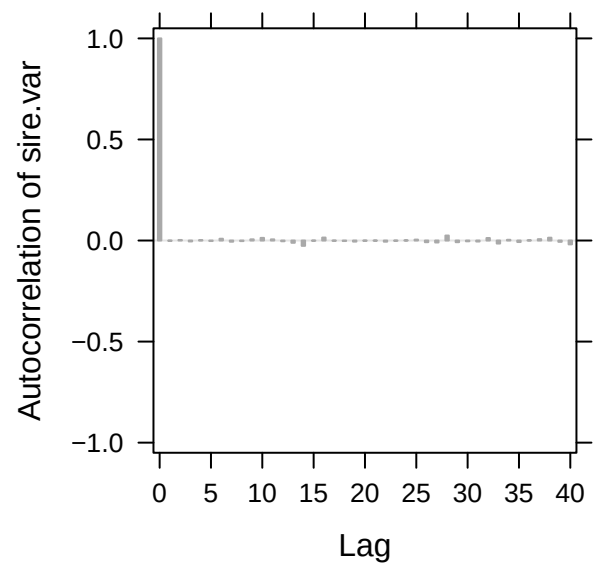
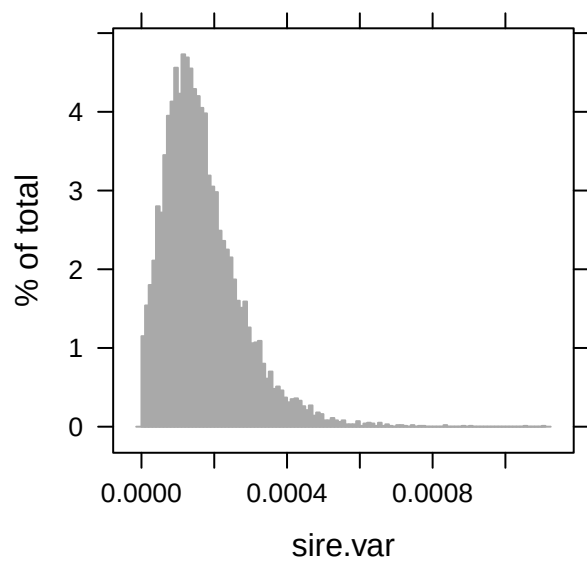
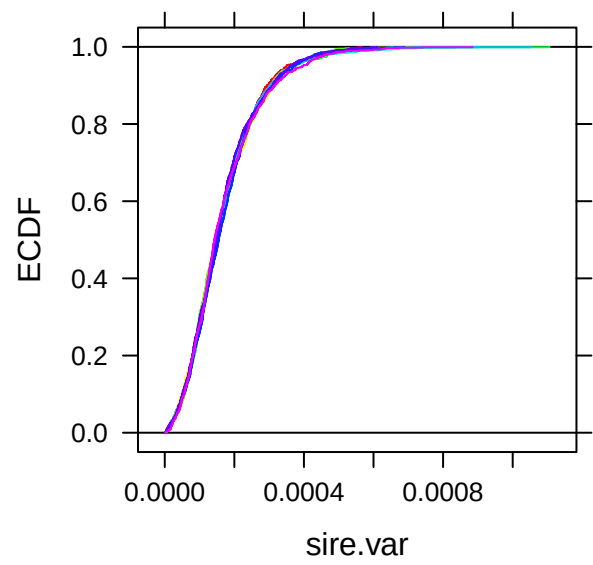
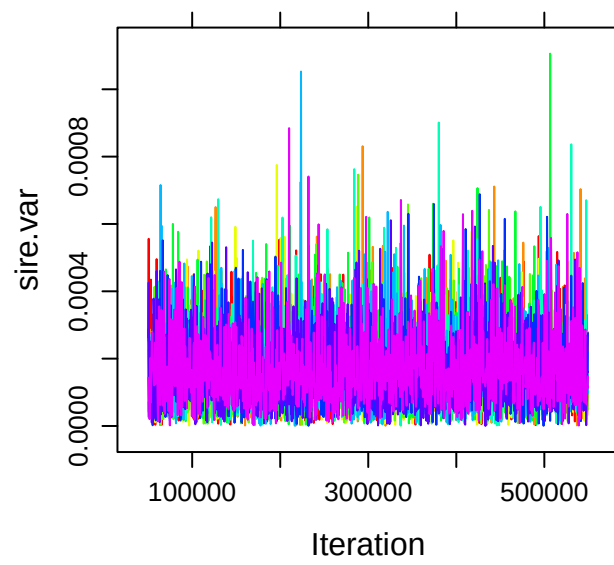
C) By-parameter posterior summaries

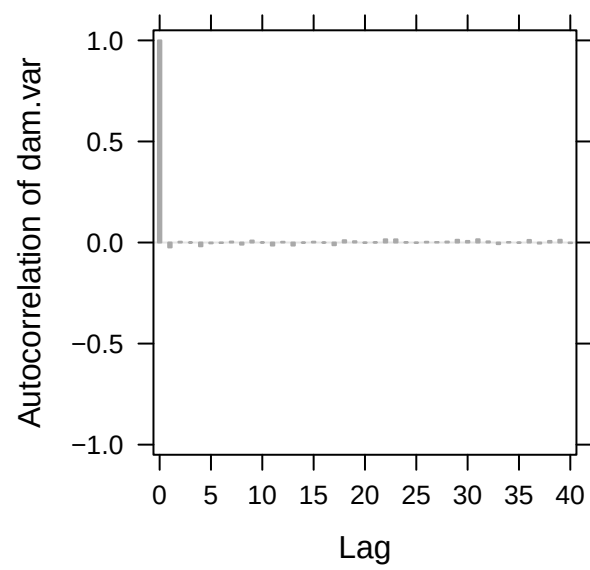
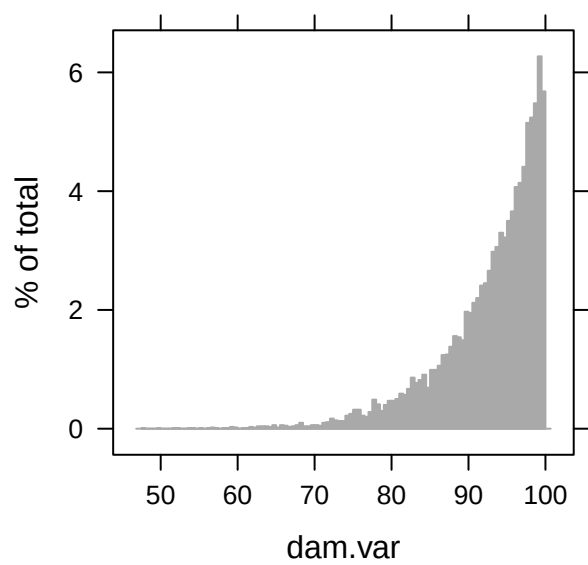
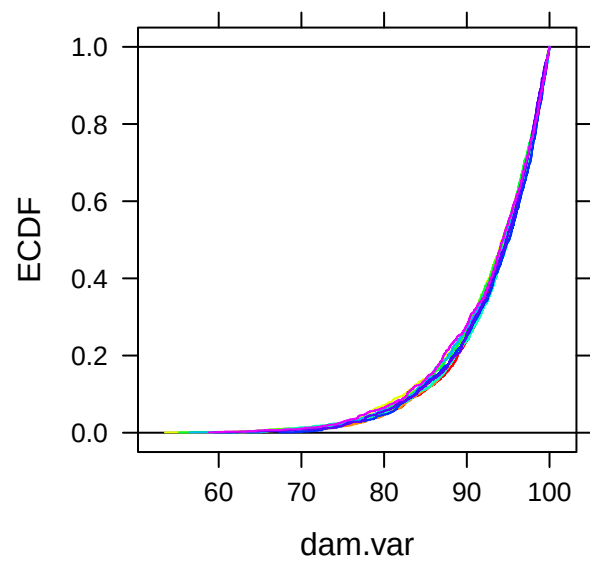
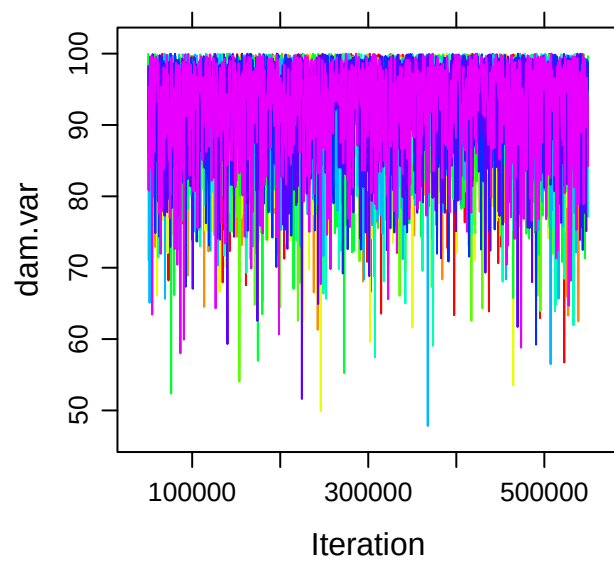
Generating summary statistics and plots (these will NOT be saved for reuse)...

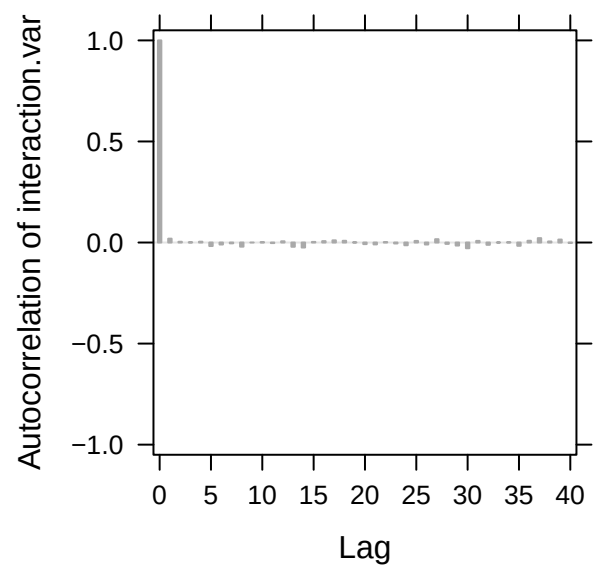
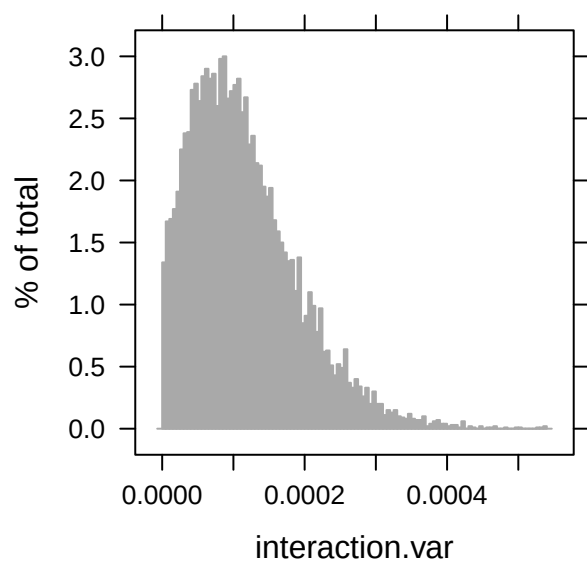
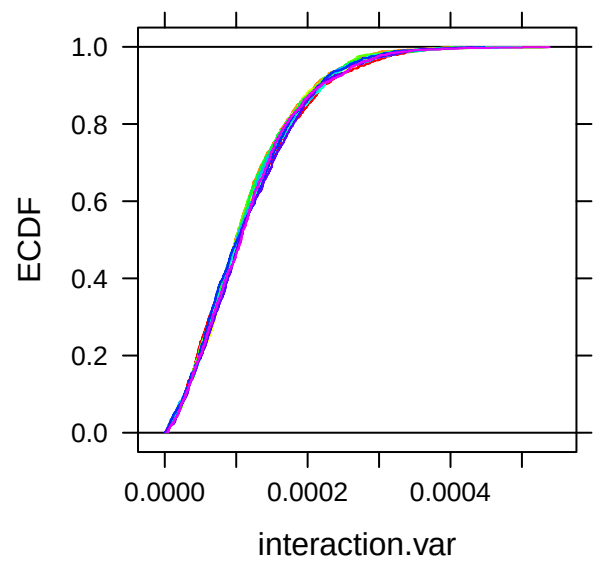
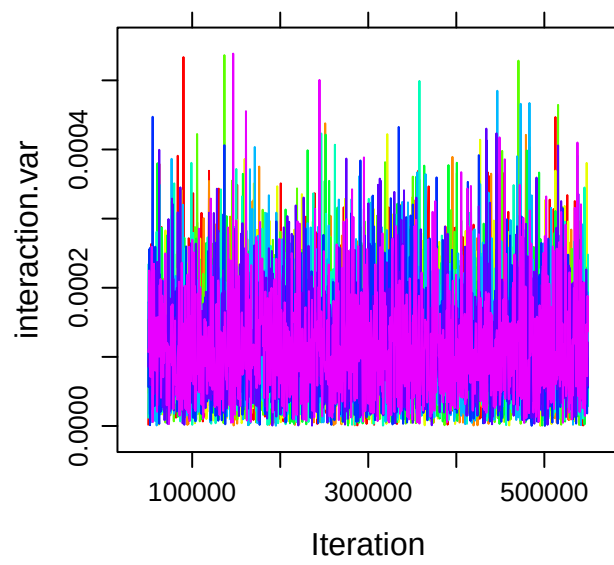
Calculating summary statistics...

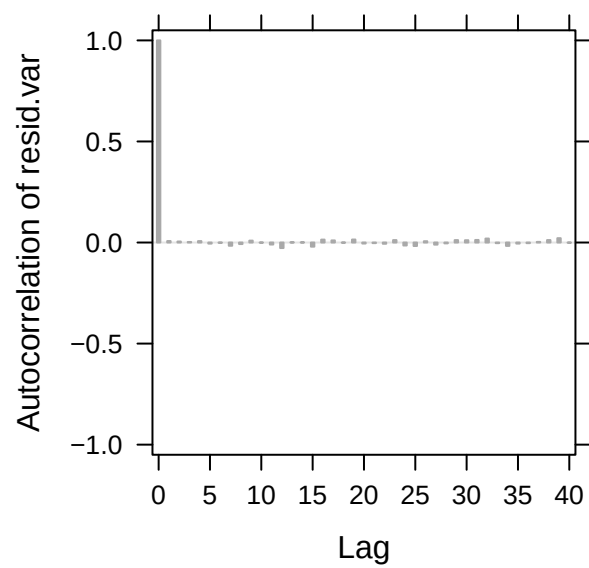
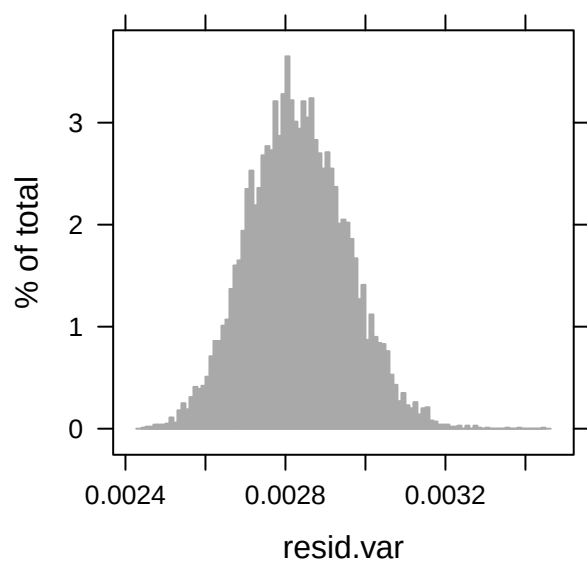
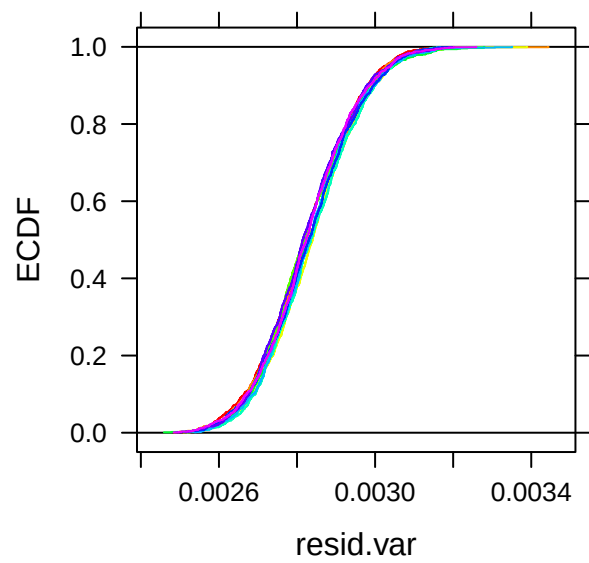
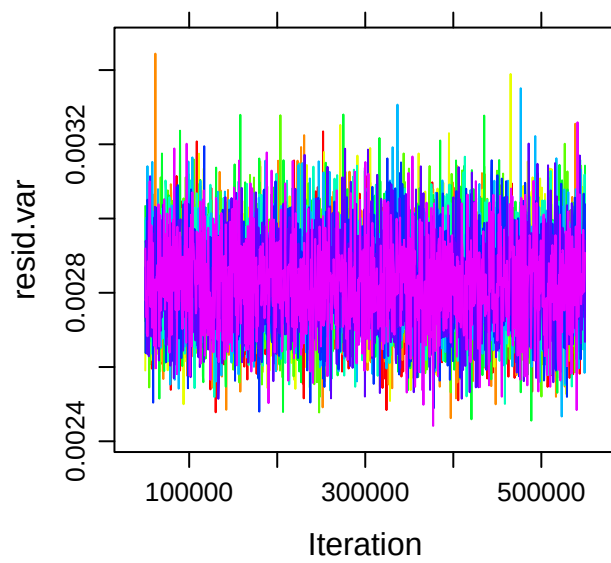
Calculating the Gelman-Rubin statistic for 21 variables....

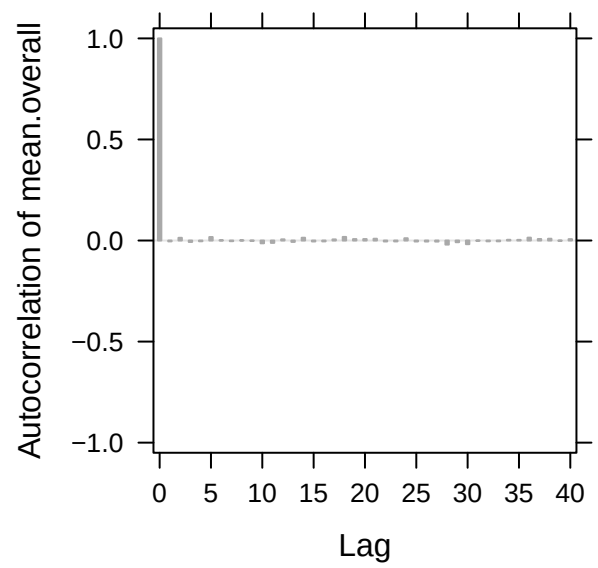
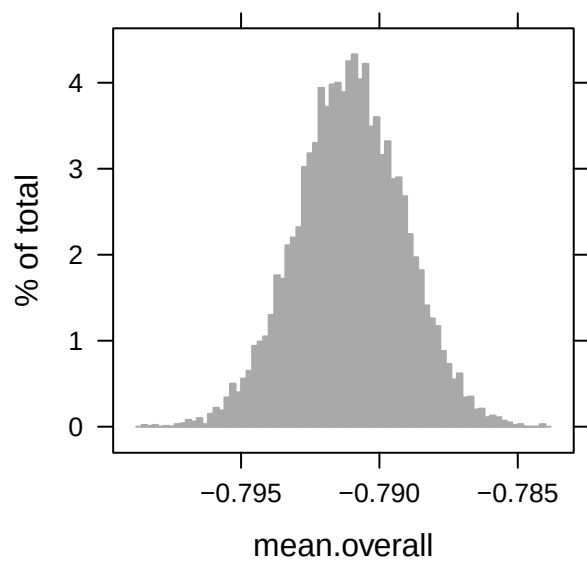
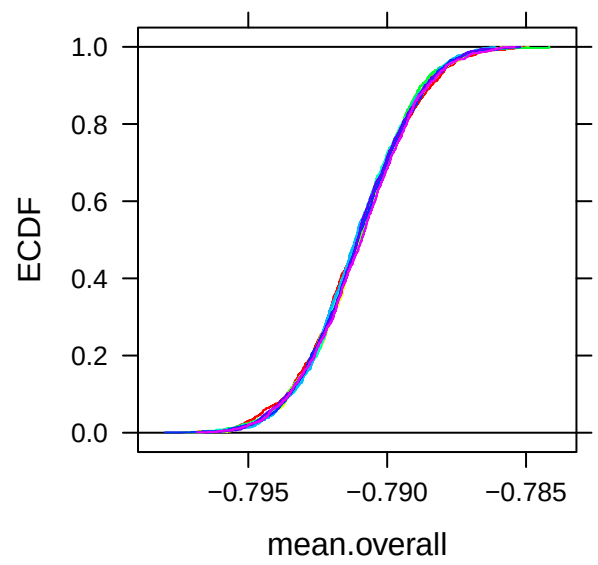
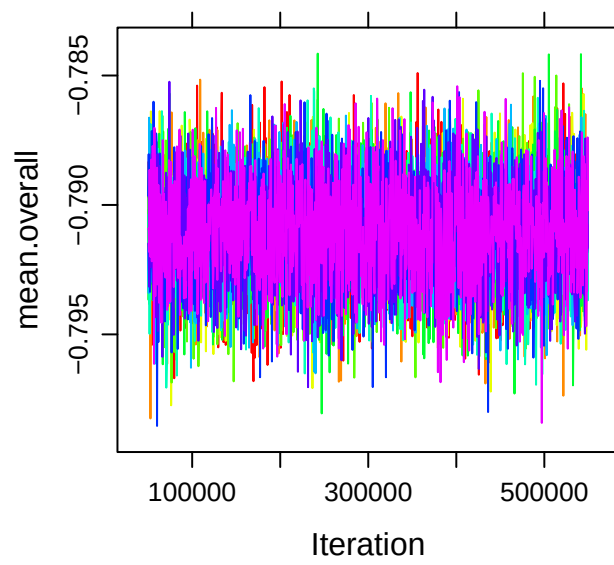


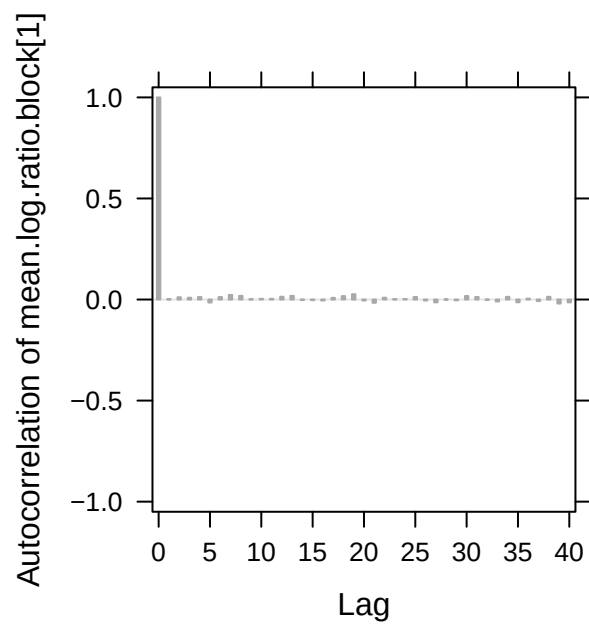
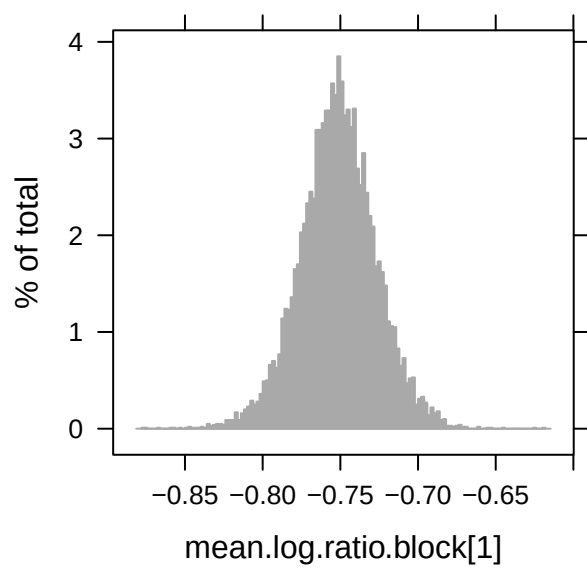
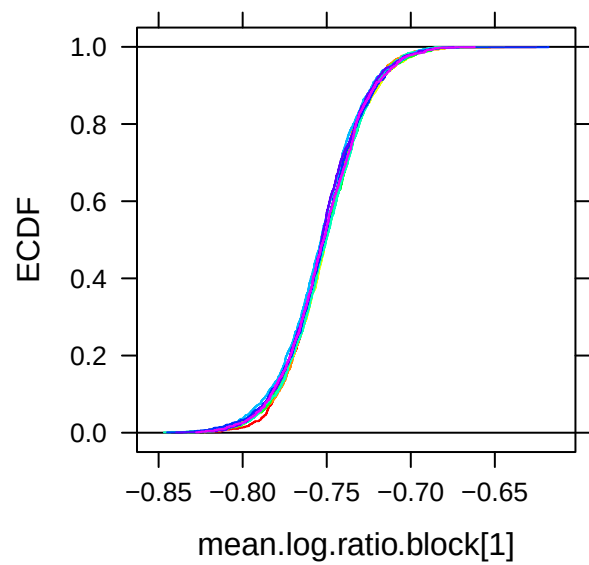
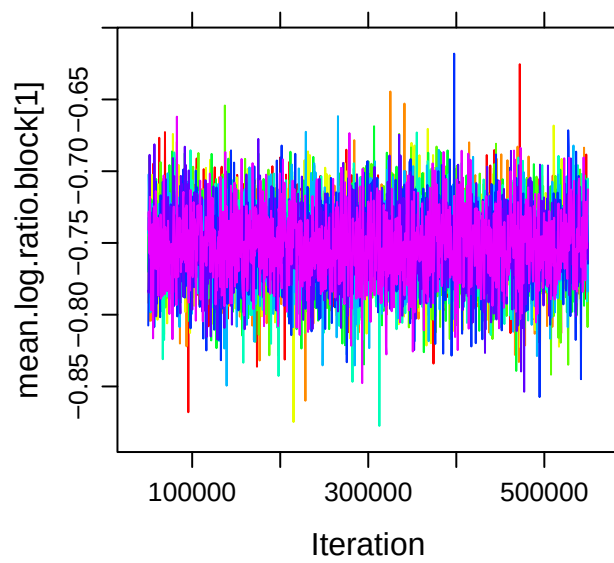


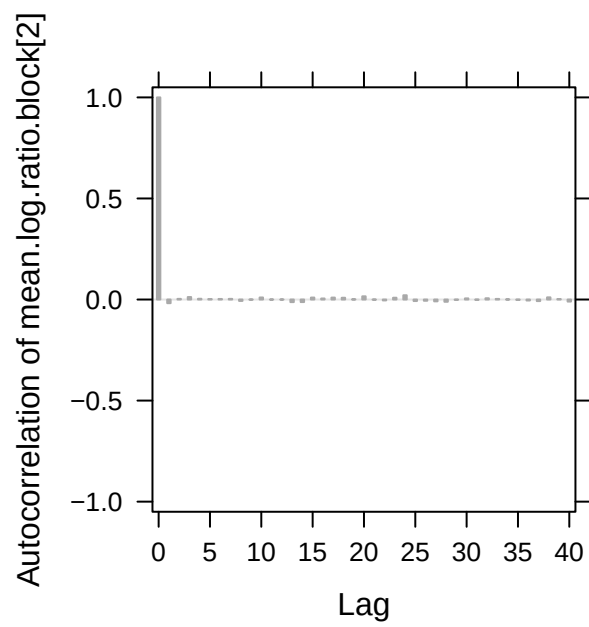
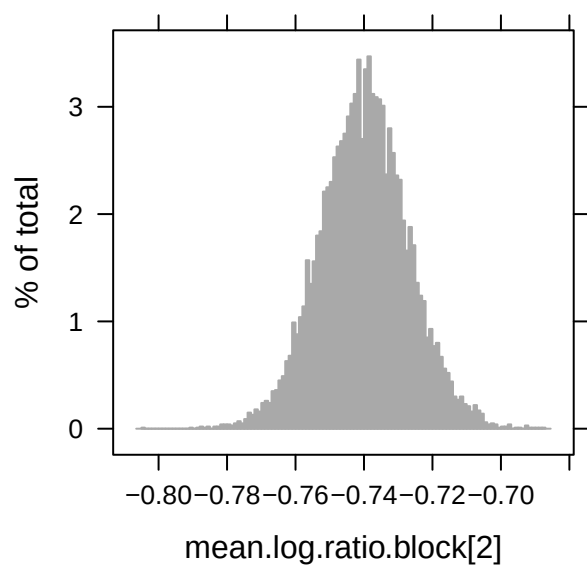
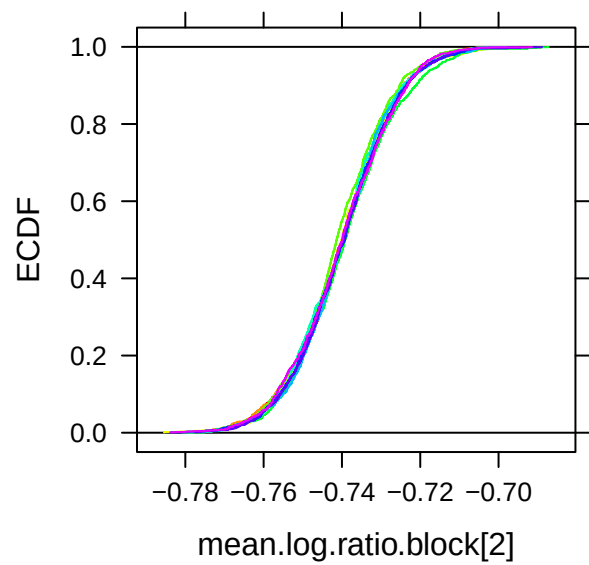
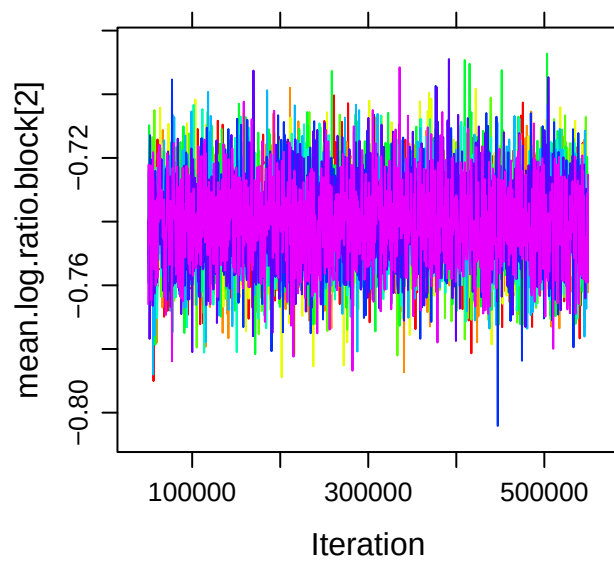


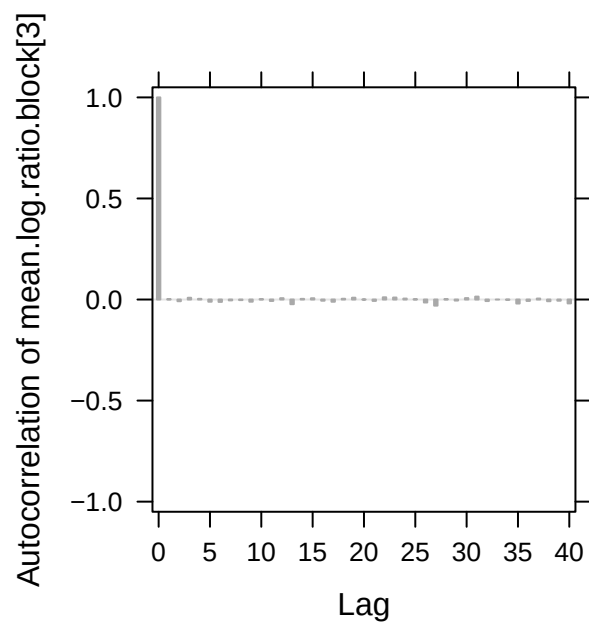
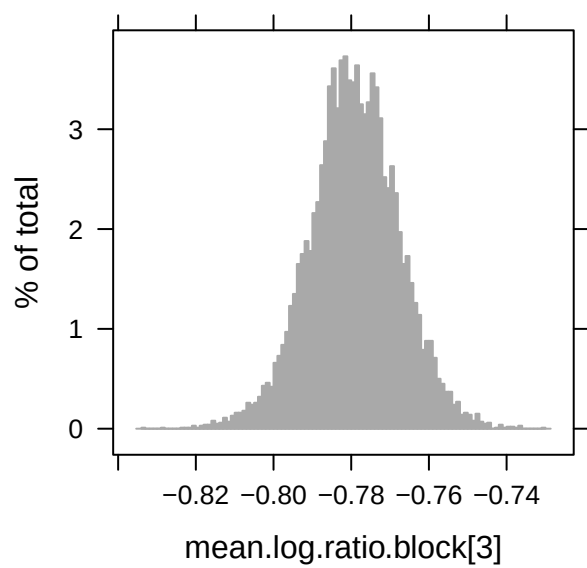
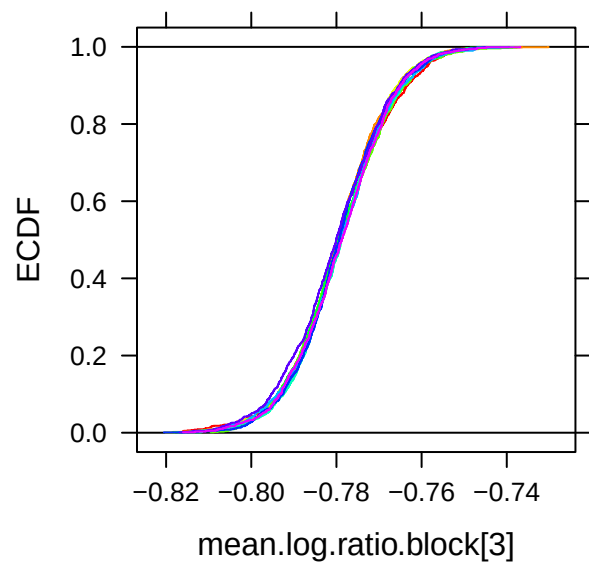
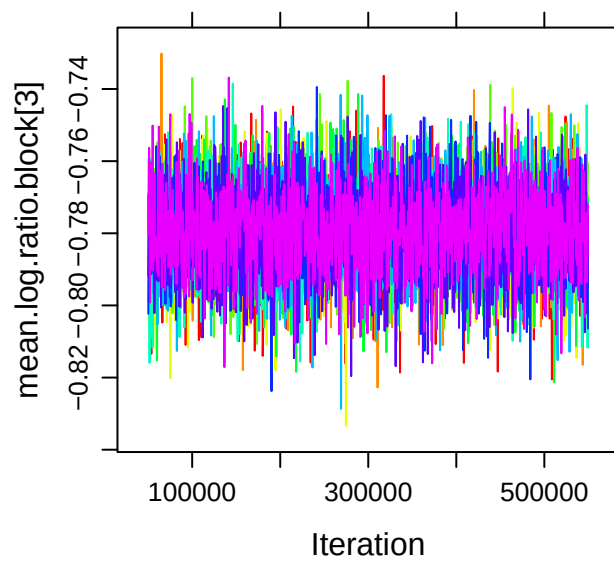


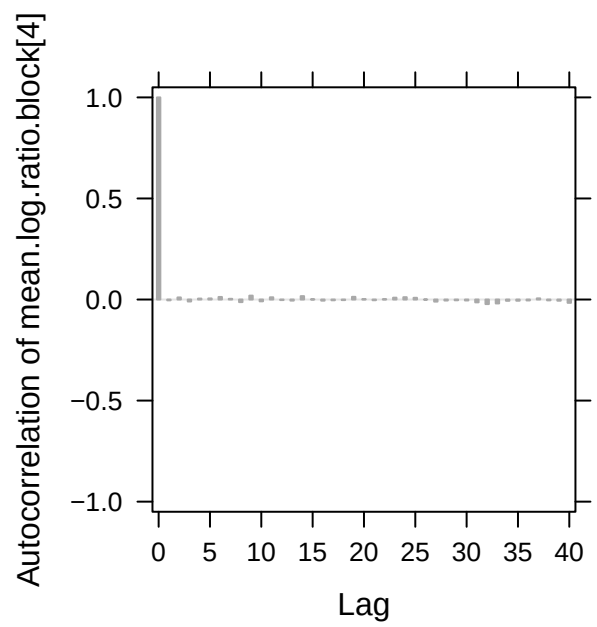
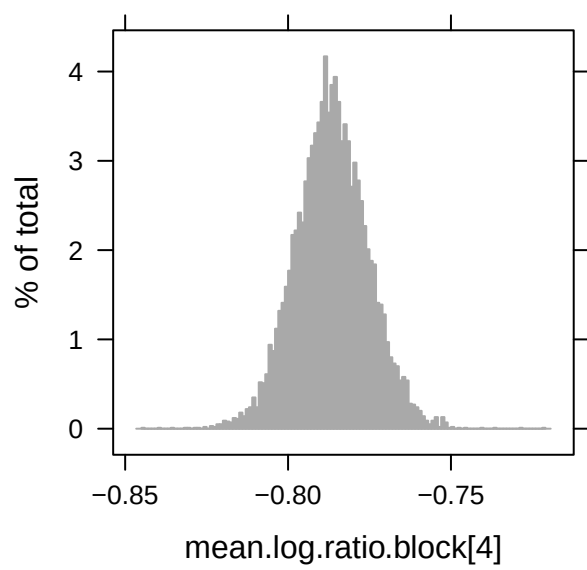
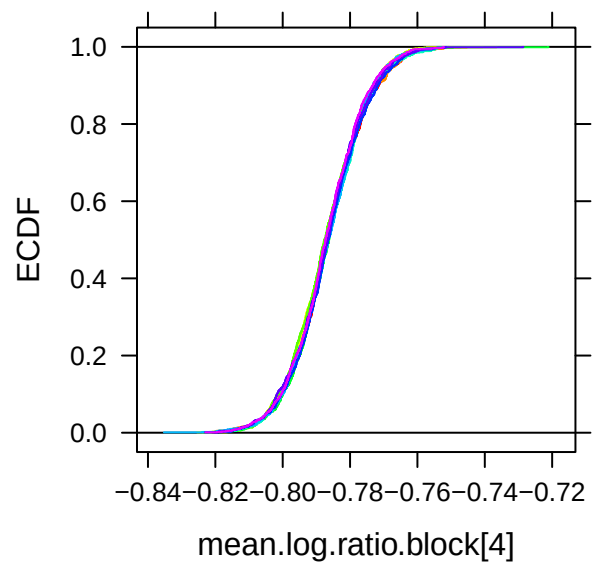
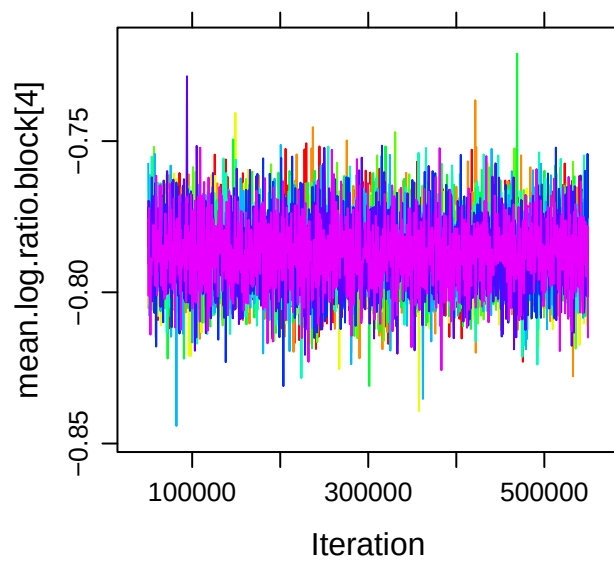


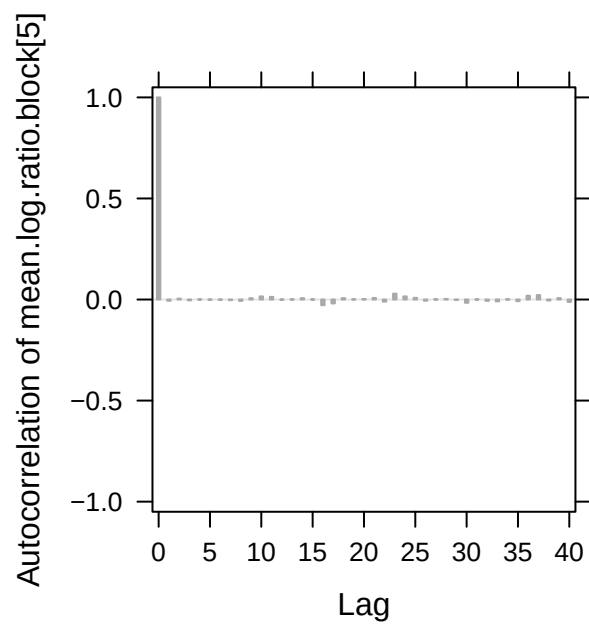
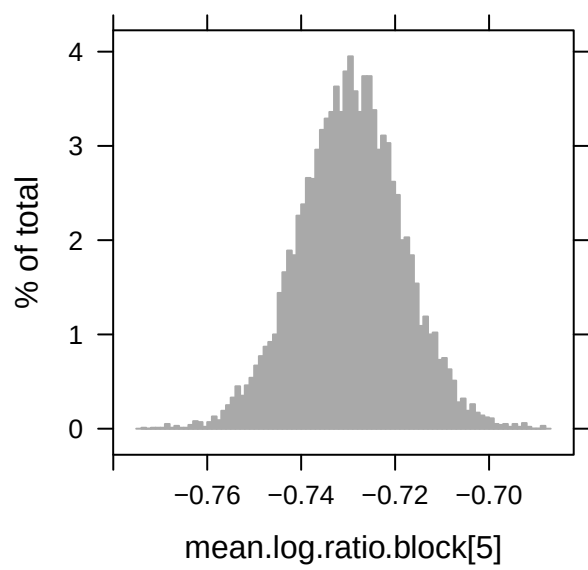
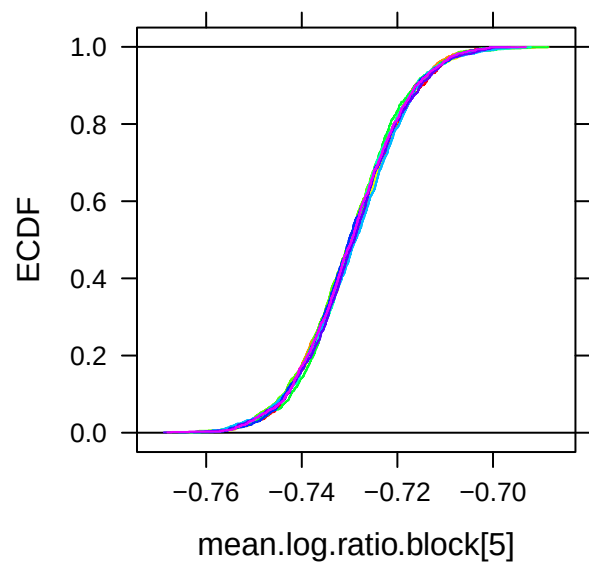
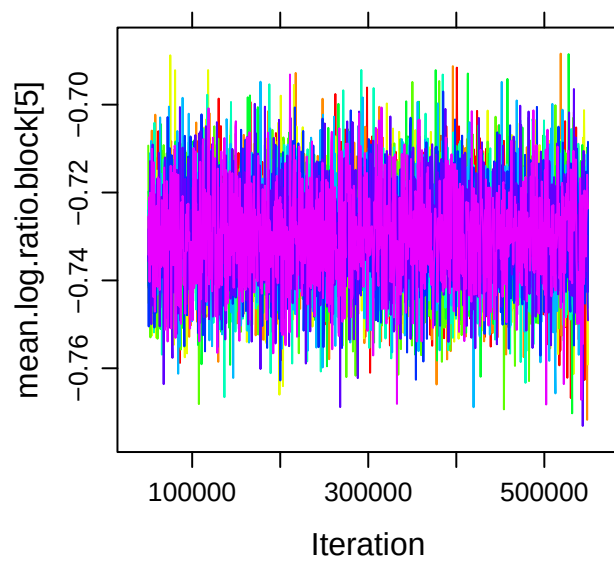


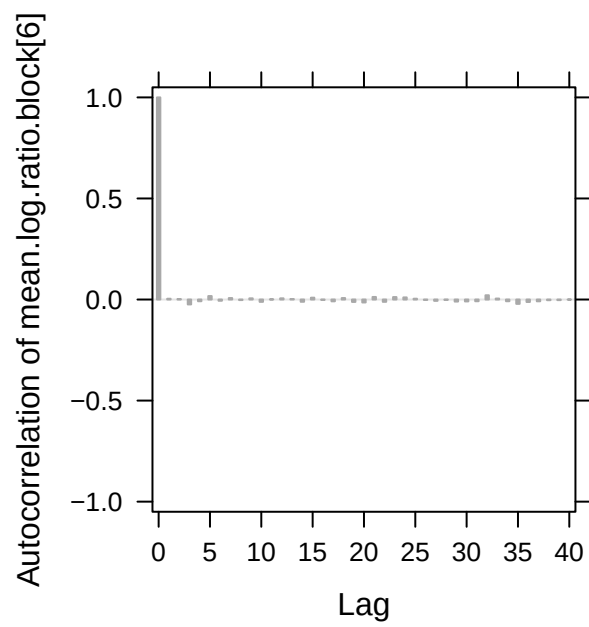
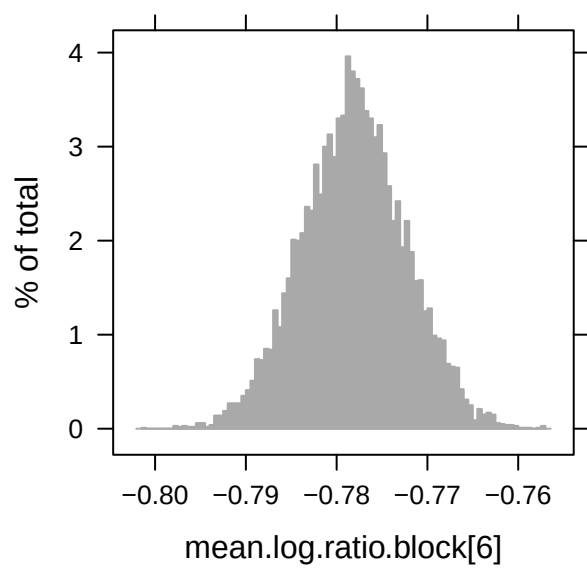
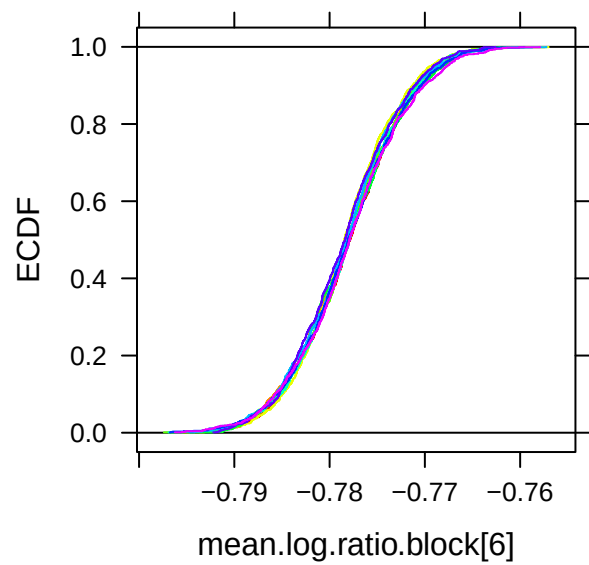
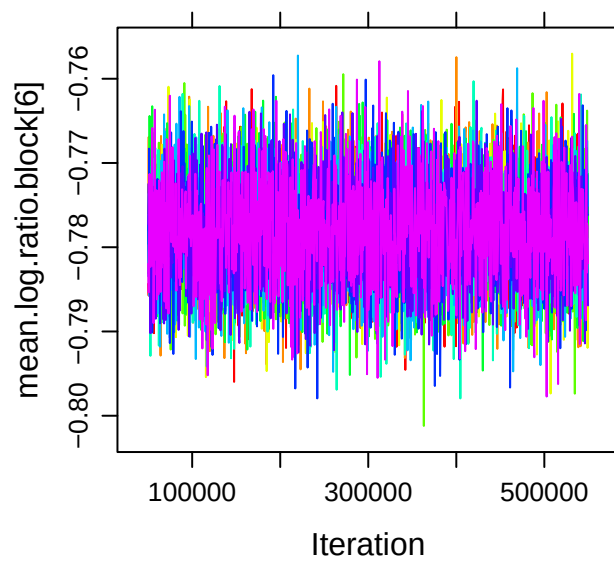


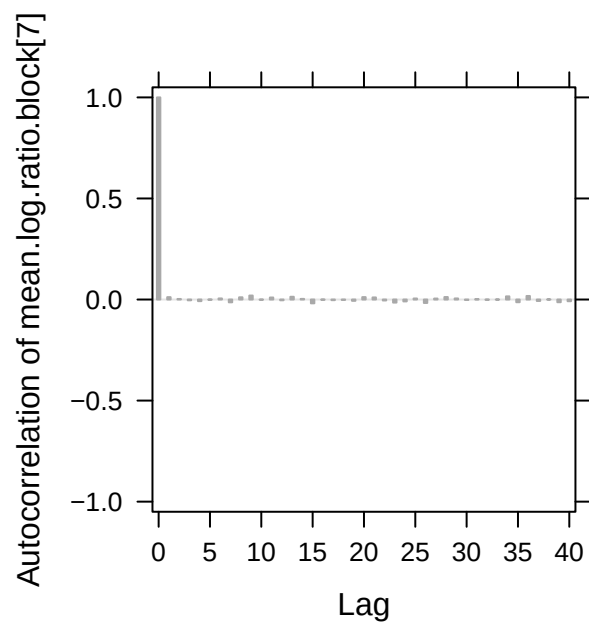
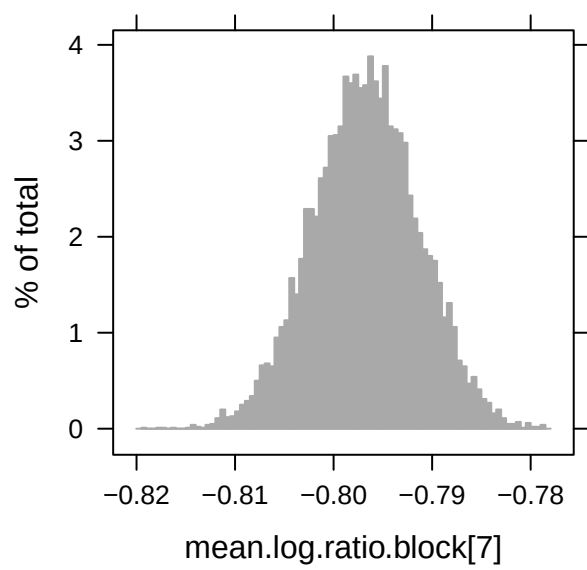
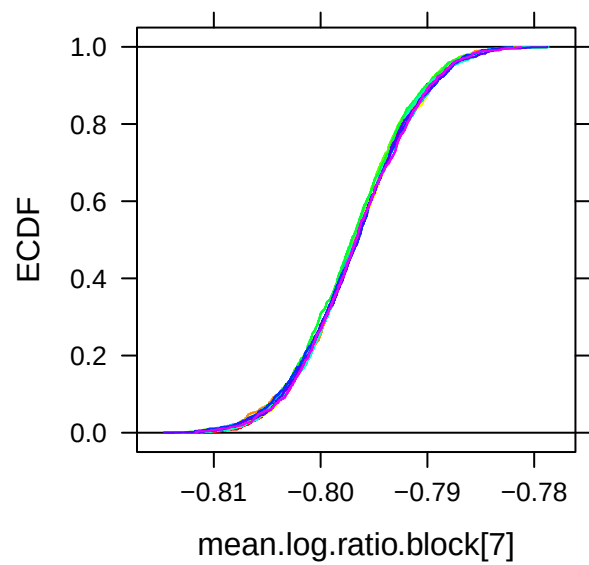
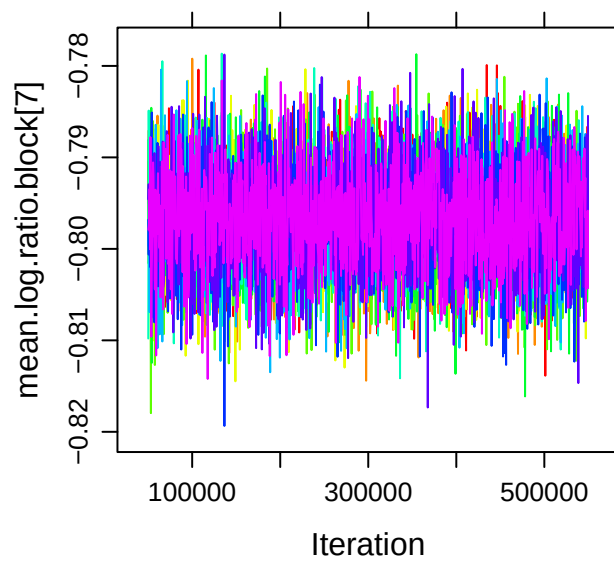


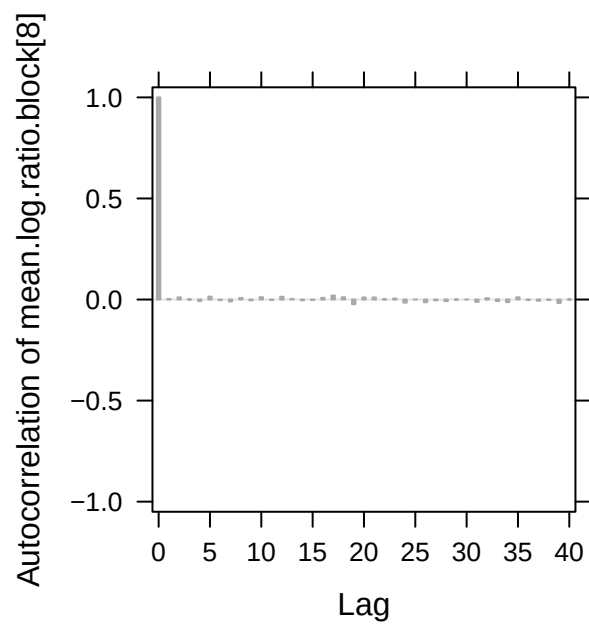
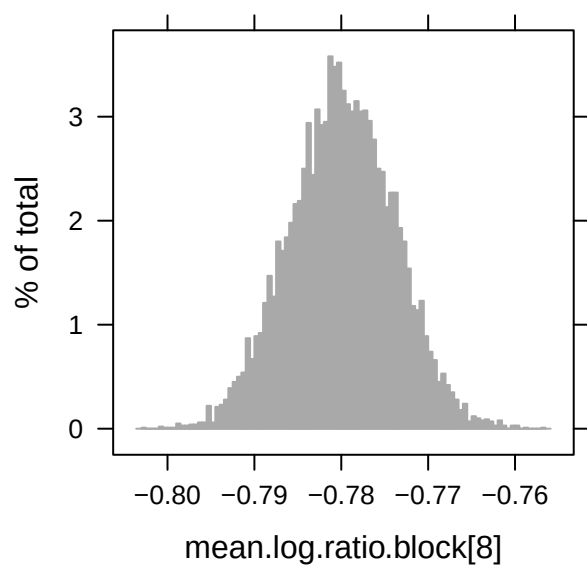
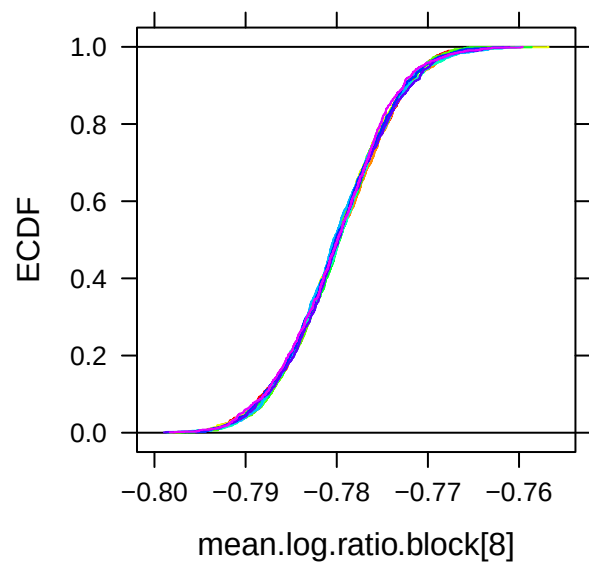
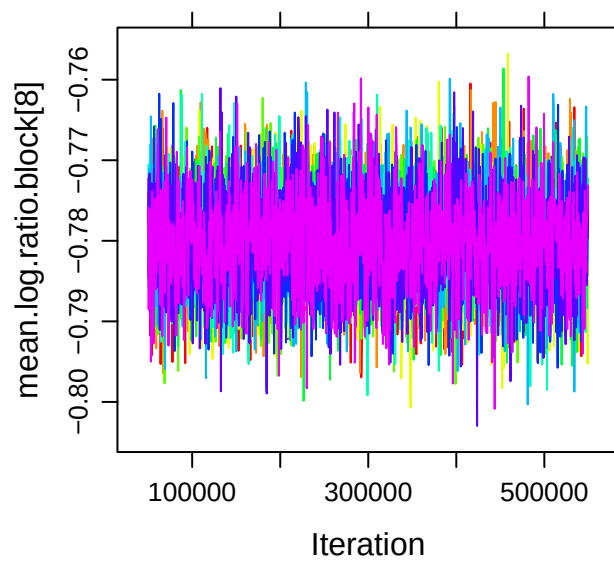


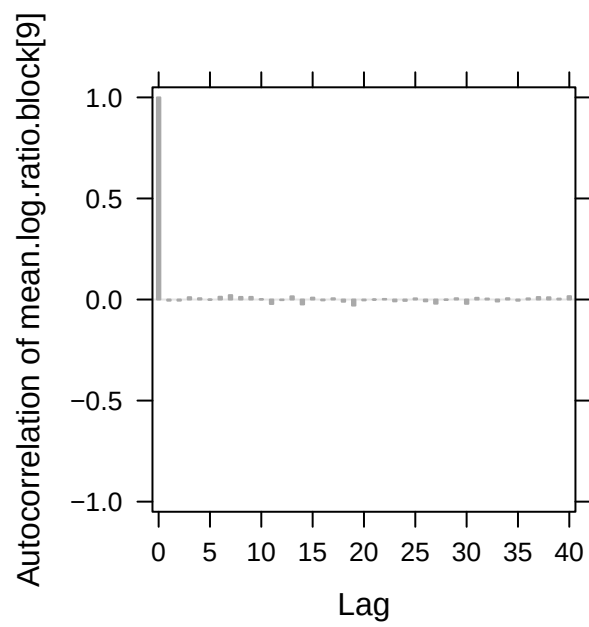
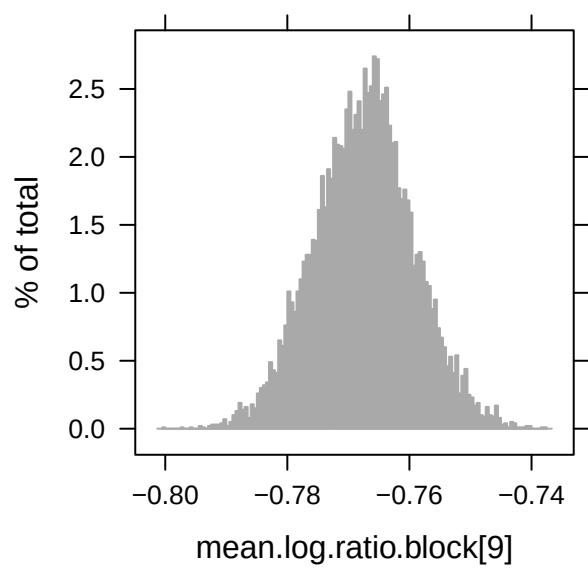
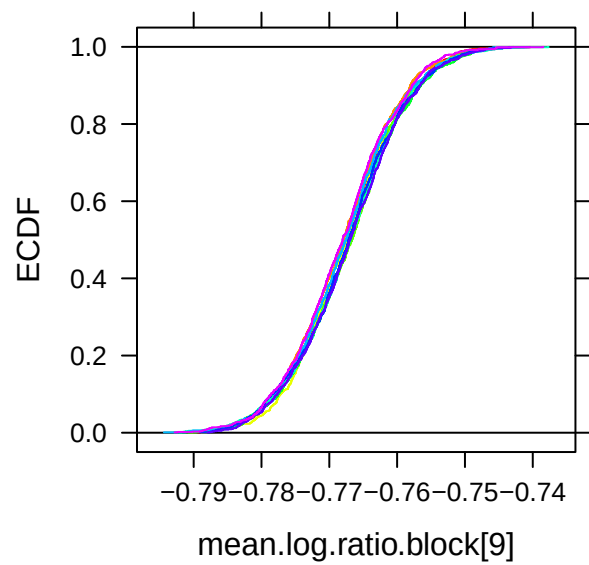
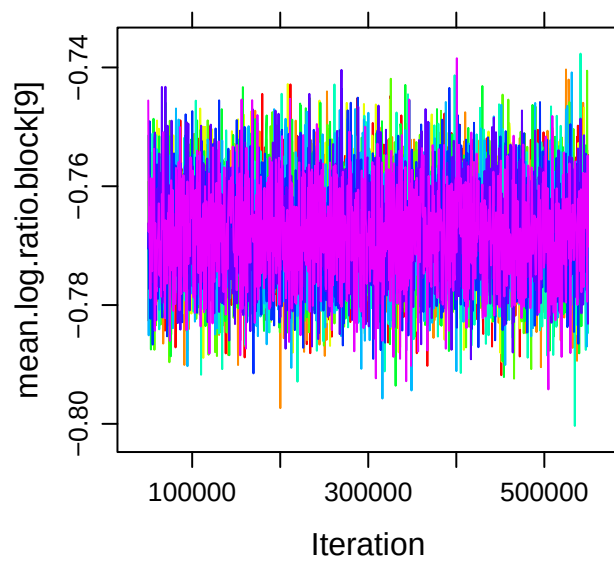


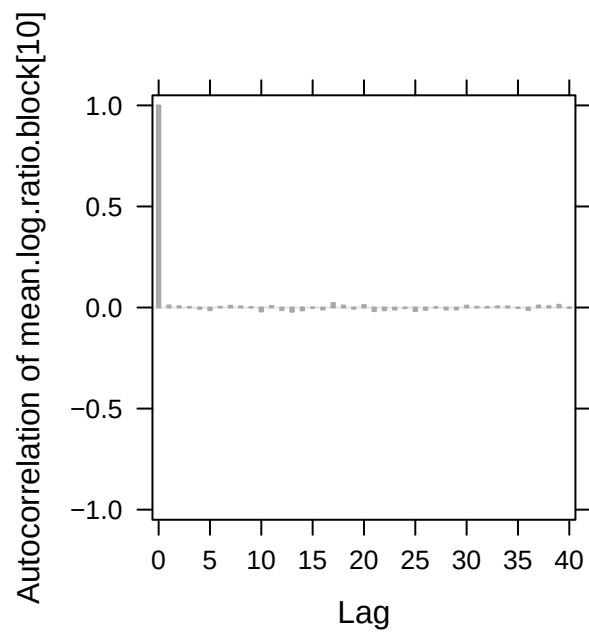
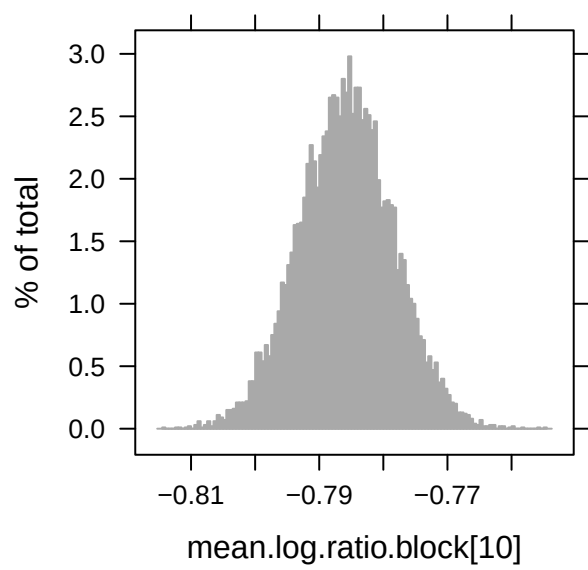
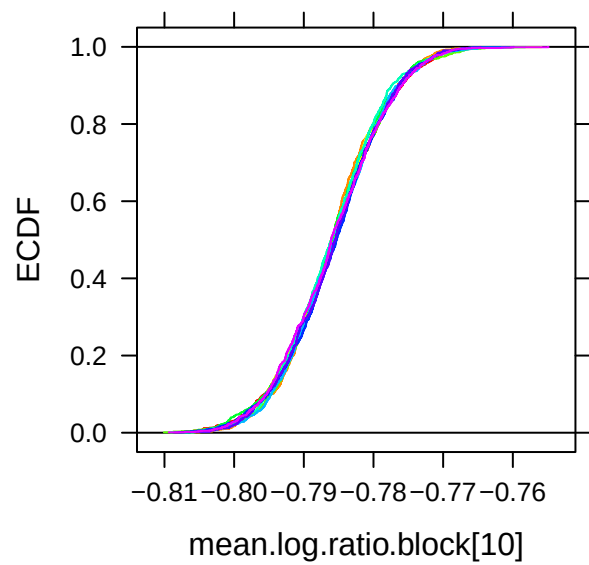
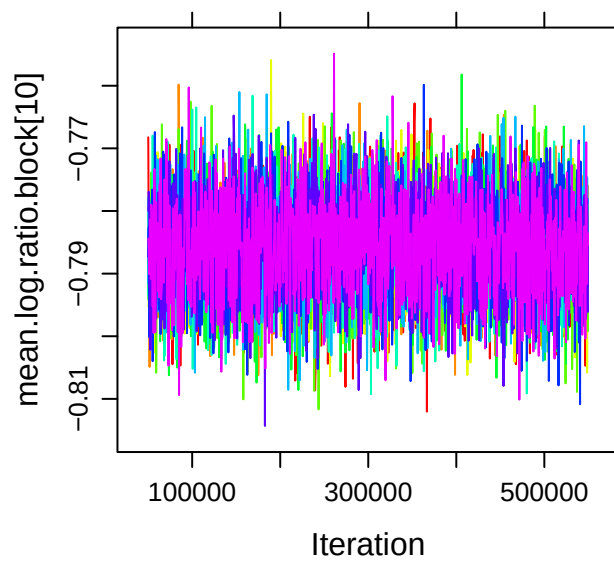


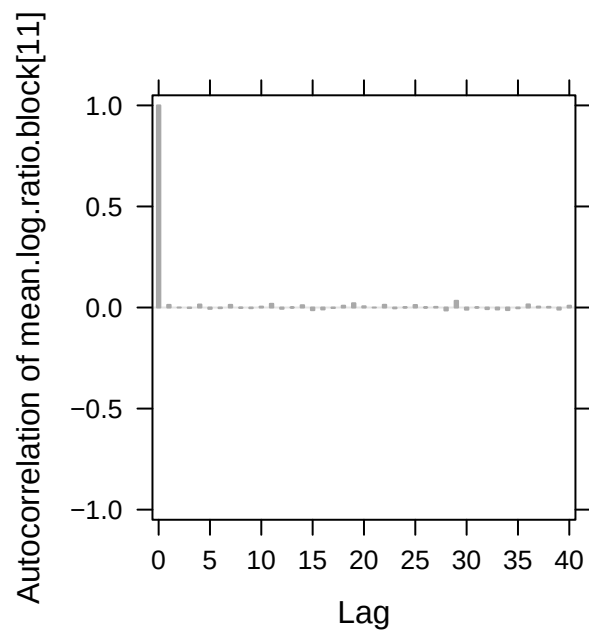
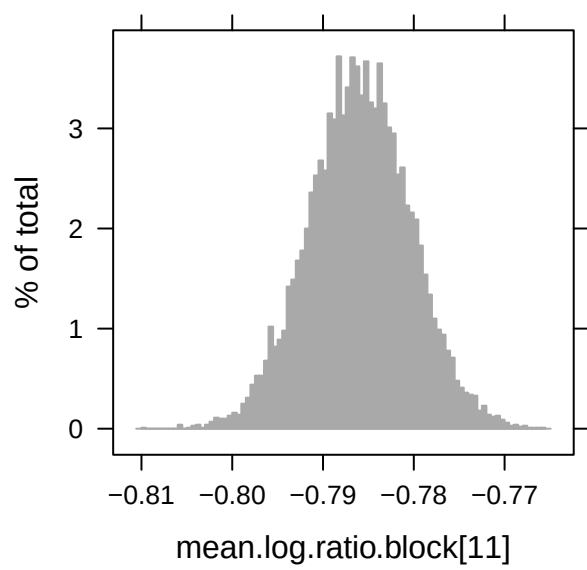
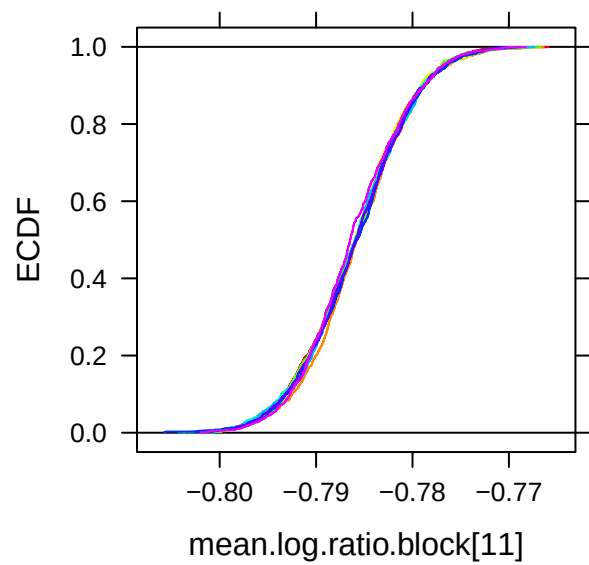
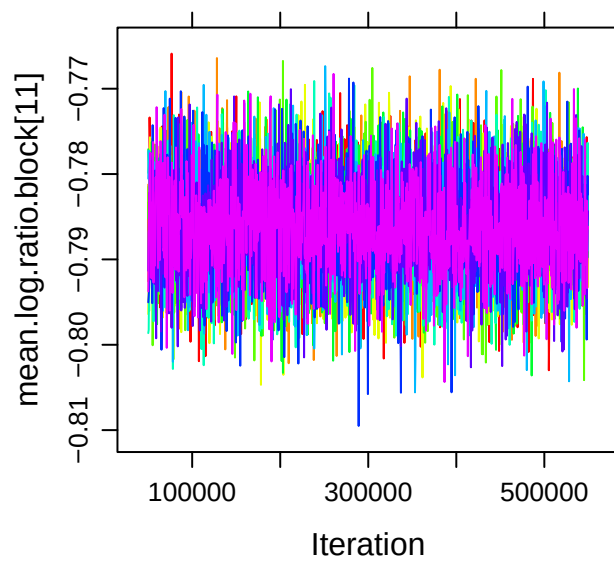


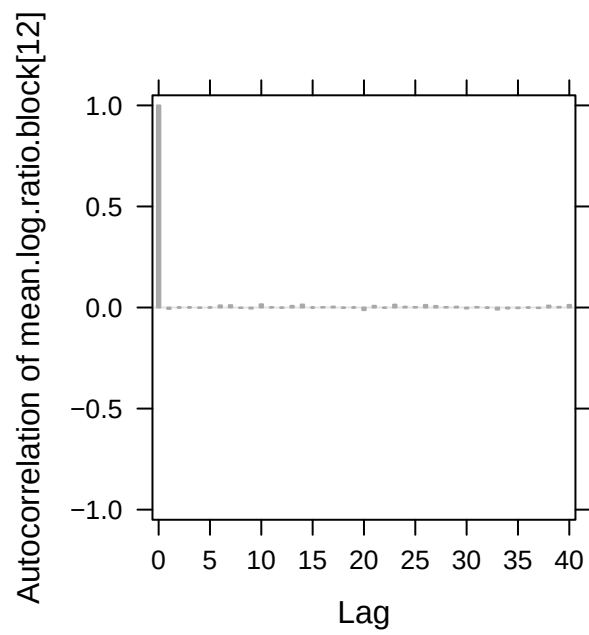
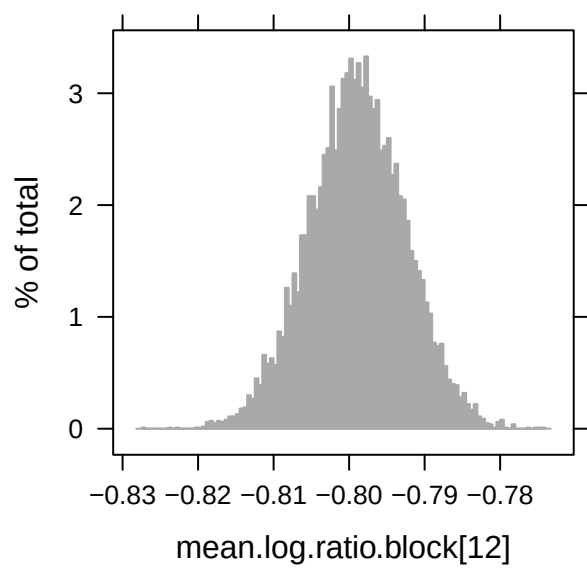
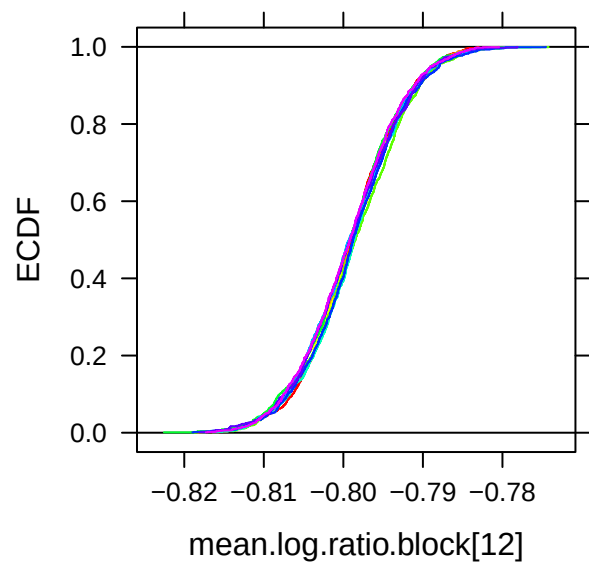
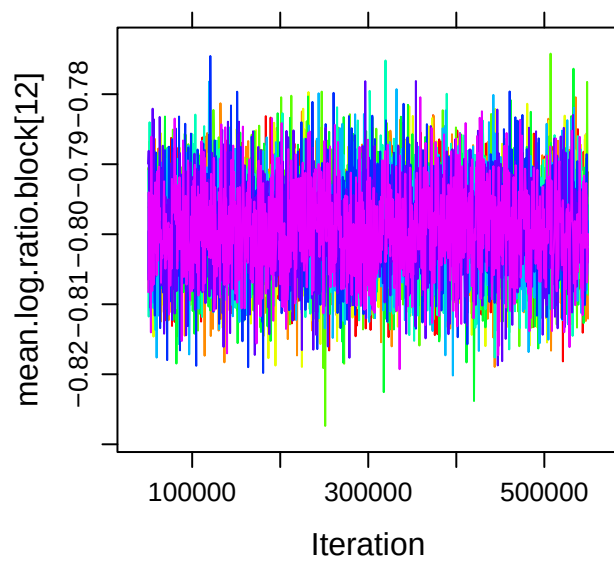


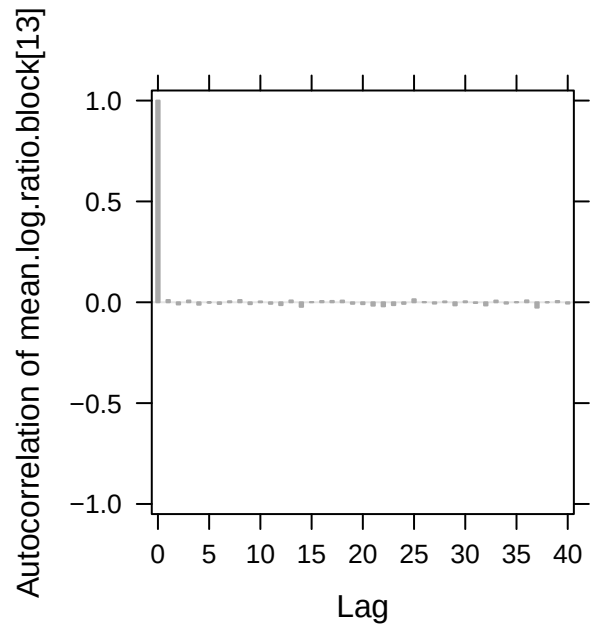
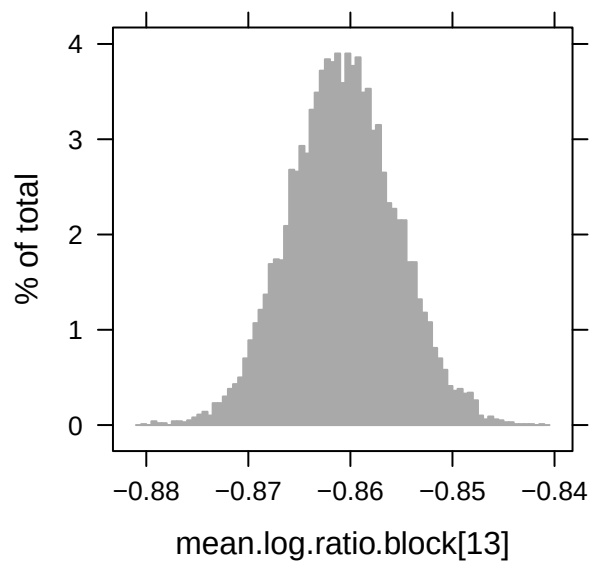
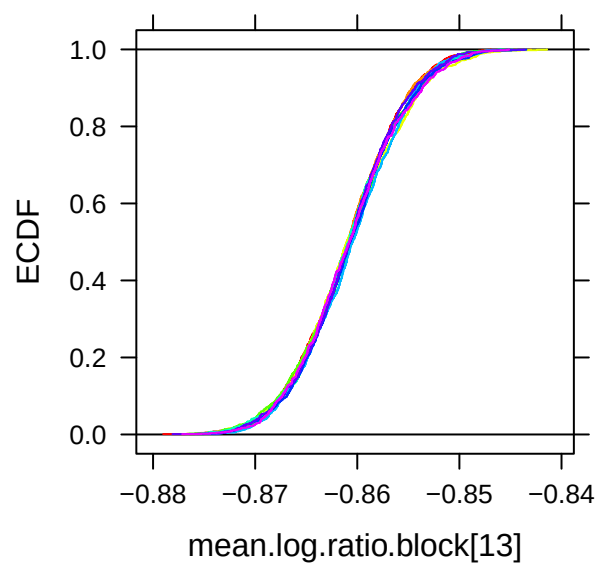
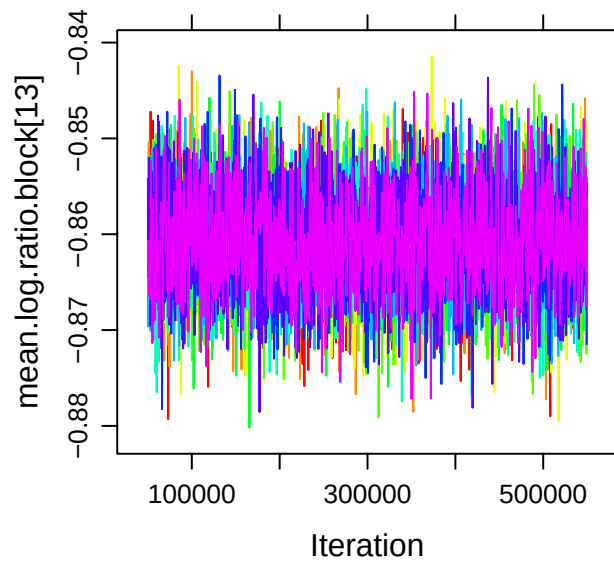


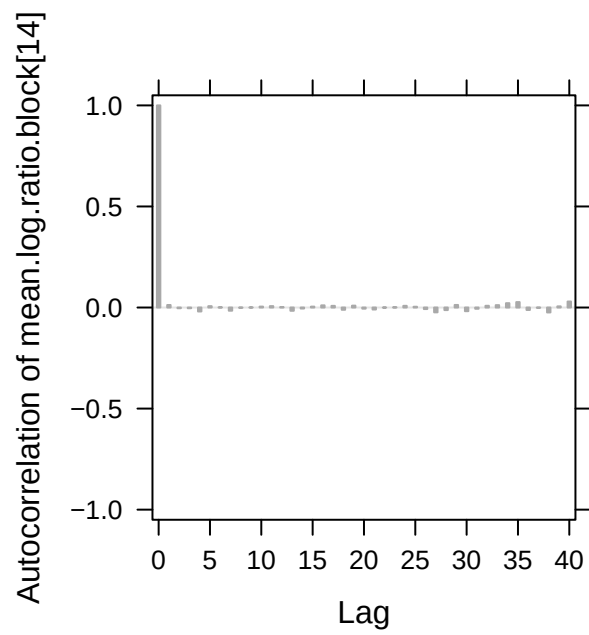
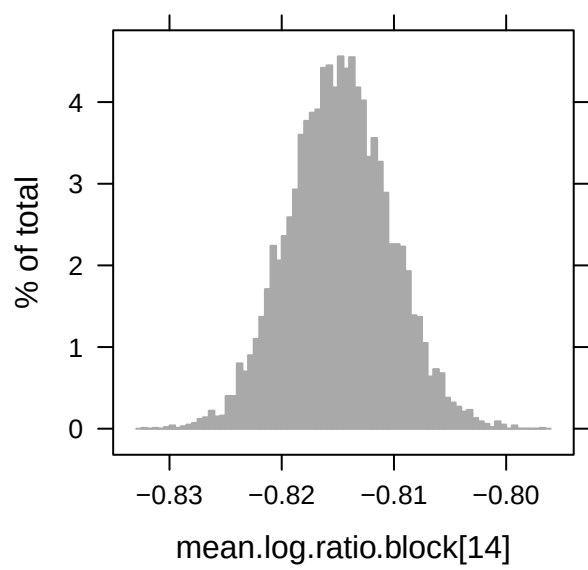
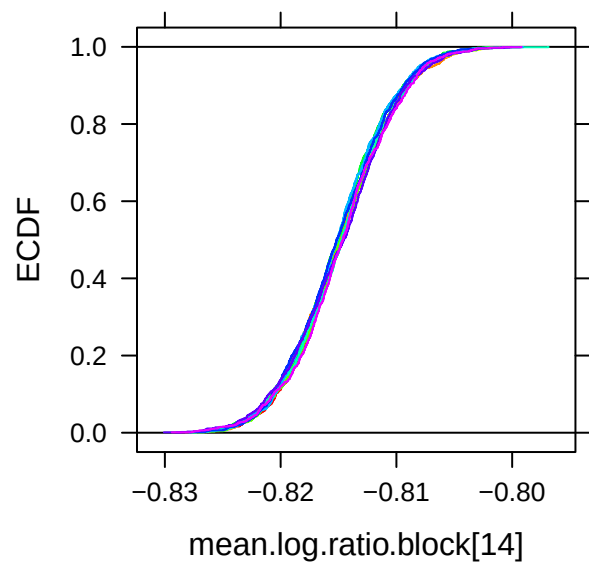
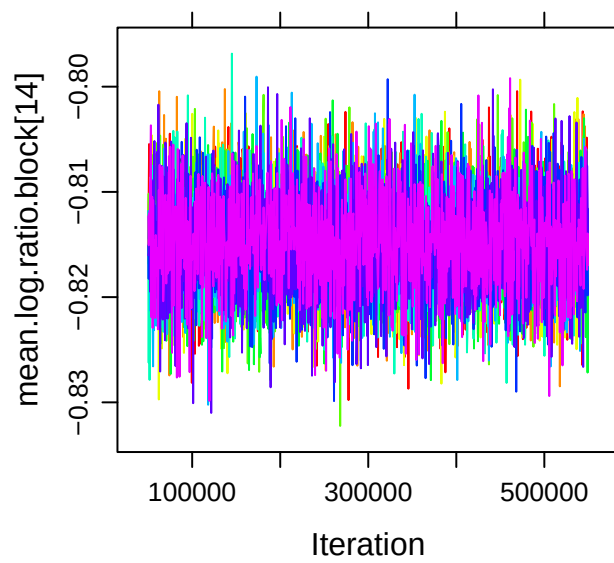


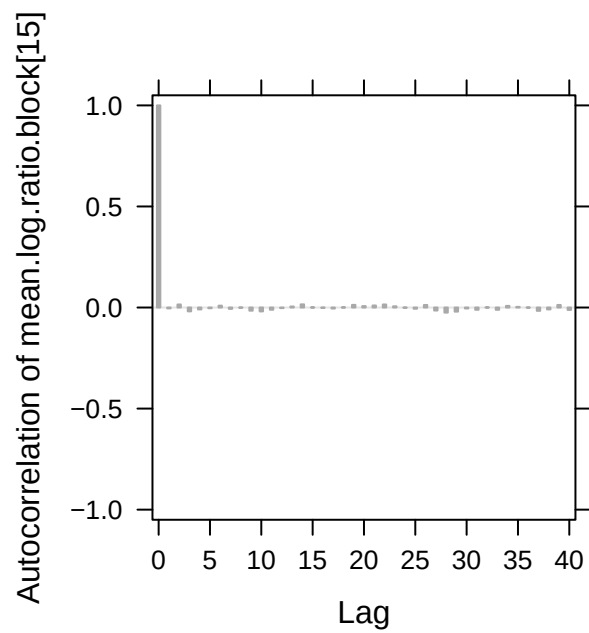
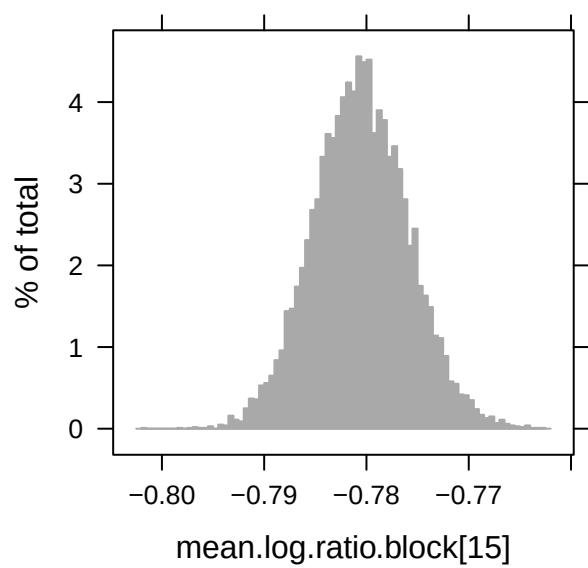
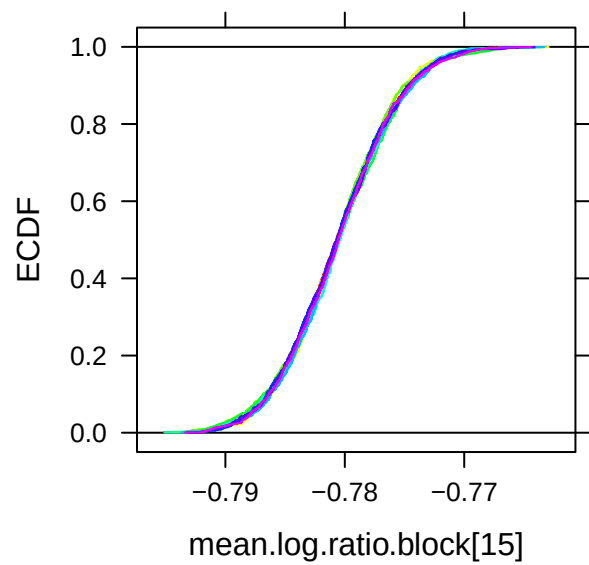
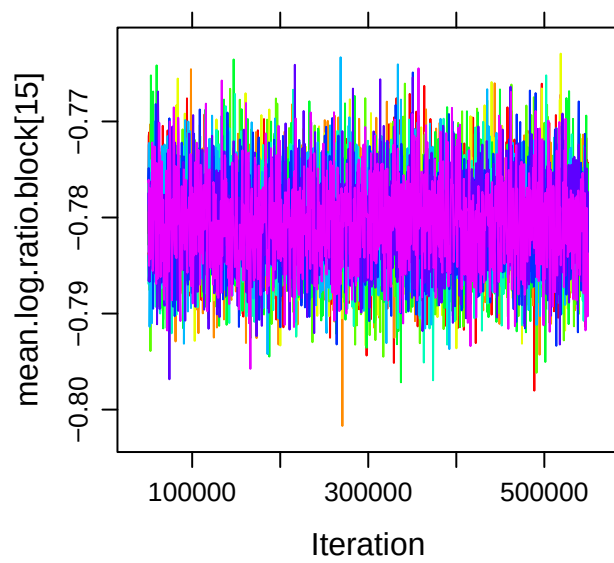


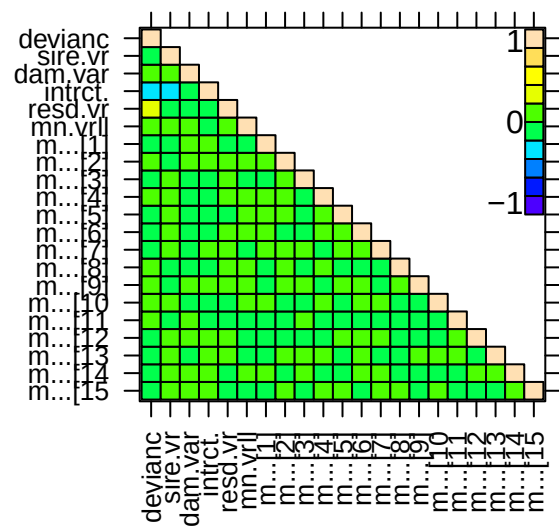












Model III

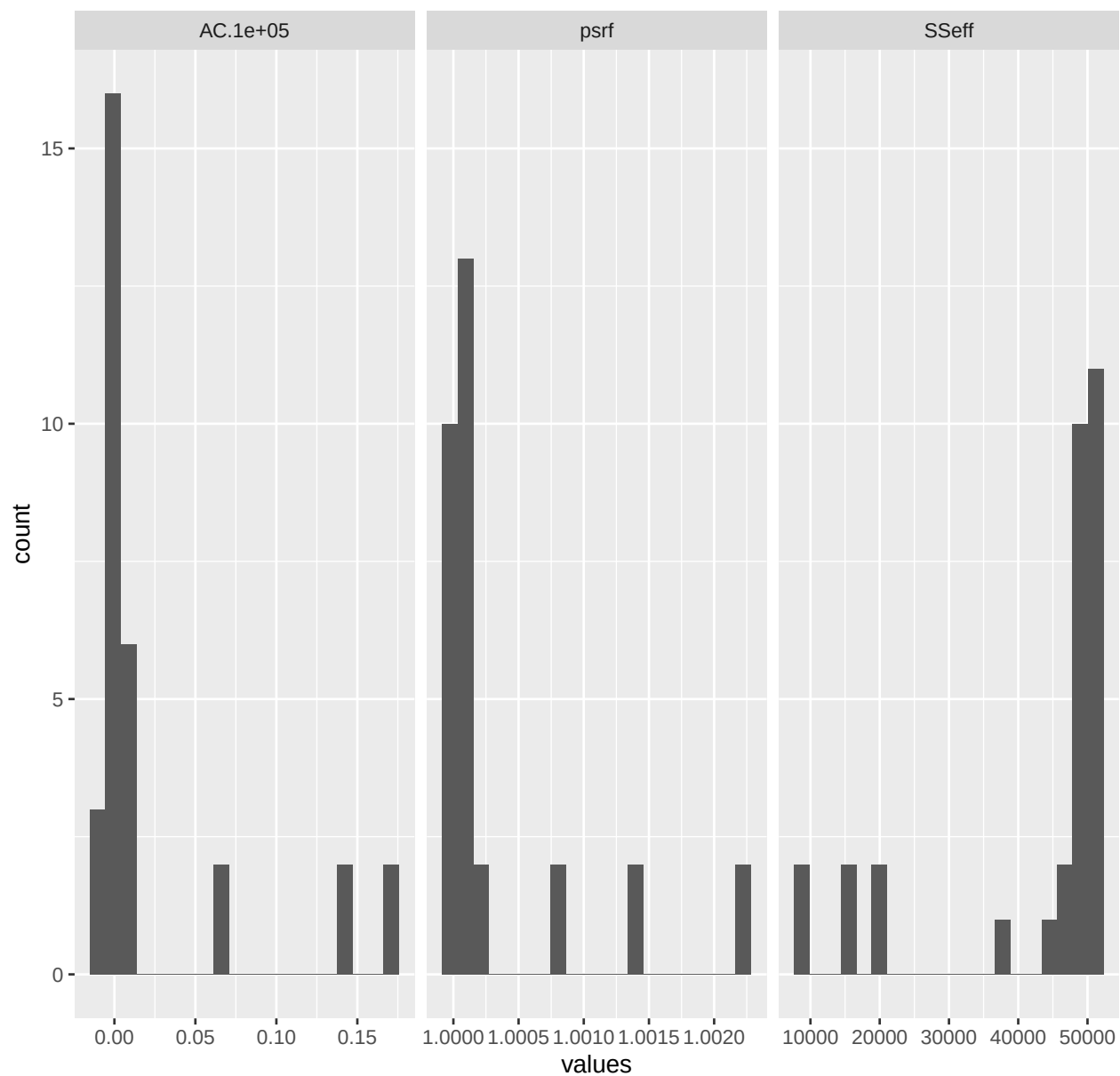
Number of chains: 10

Burnin: 80100

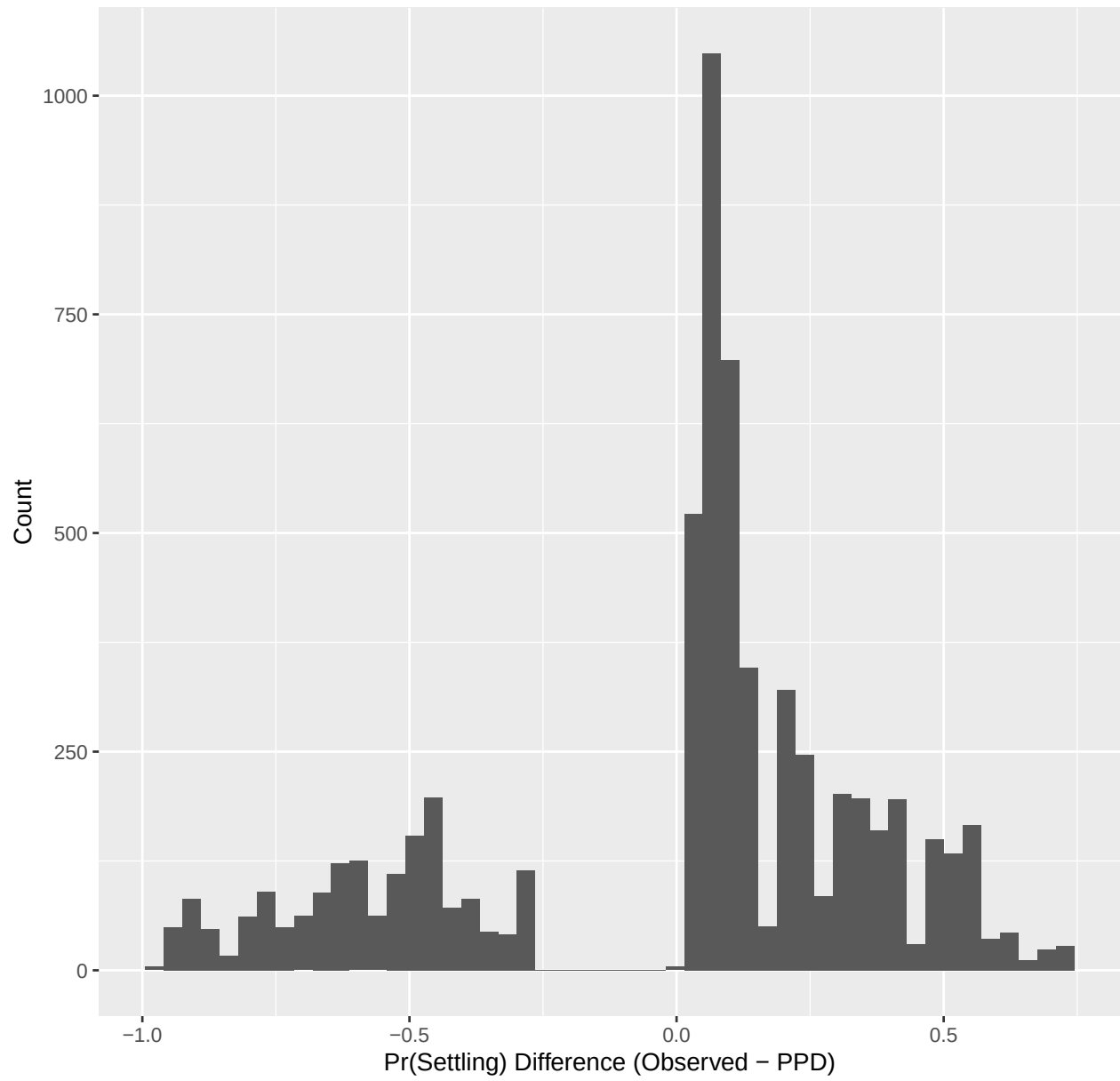
Samples per chain: 5000

Total samples: 50000

A) Posterior distribution diagnostics



B) Posterior predictive check distribution



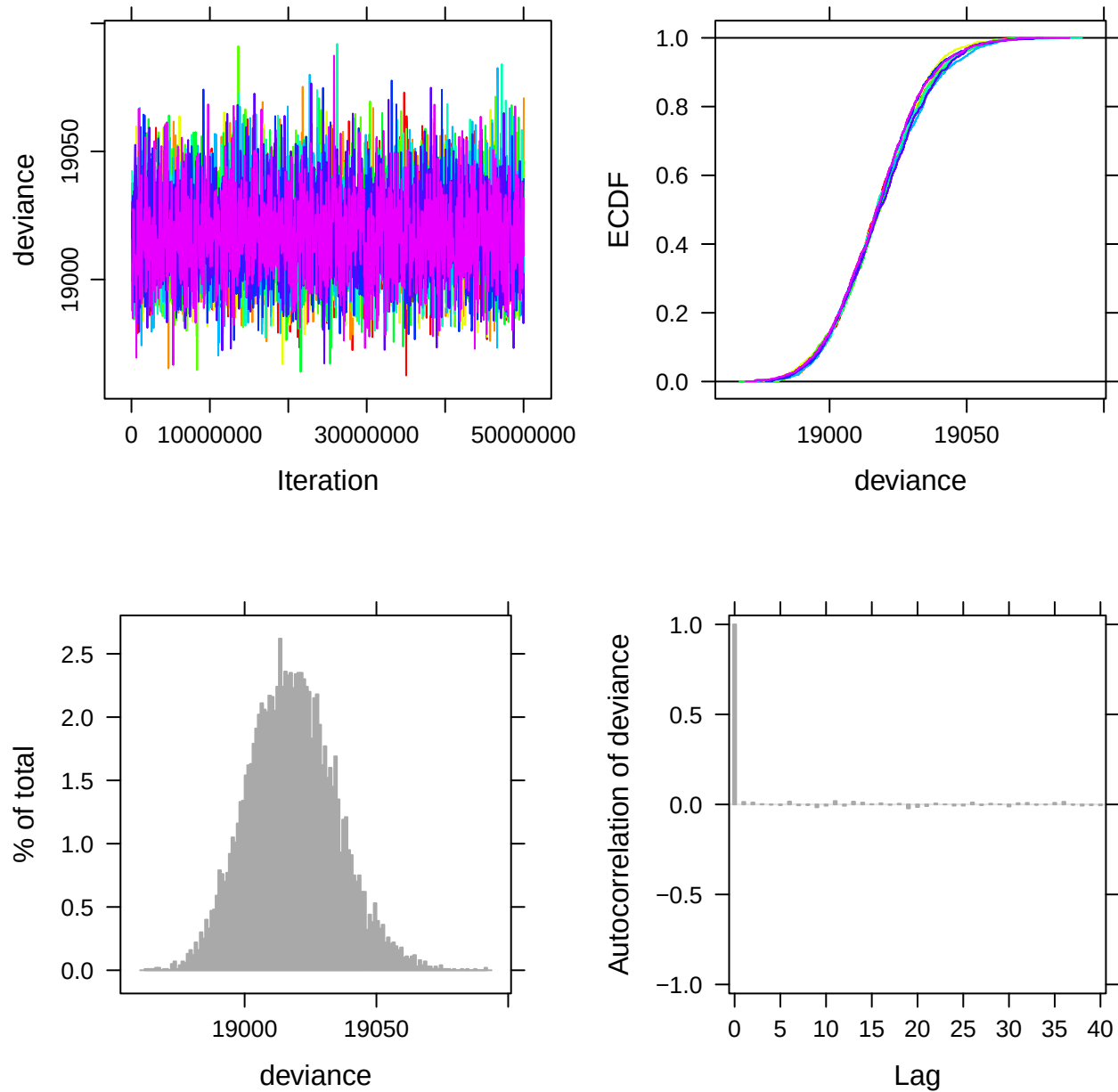
C) By-parameter posterior summaries

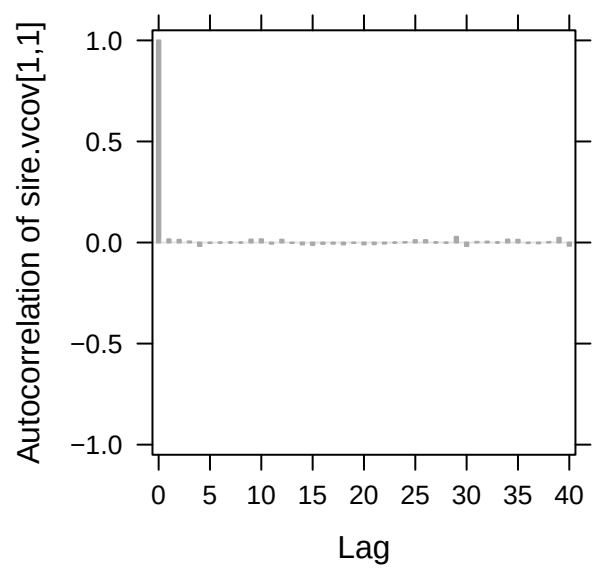
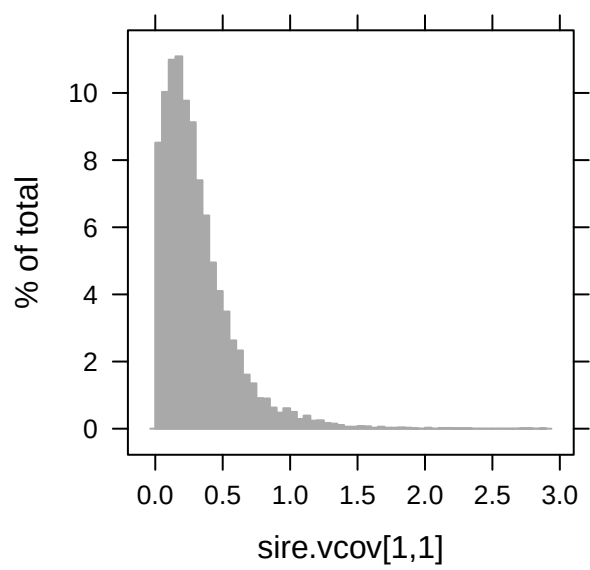
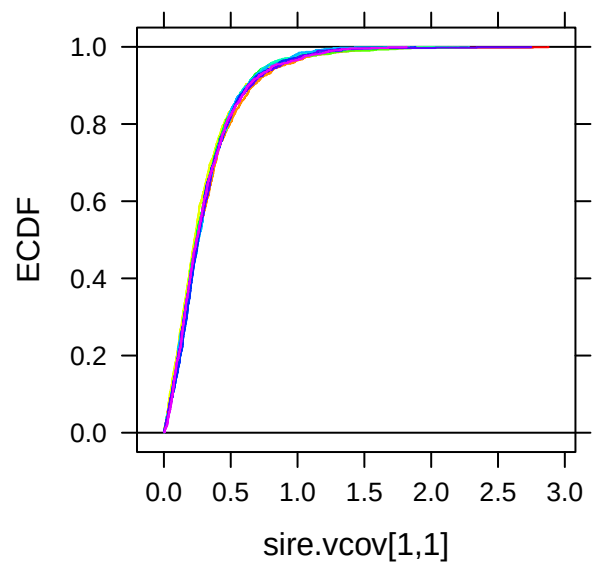
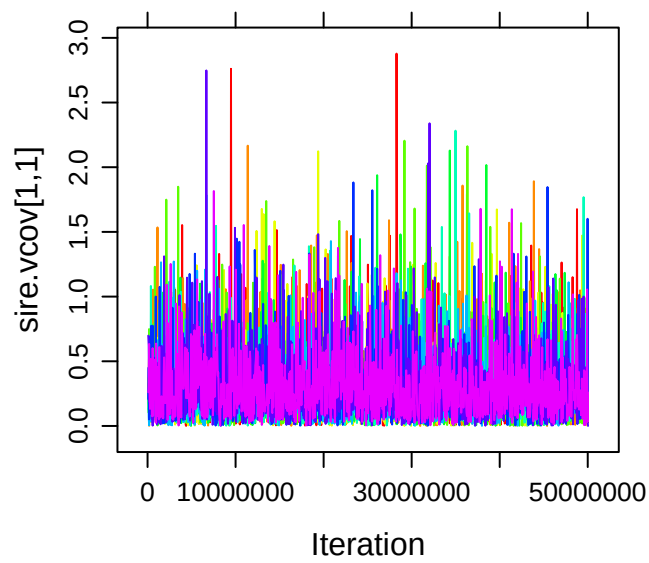
Generating summary statistics and plots (these will NOT be saved for reuse)...

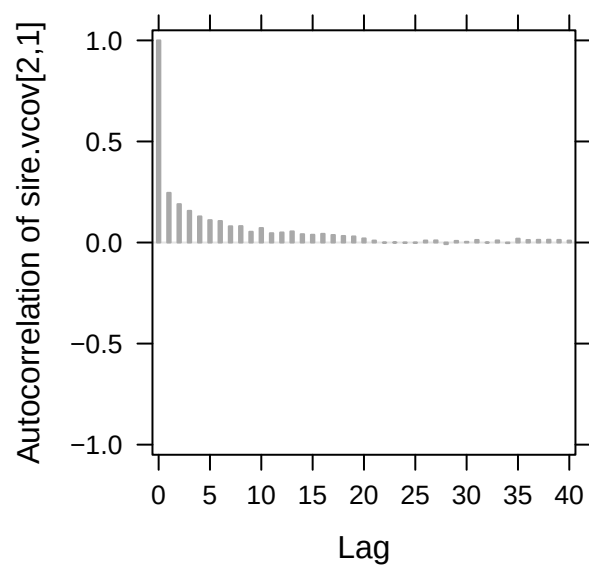
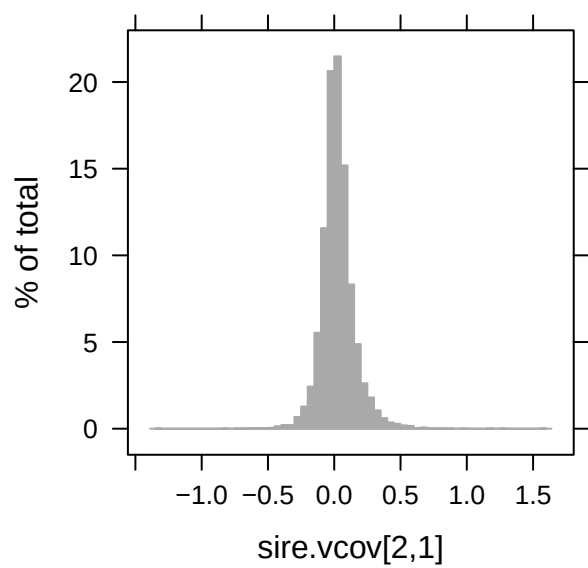
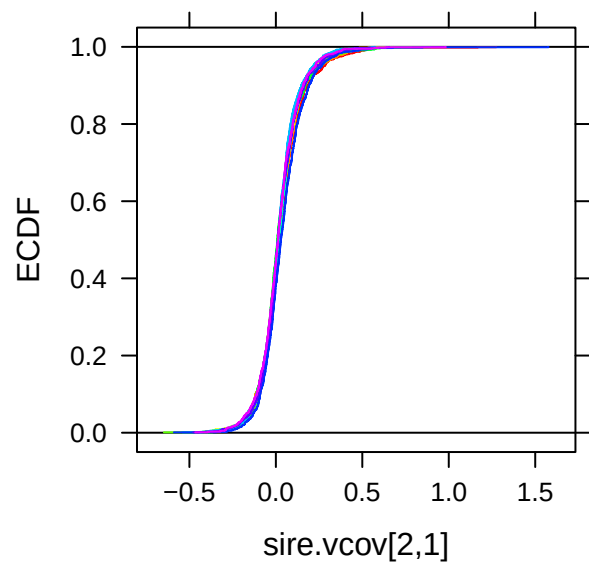
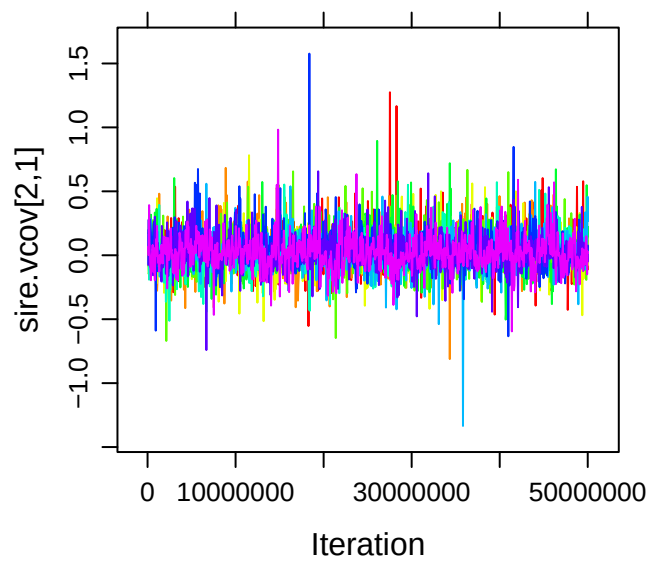
Calculating summary statistics...

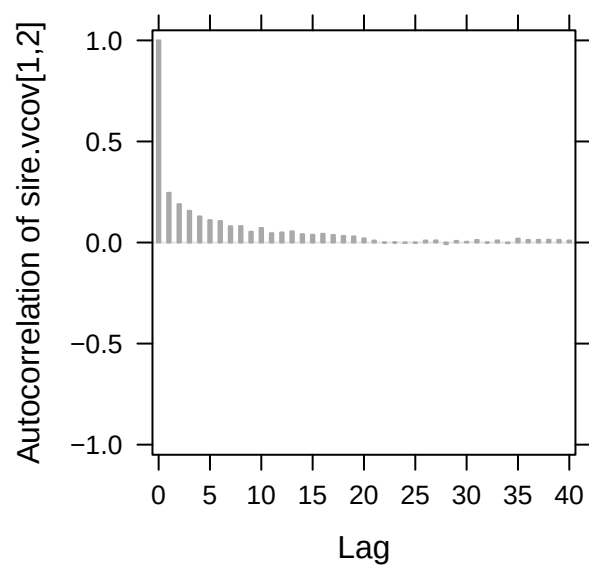
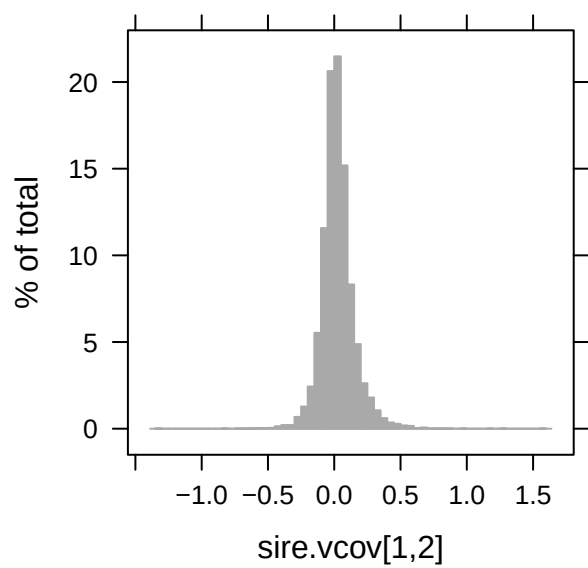
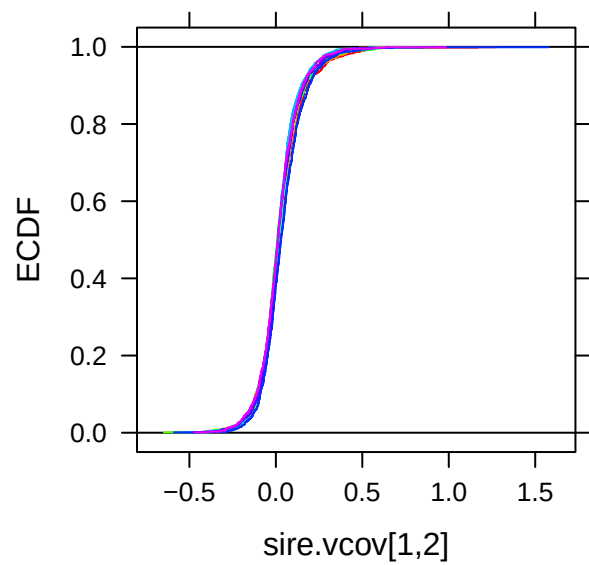
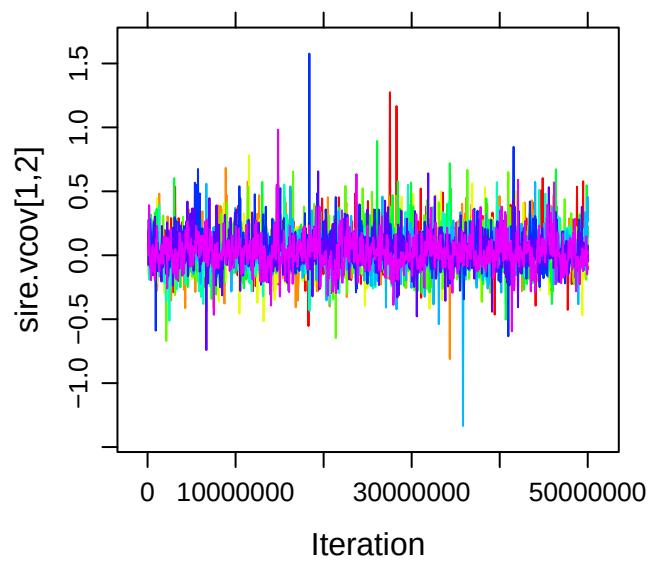
Calculating the Gelman-Rubin statistic for 31 variables....

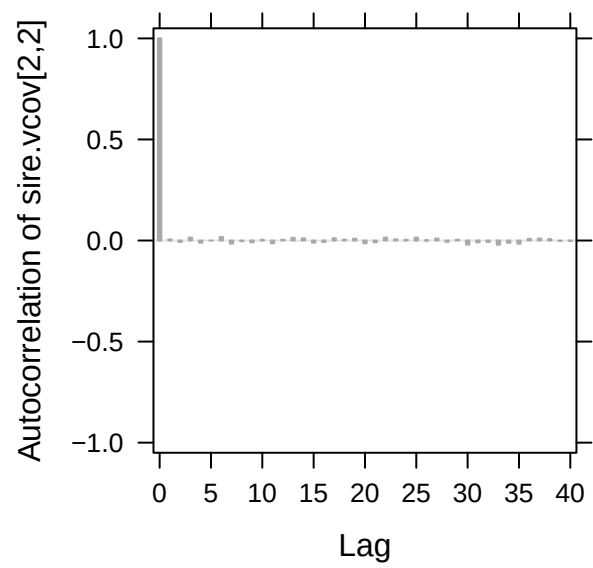
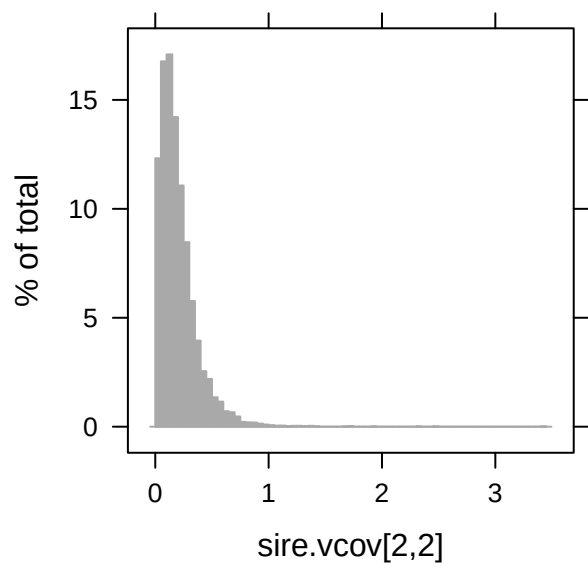
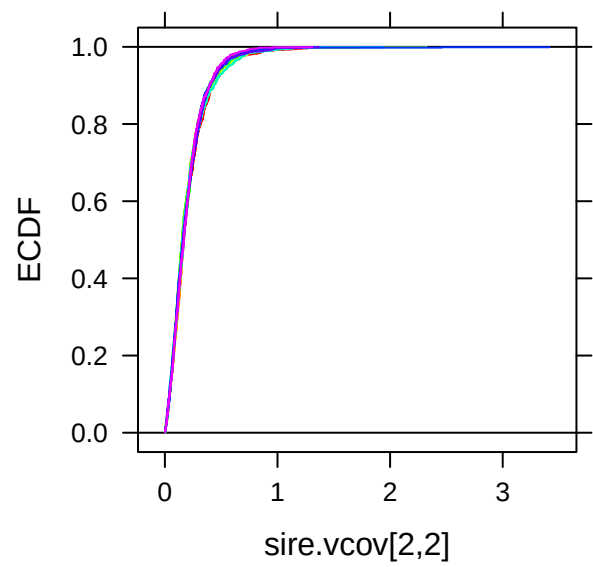
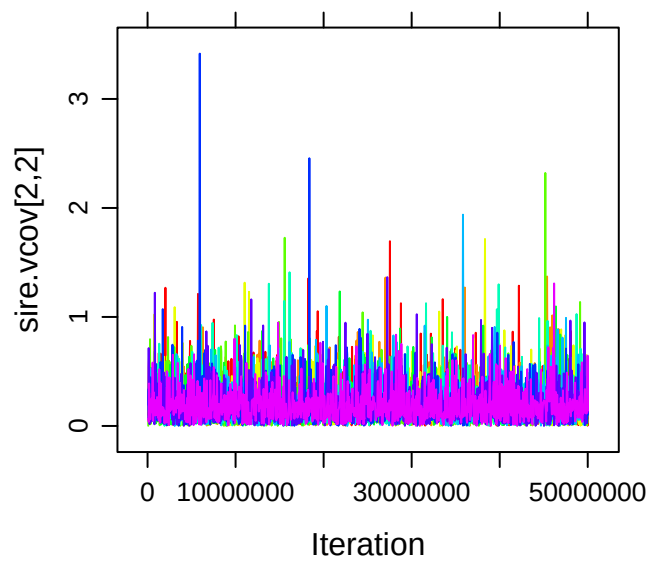
Note: Unable to calculate the multivariate psrf

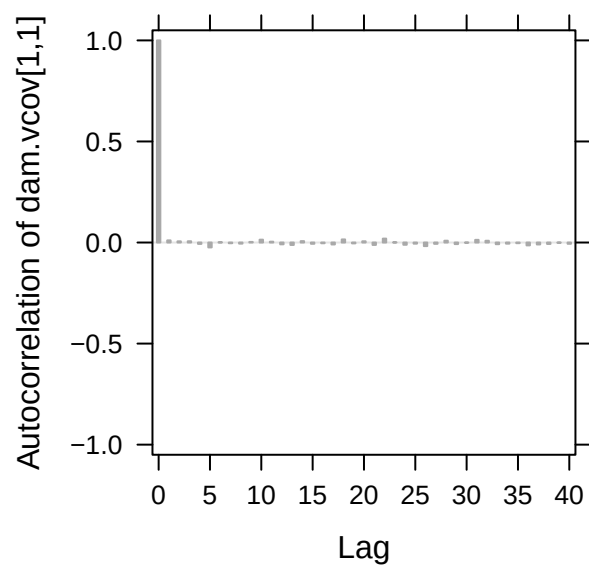
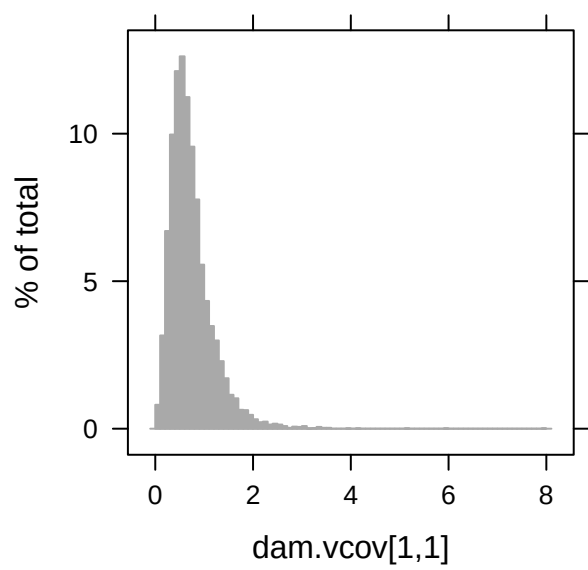
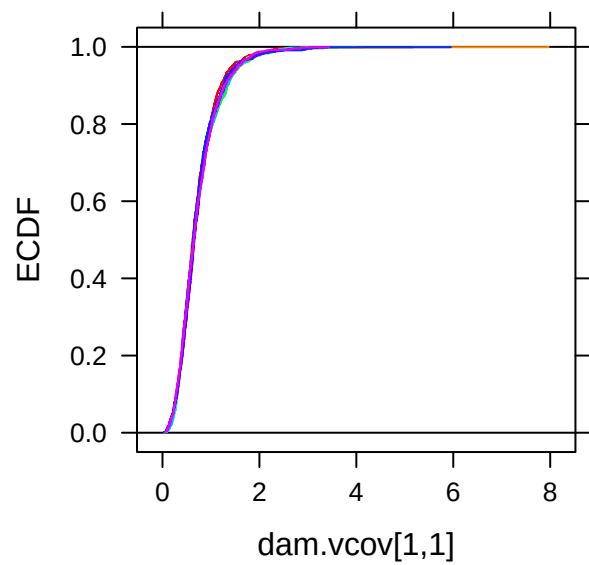
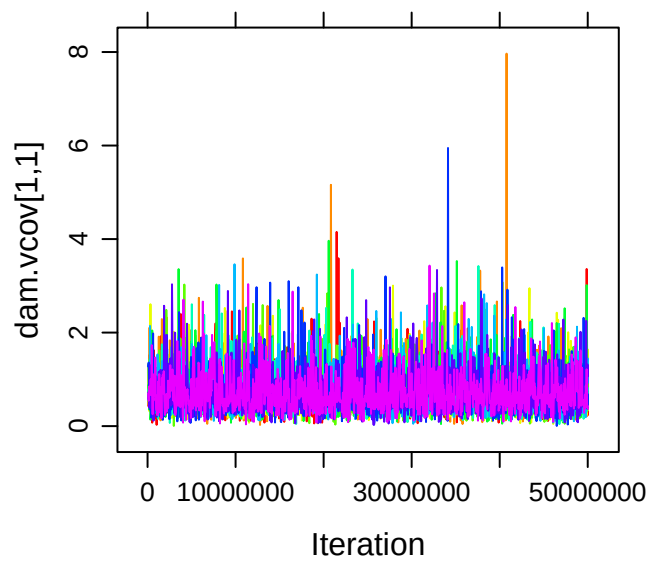


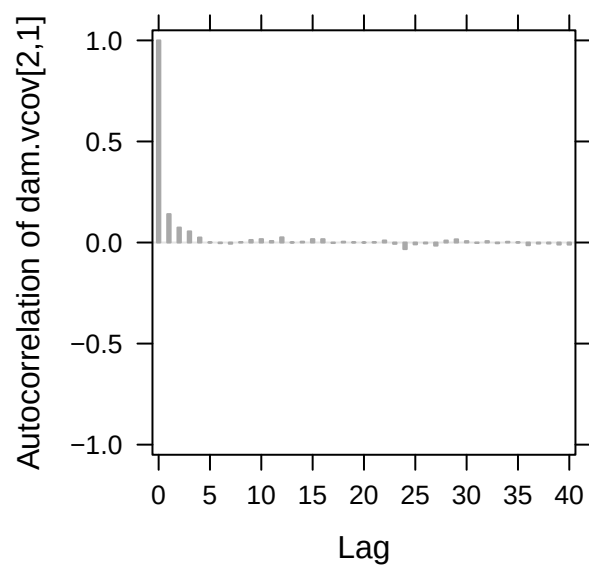
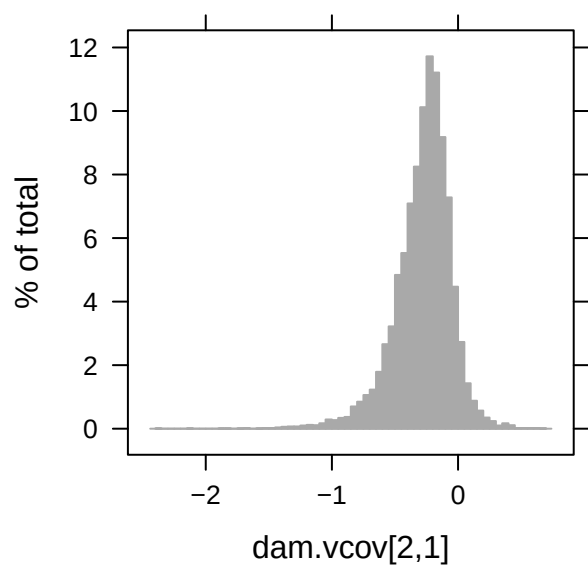
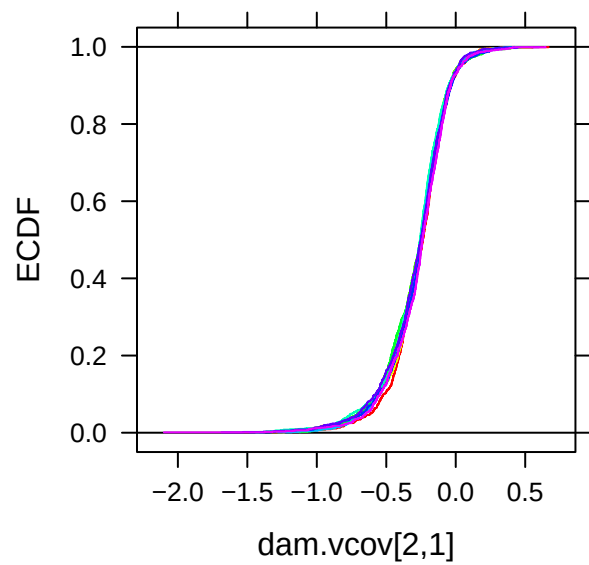
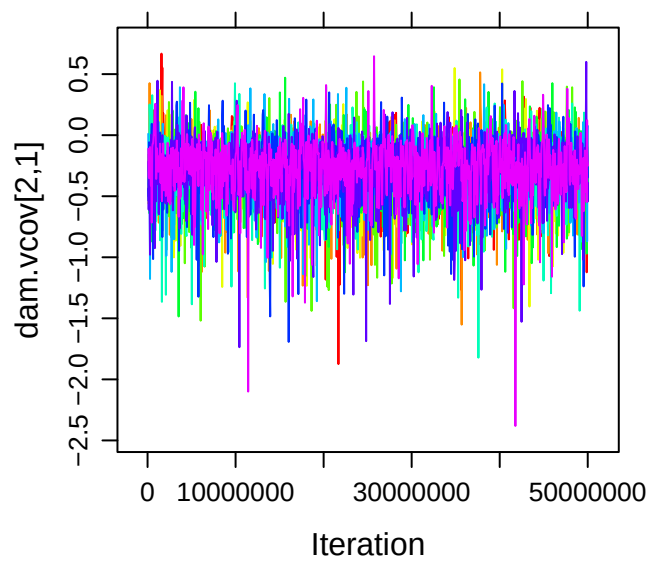


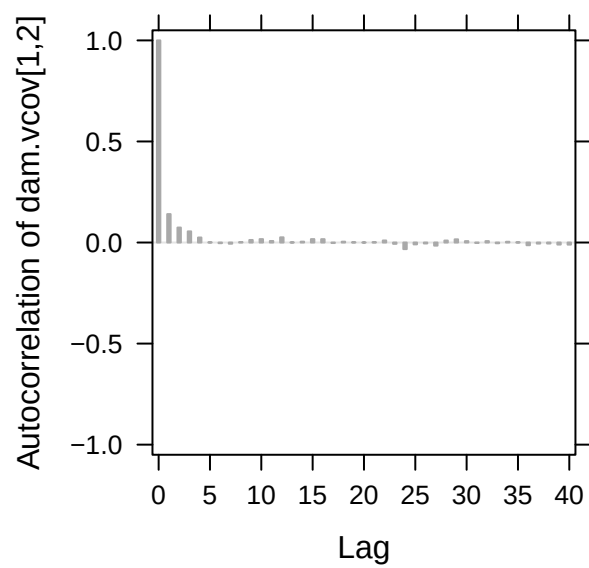
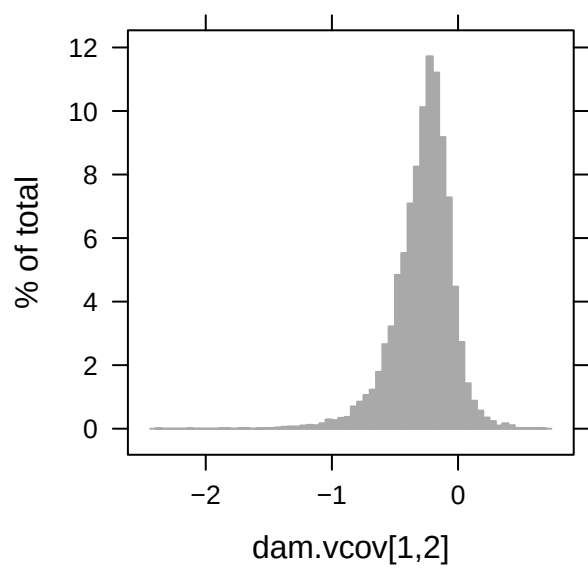
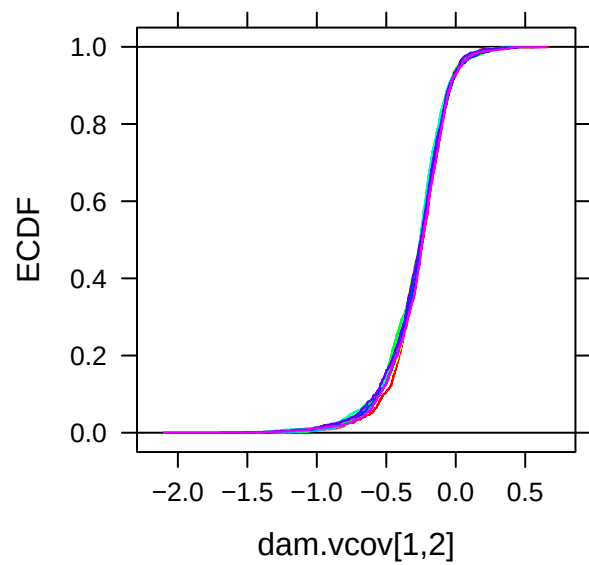
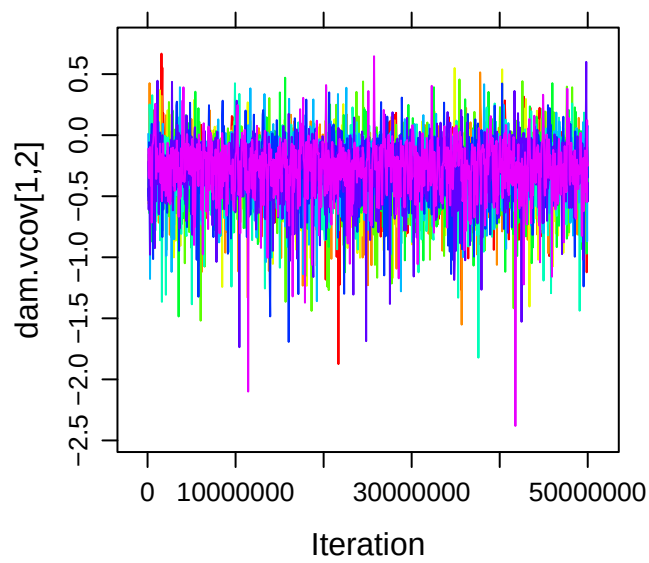


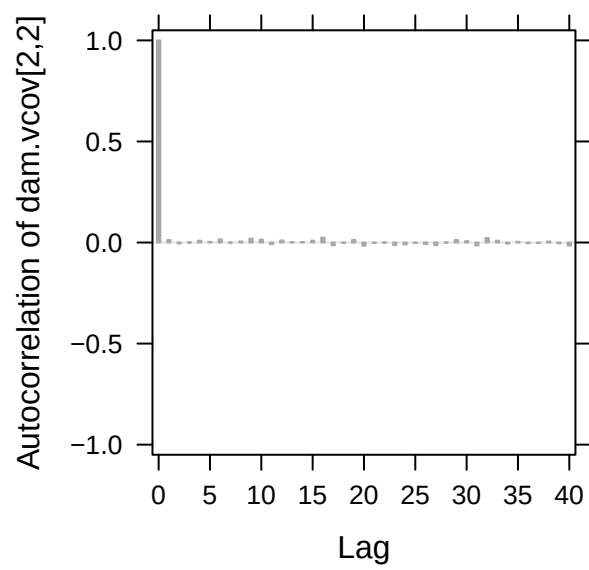
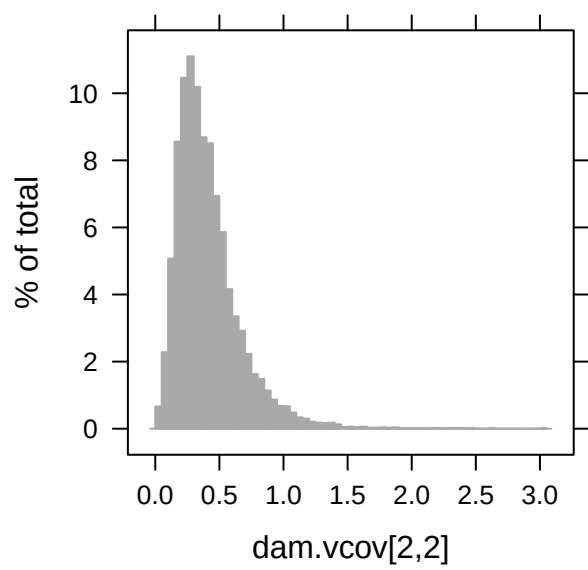
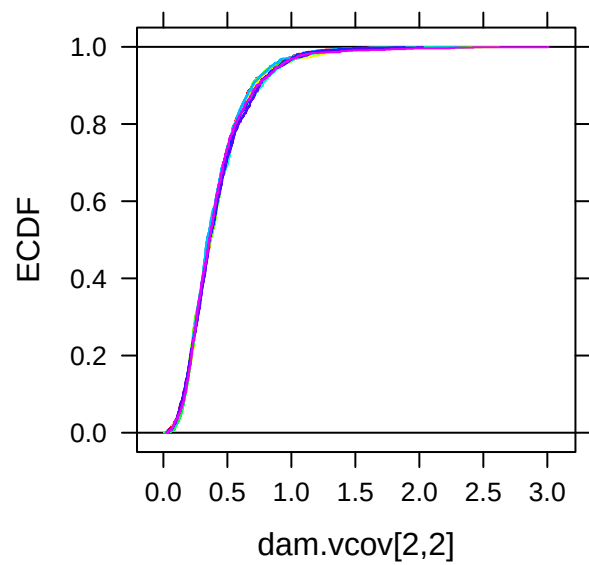
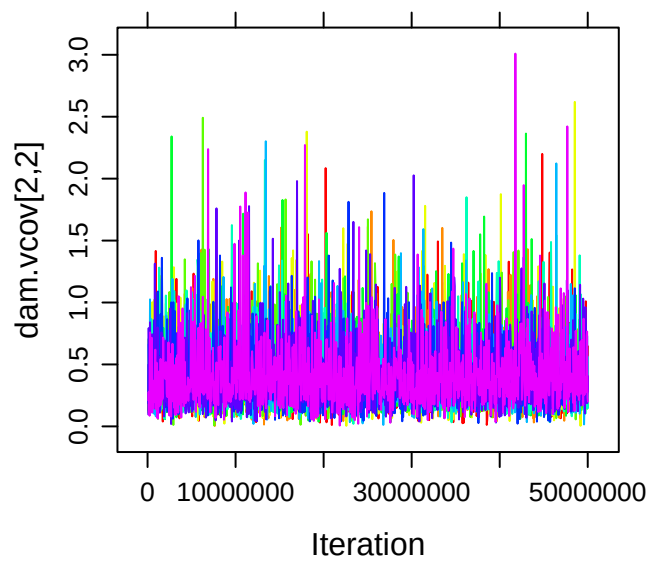


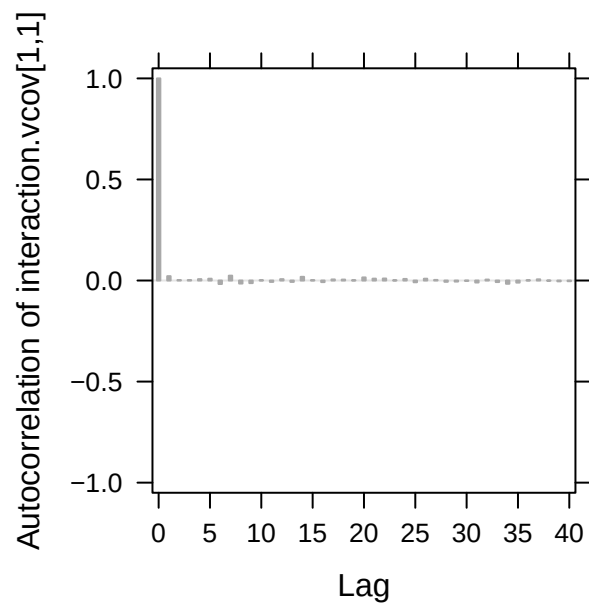
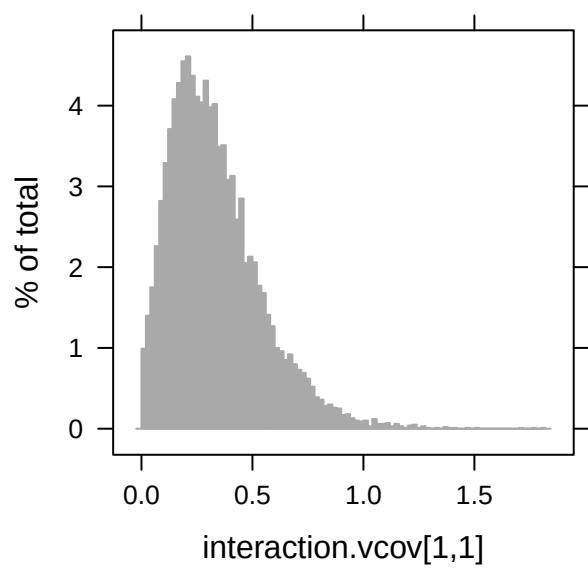
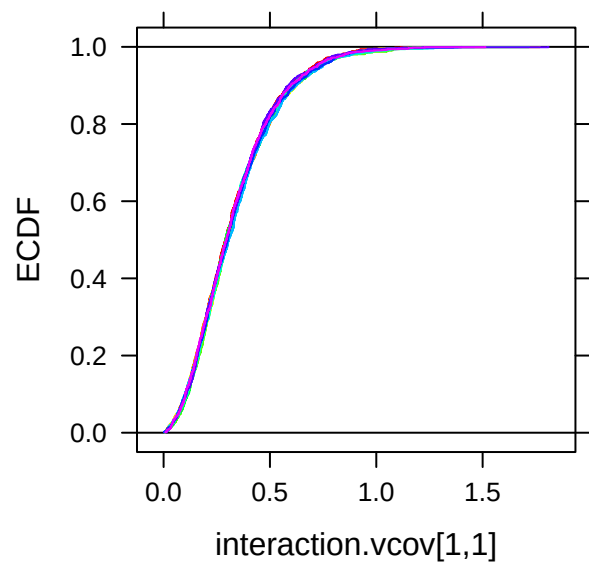
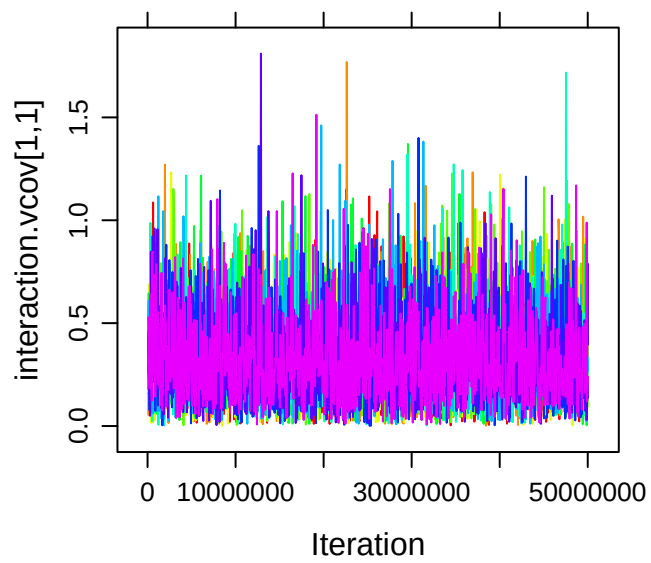


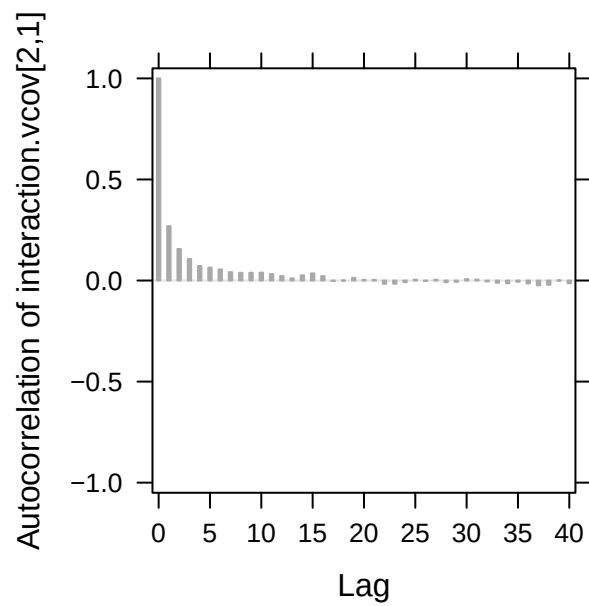
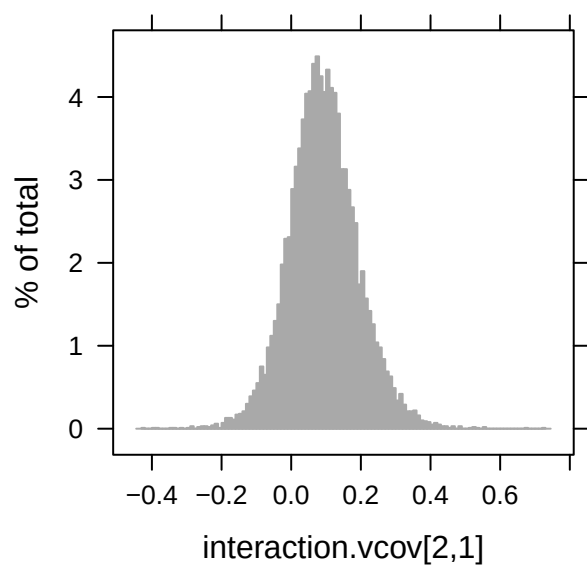
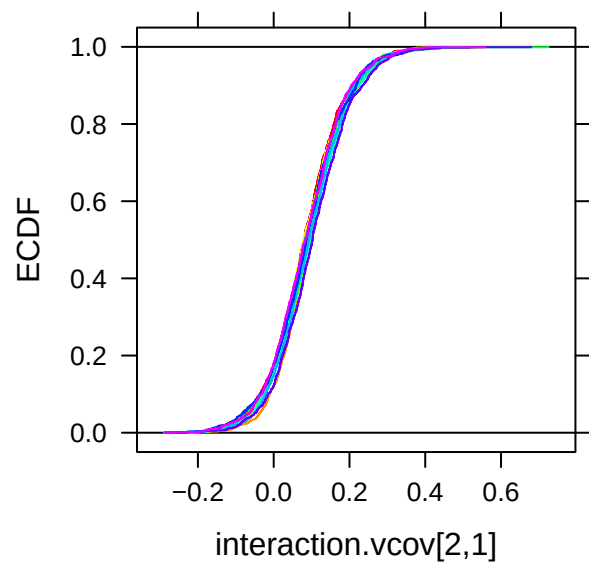
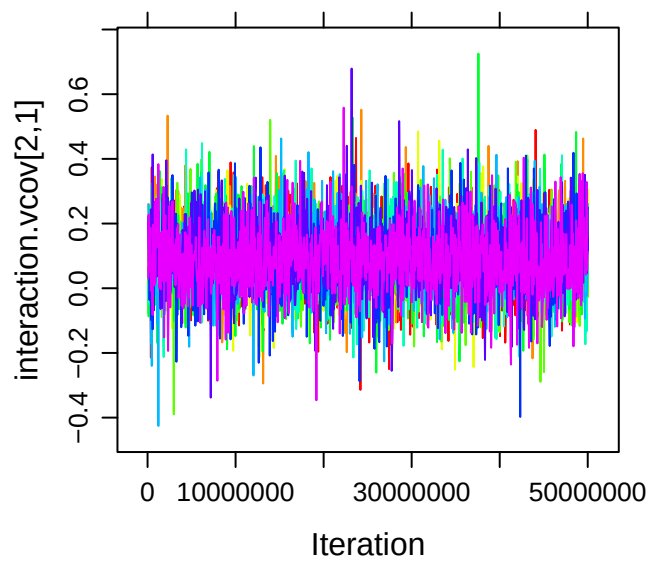


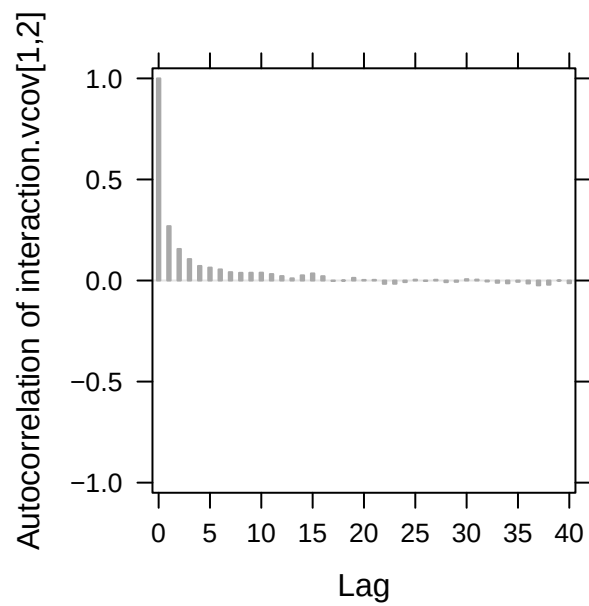
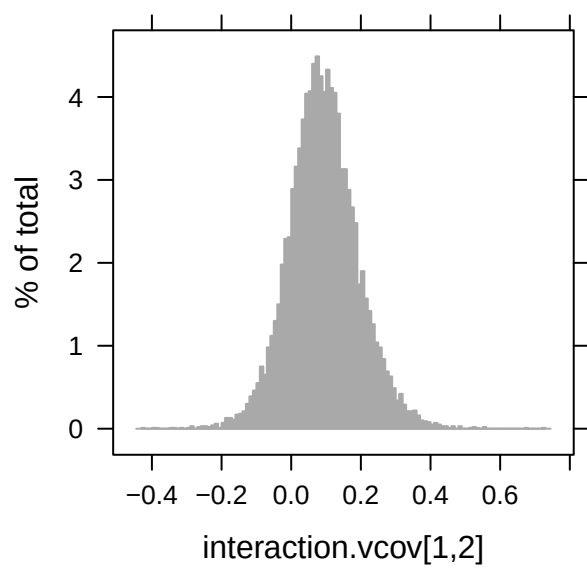
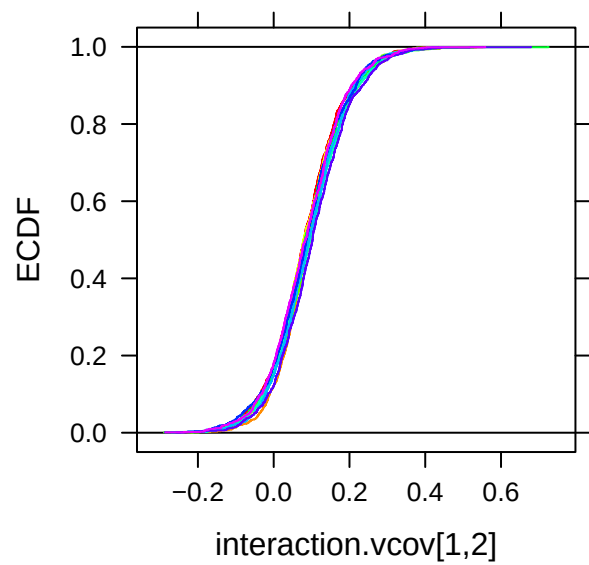
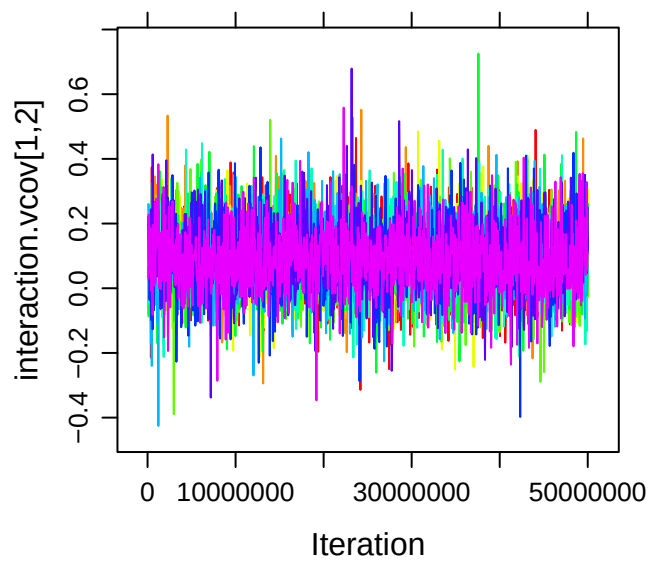


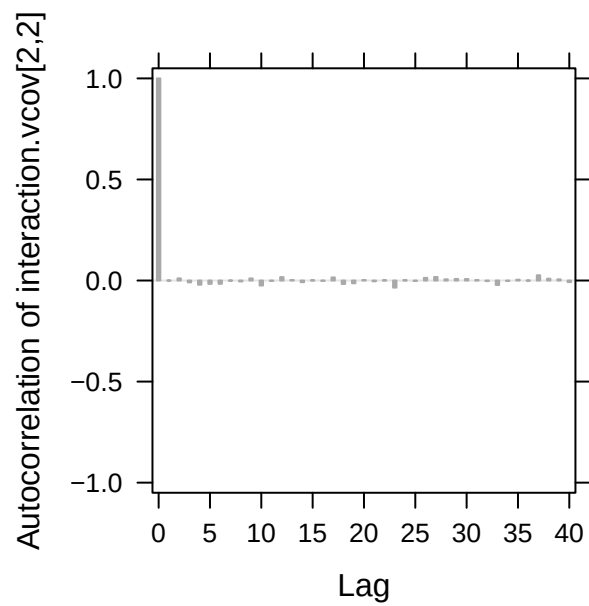
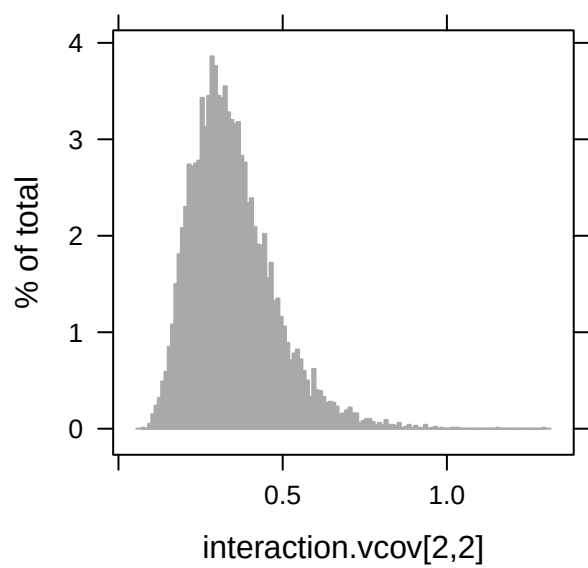
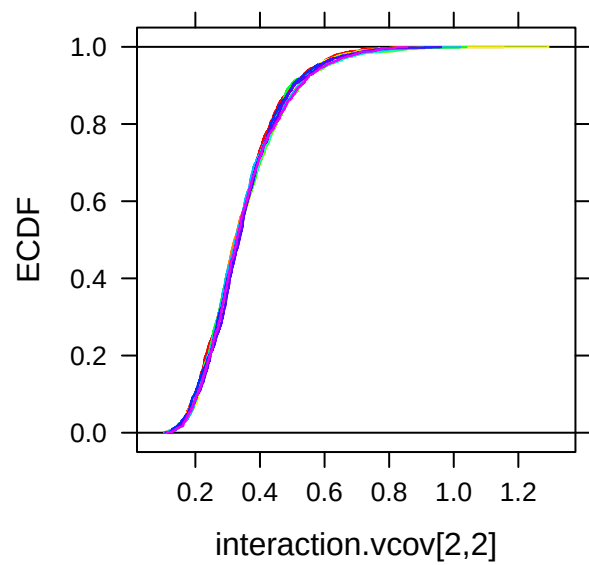
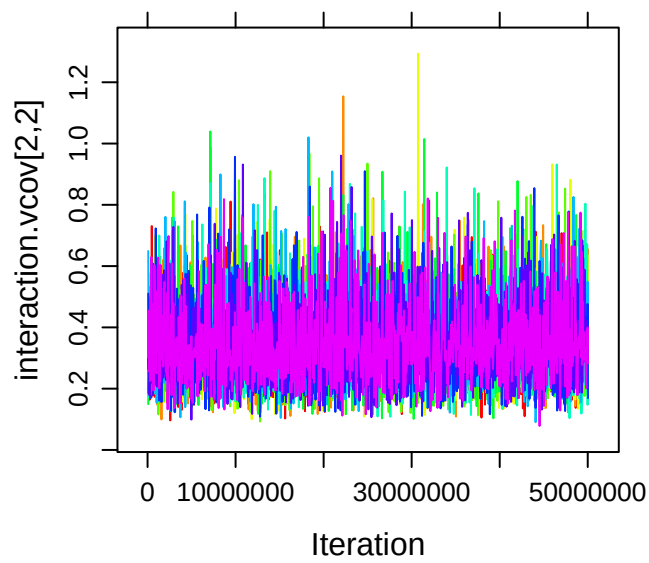


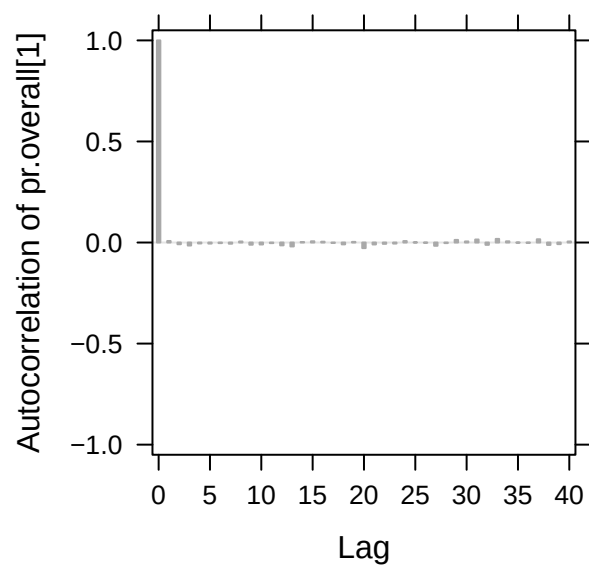
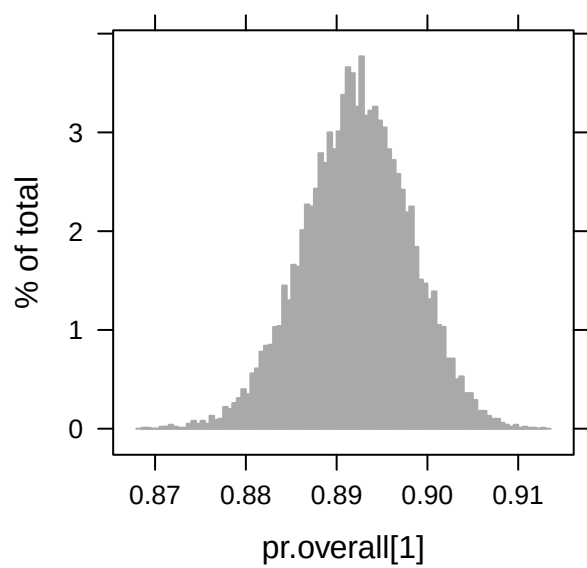
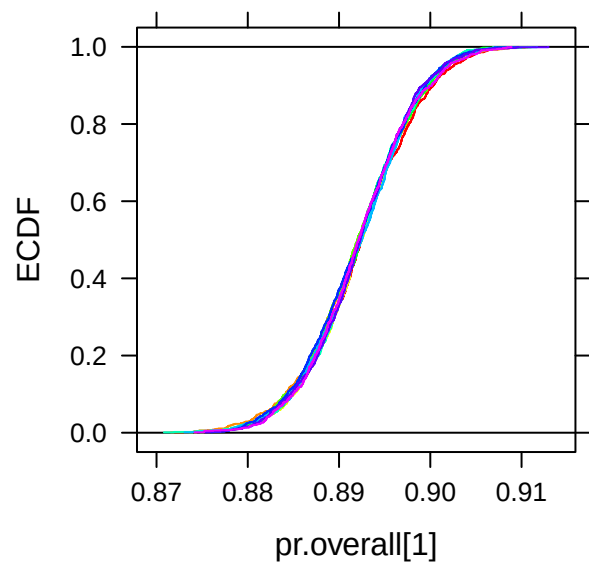
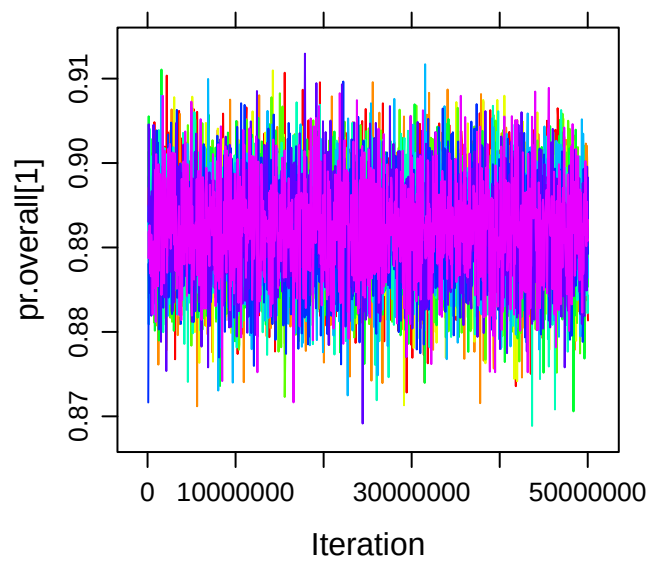


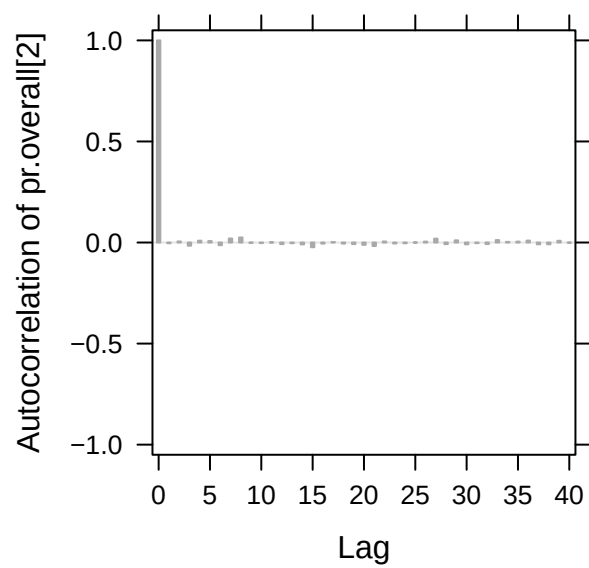
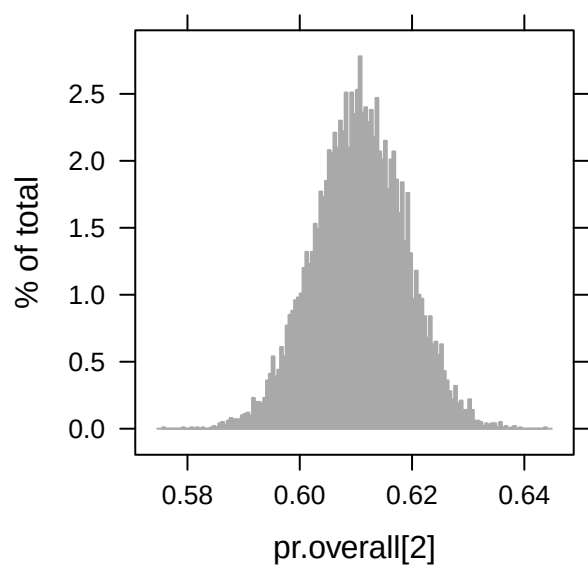
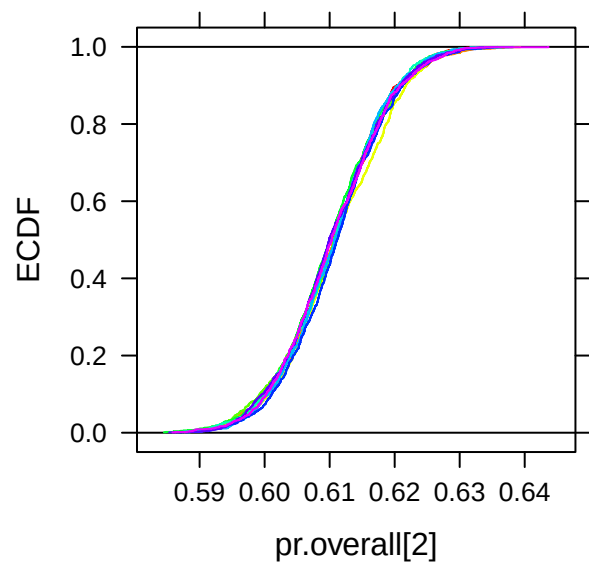
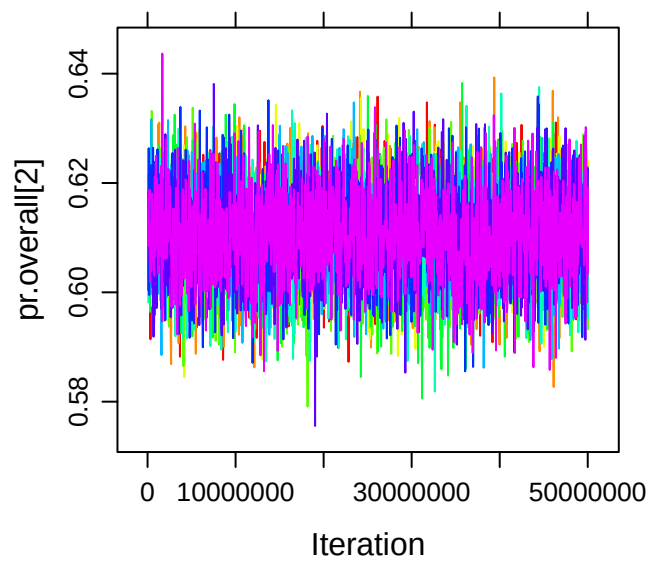


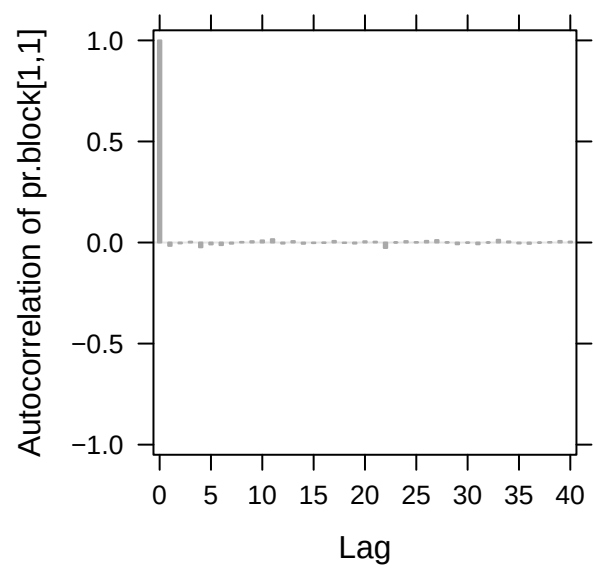
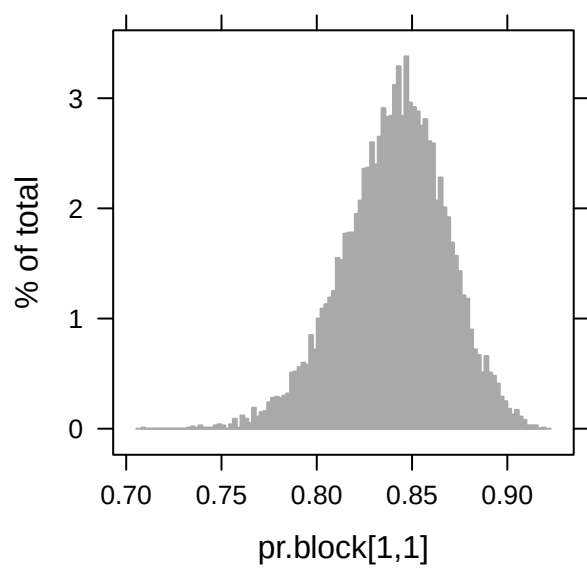
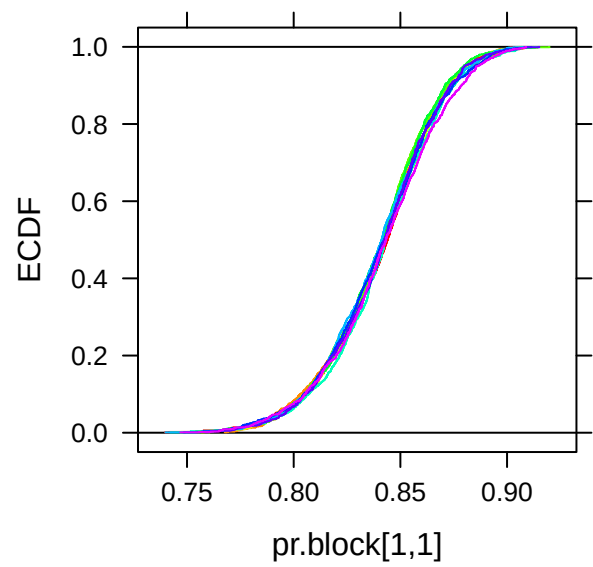
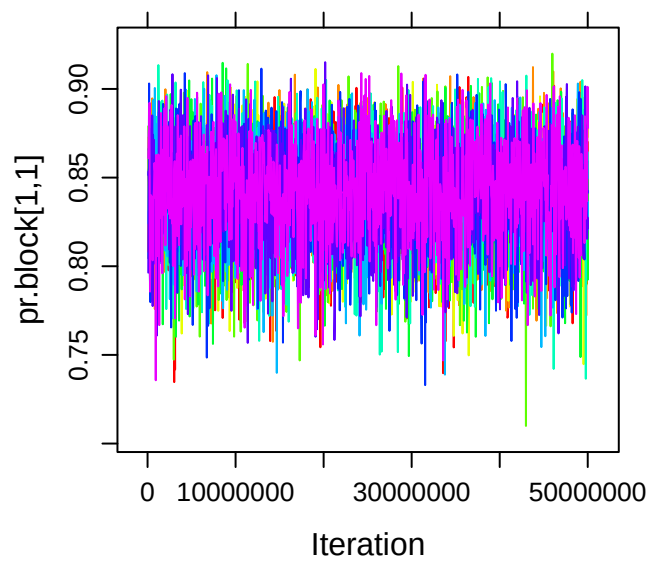


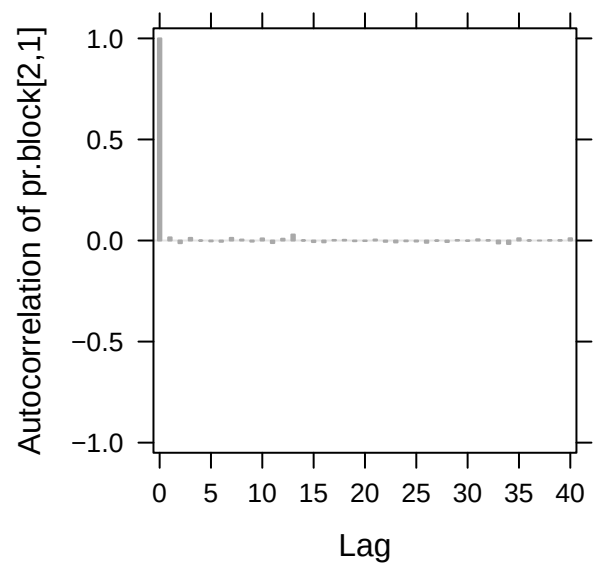
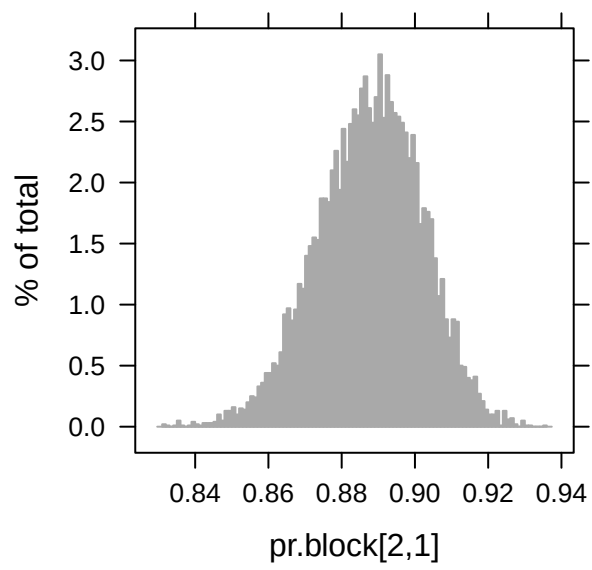
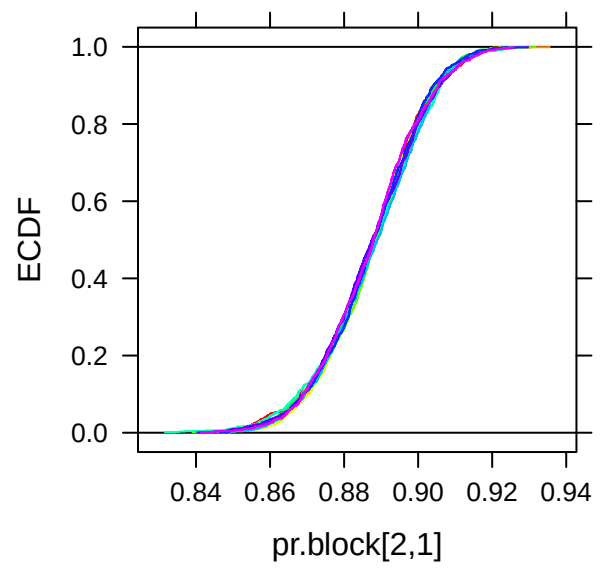
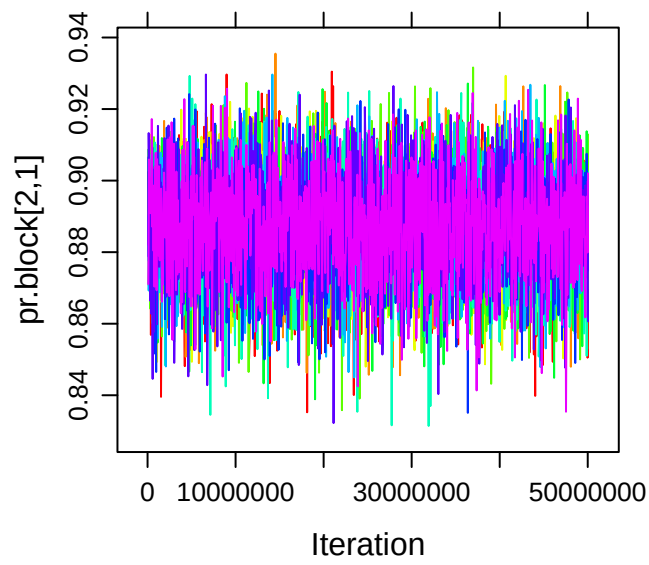


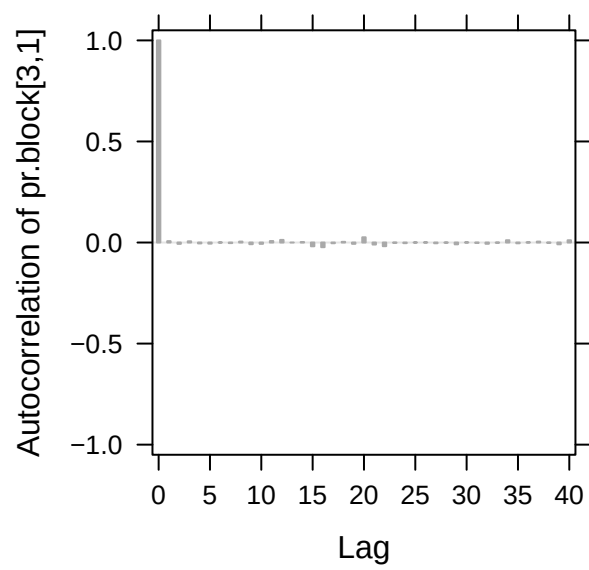
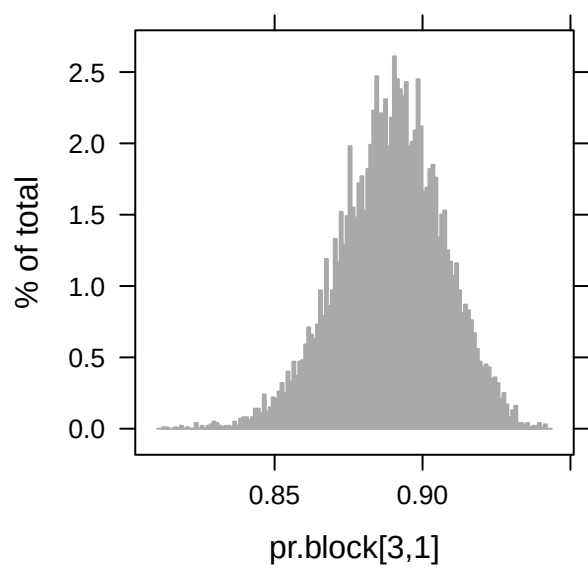
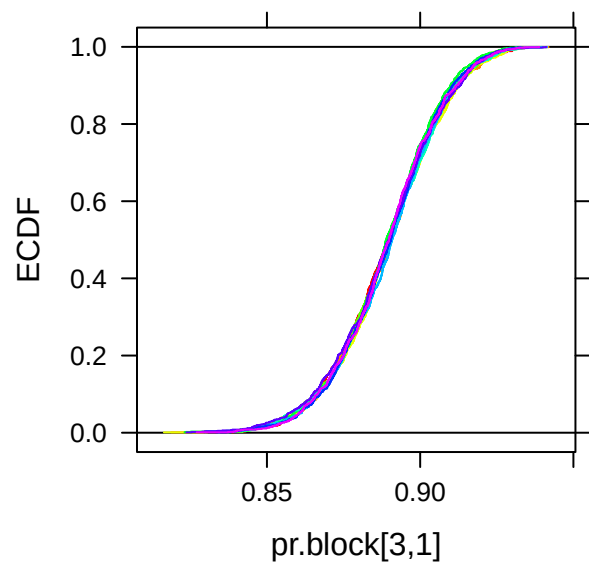
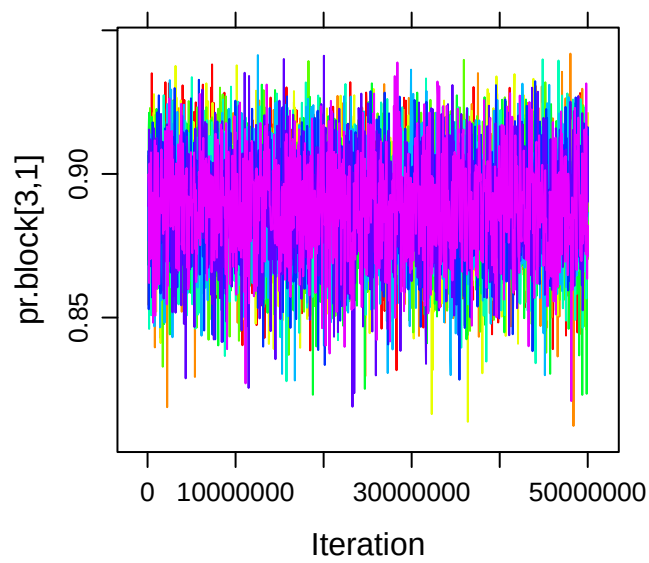


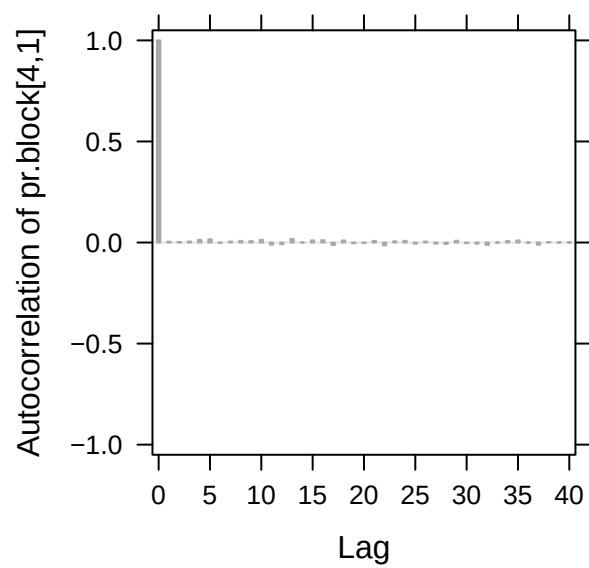
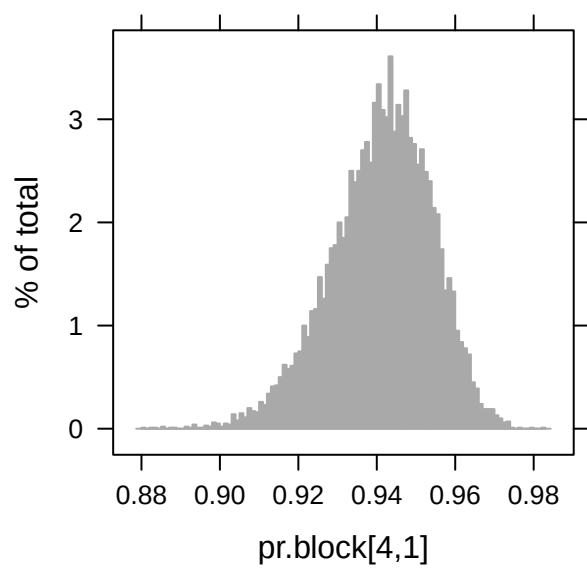
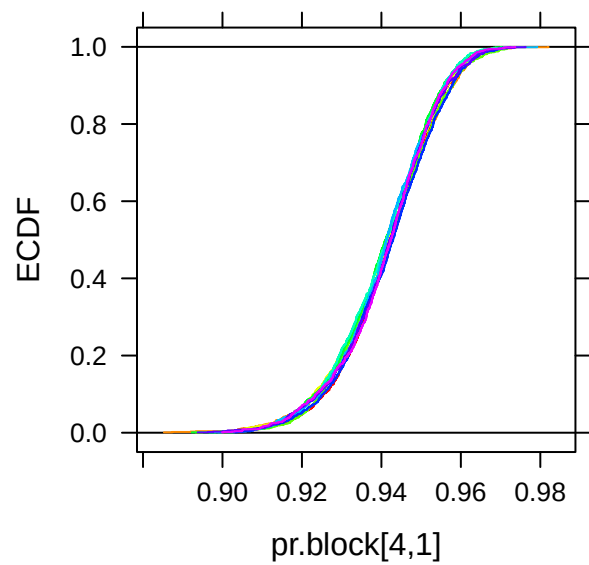
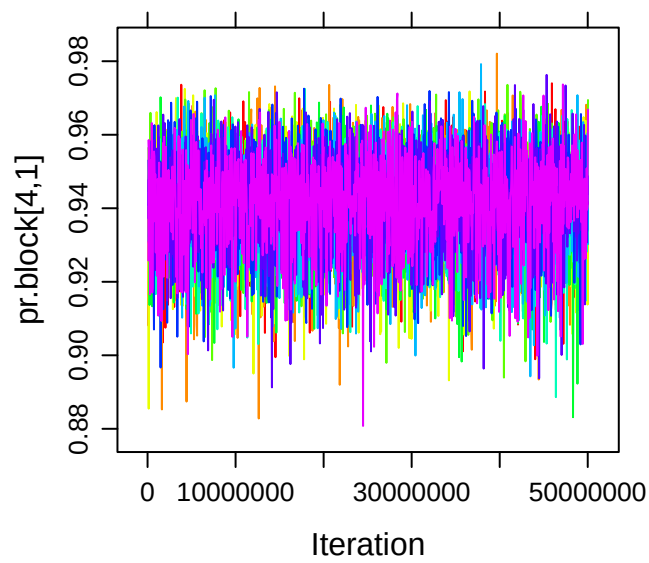


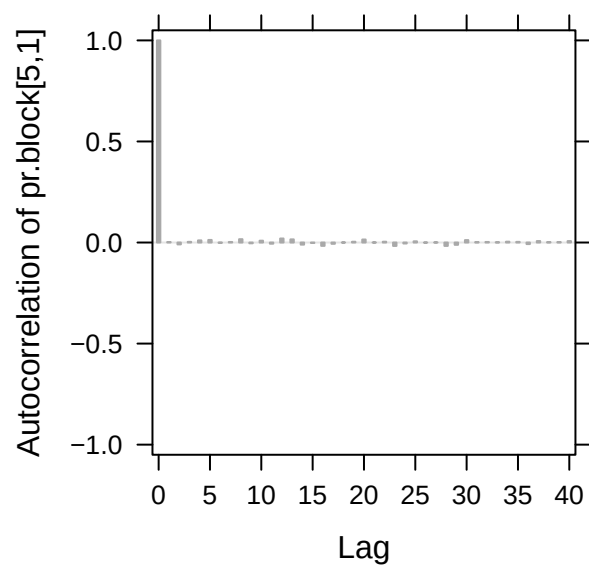
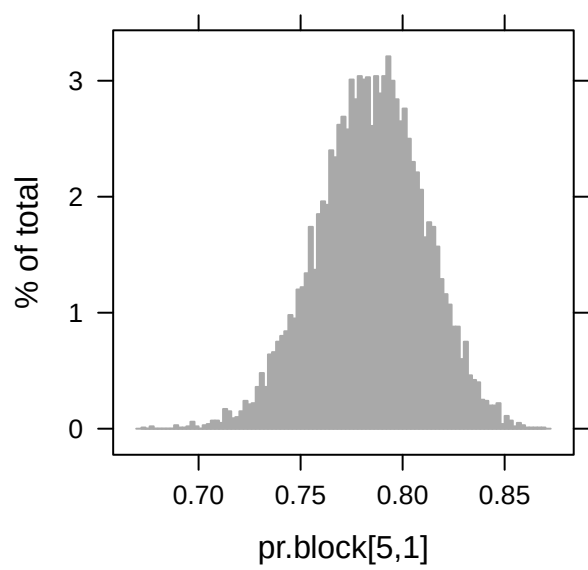
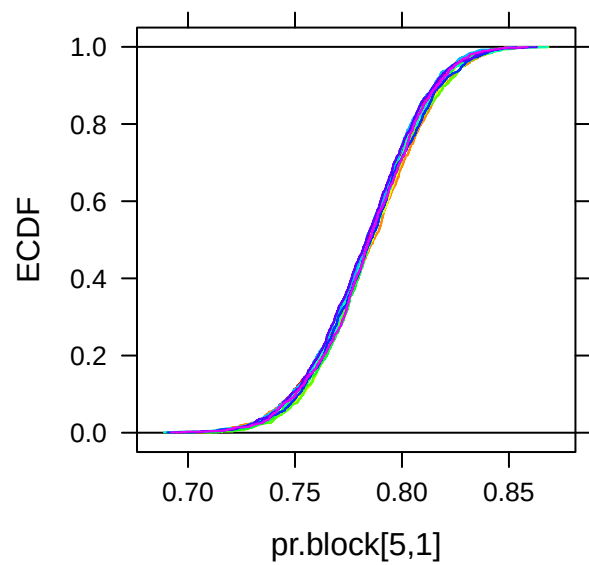
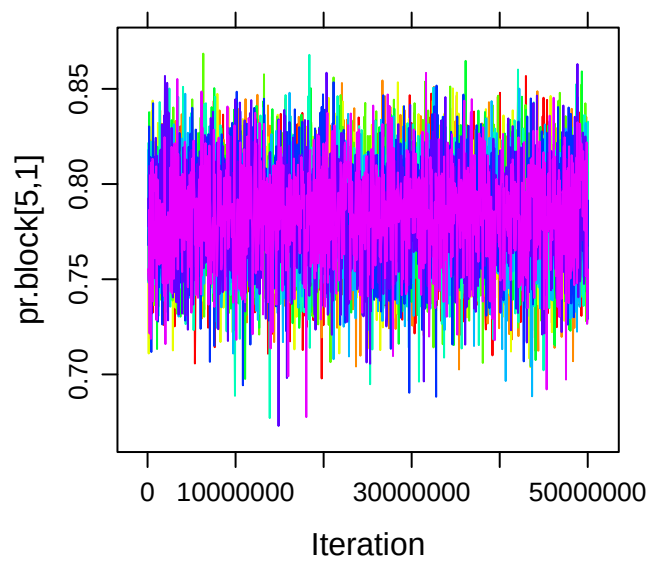


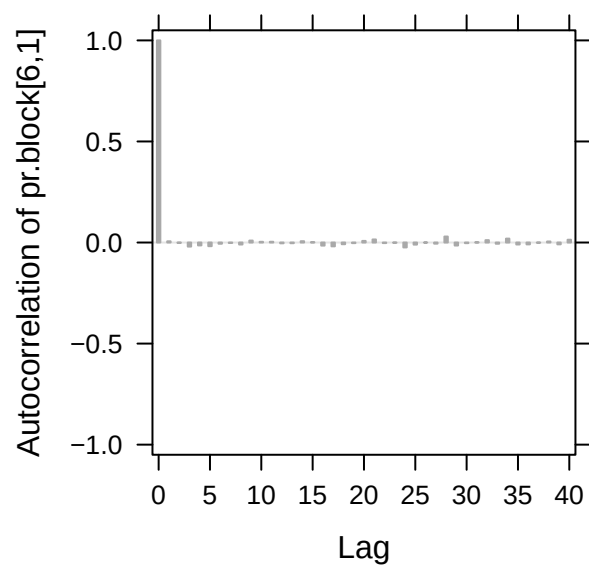
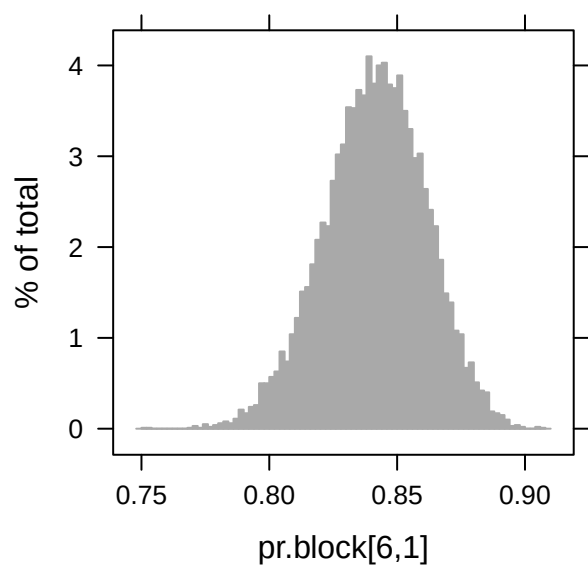
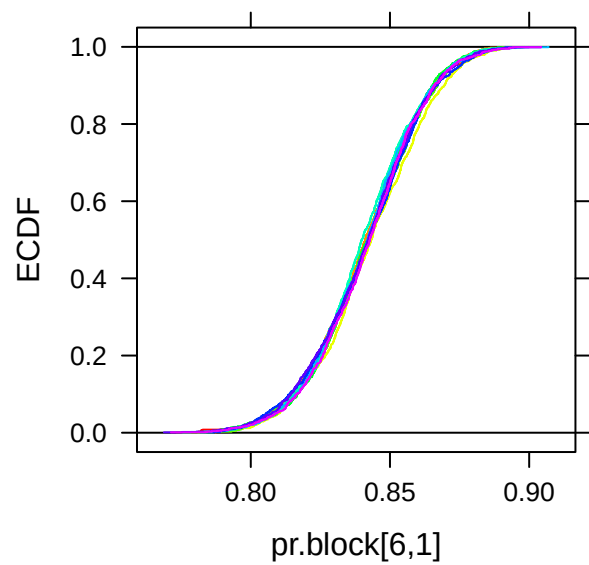
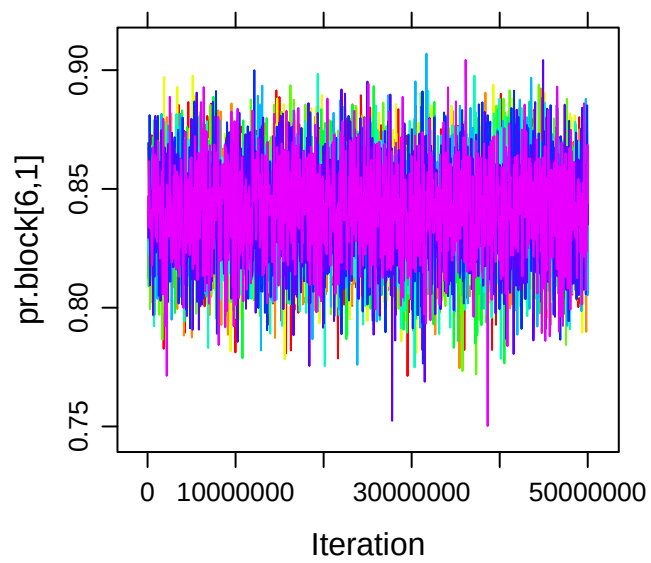


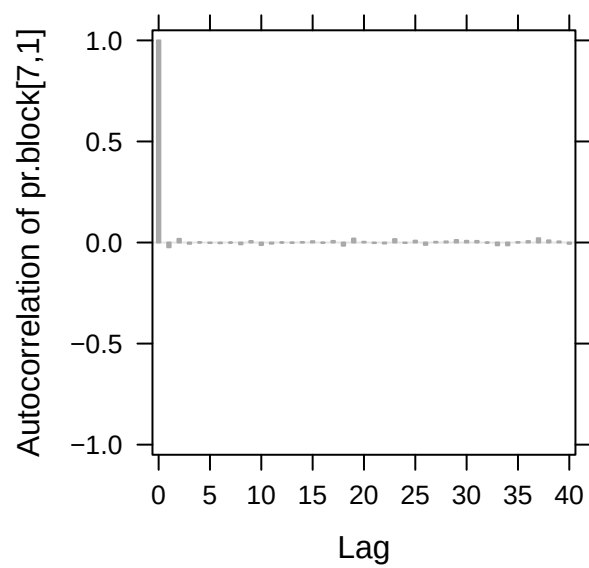
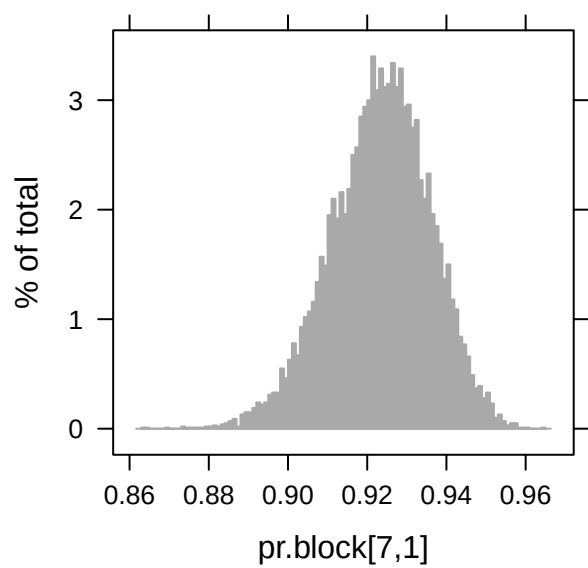
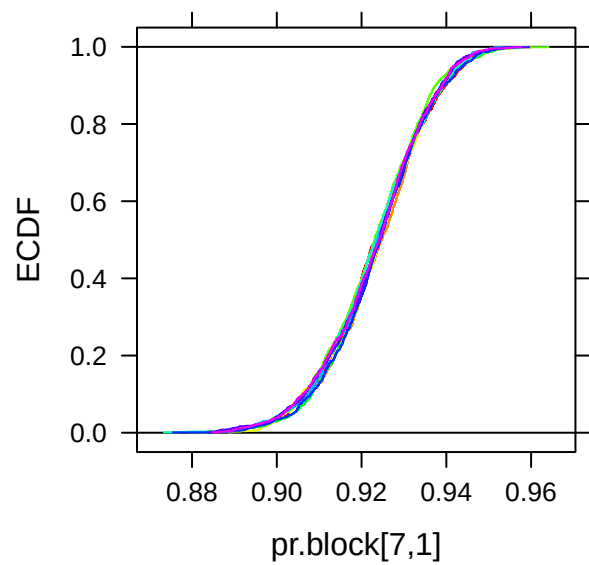
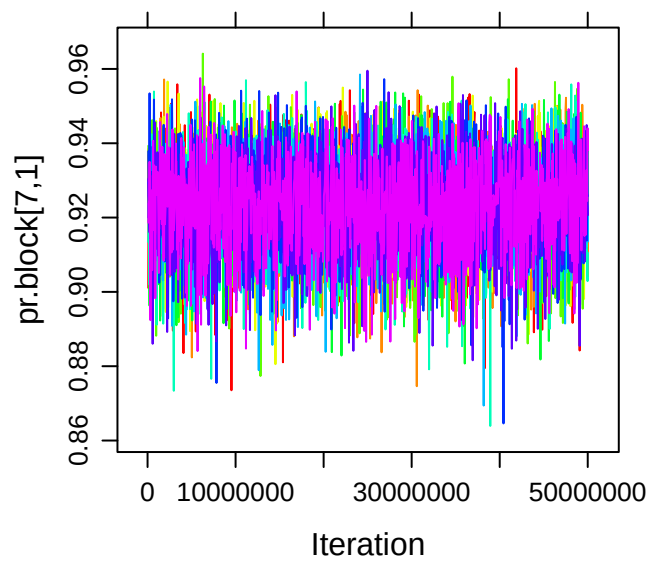


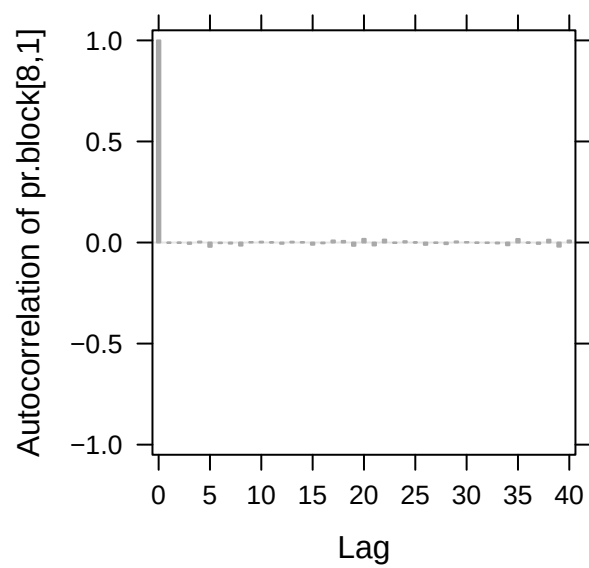
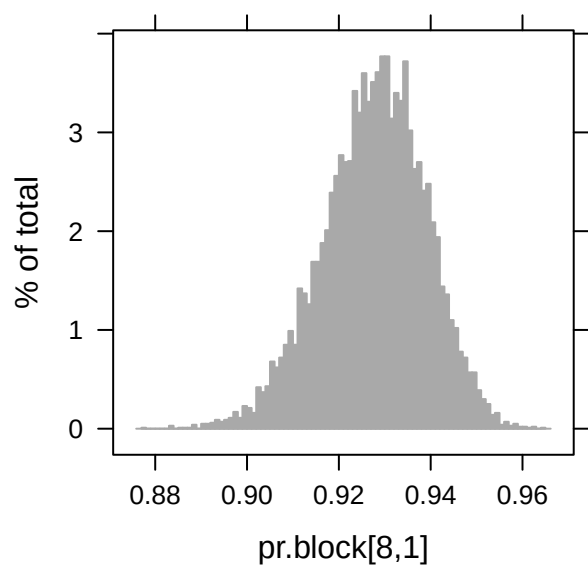
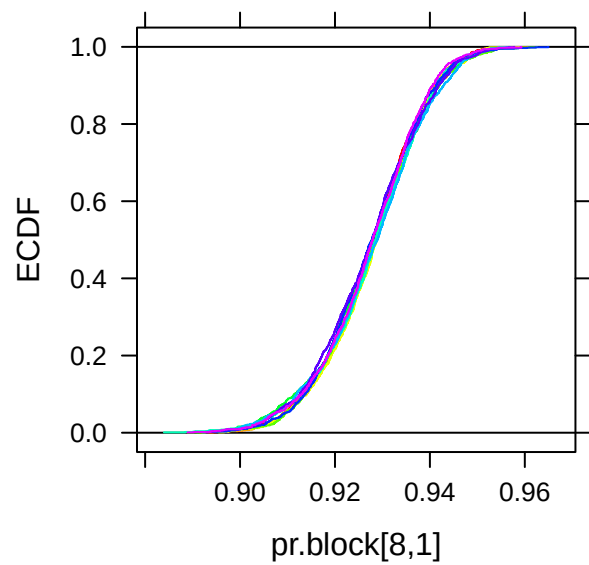
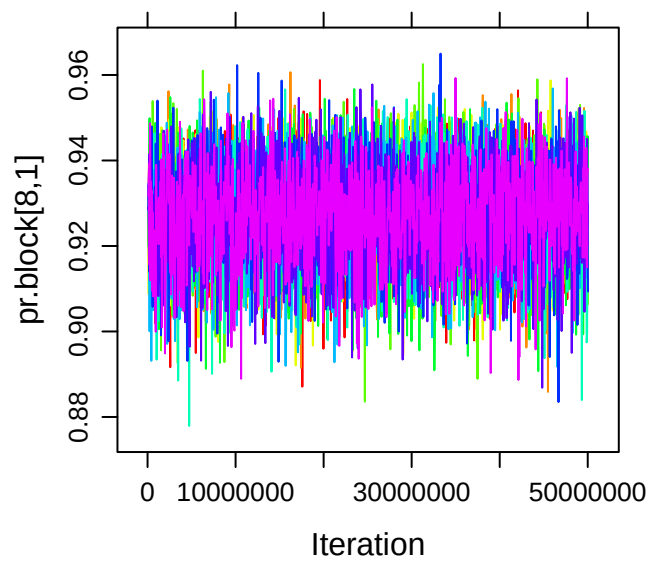


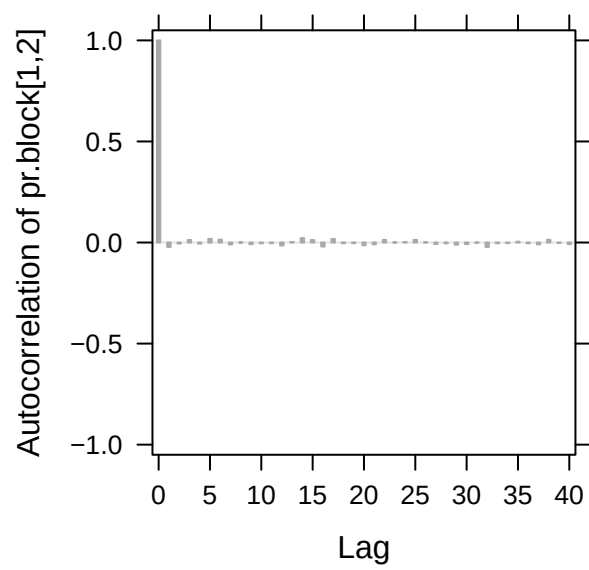
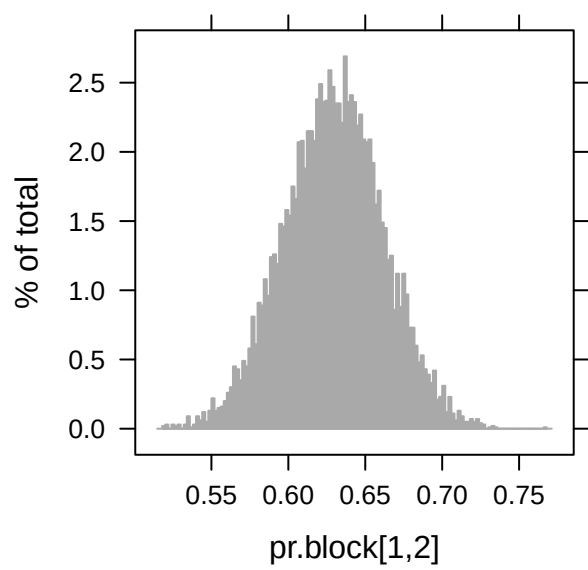
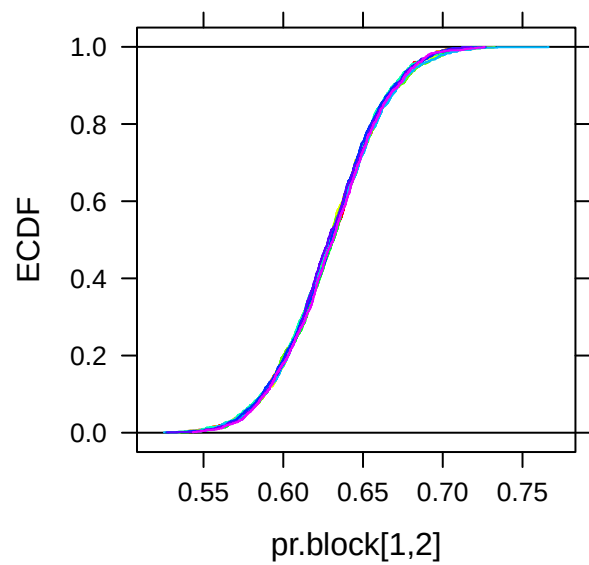
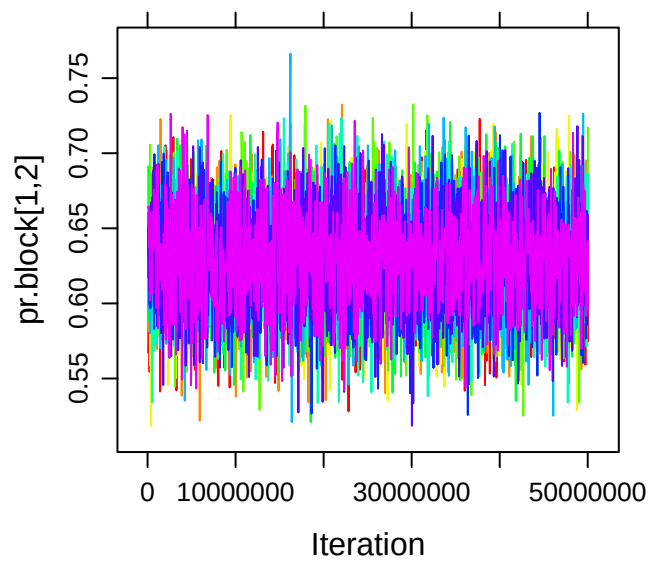


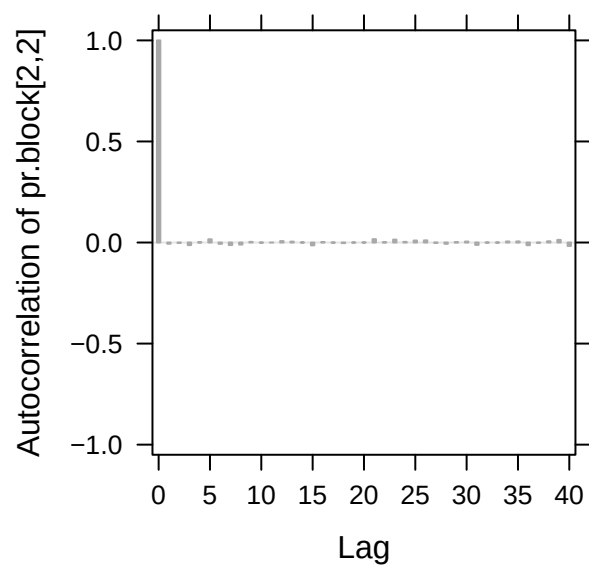
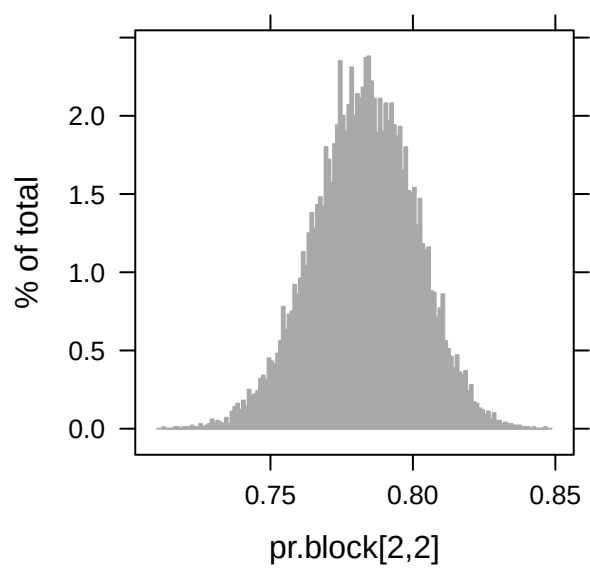
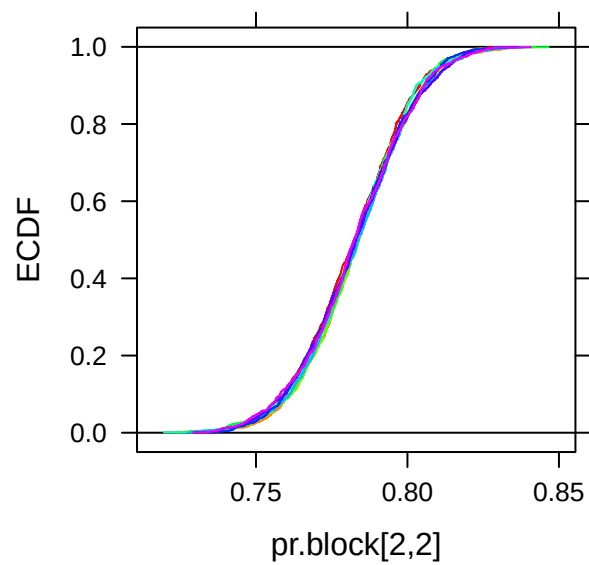
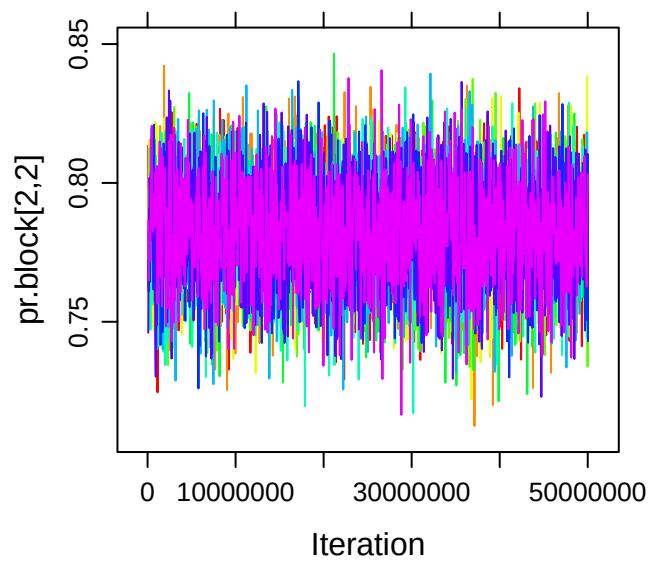


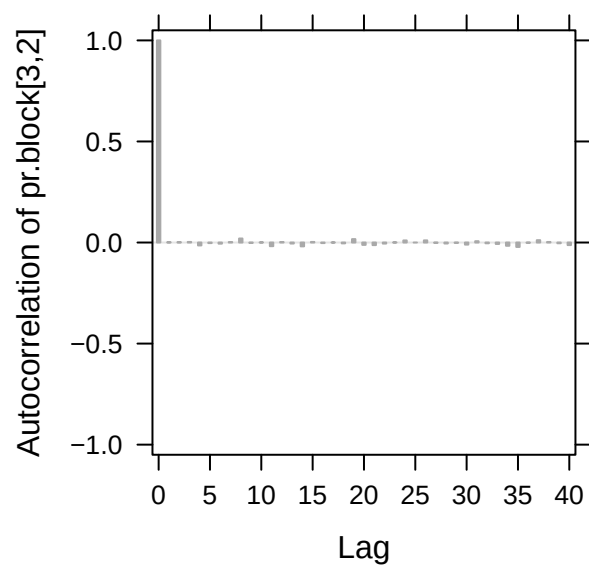
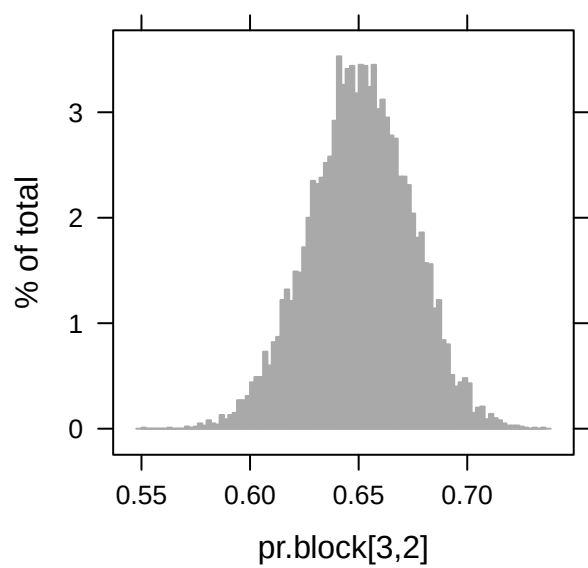
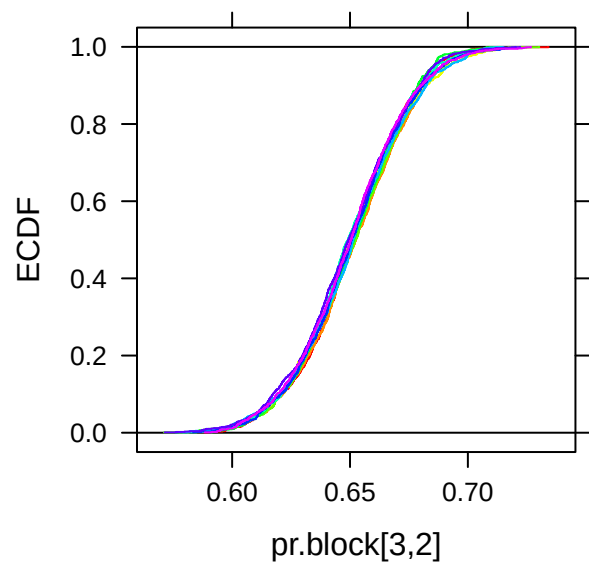
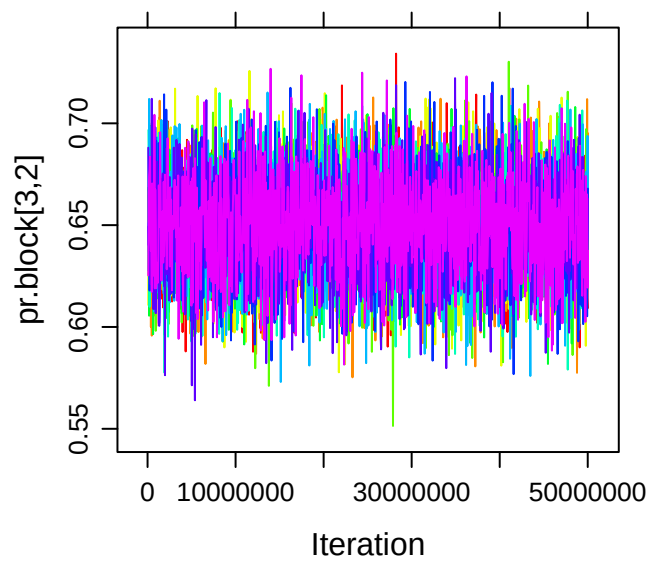


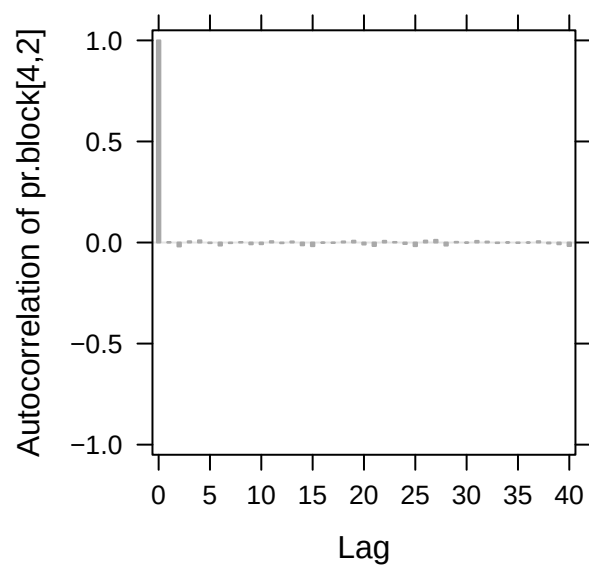
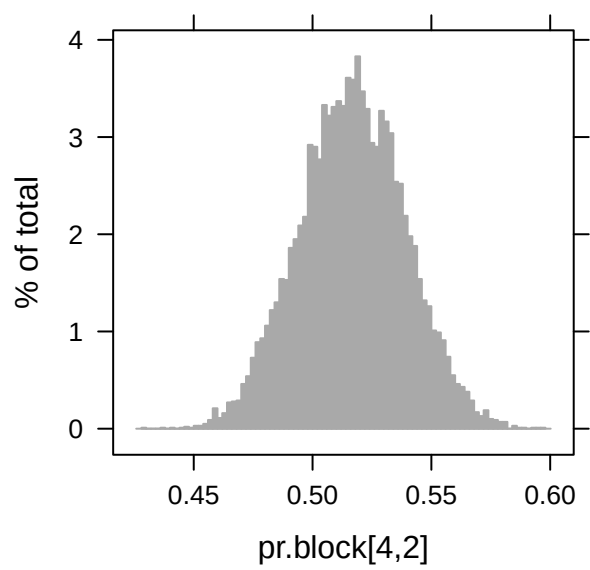
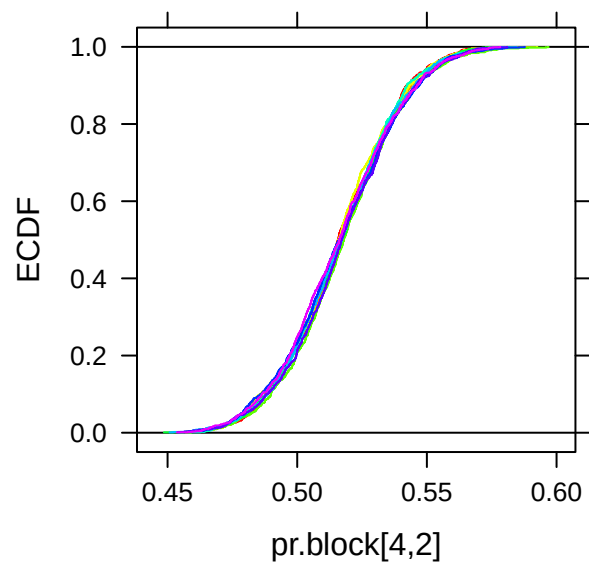
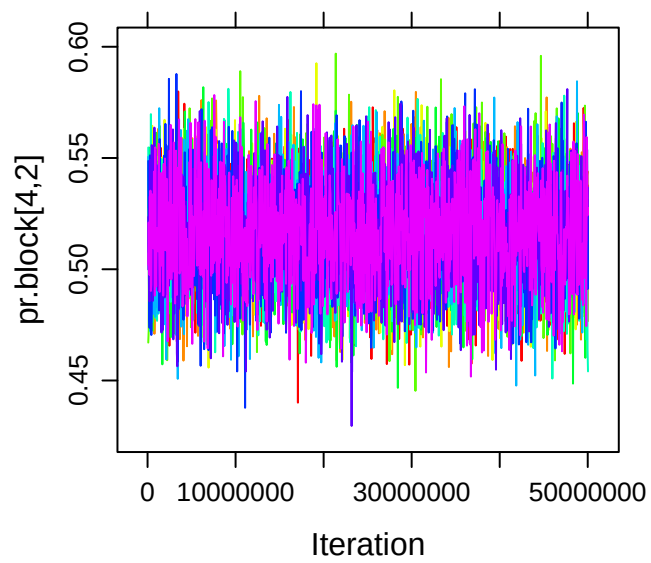


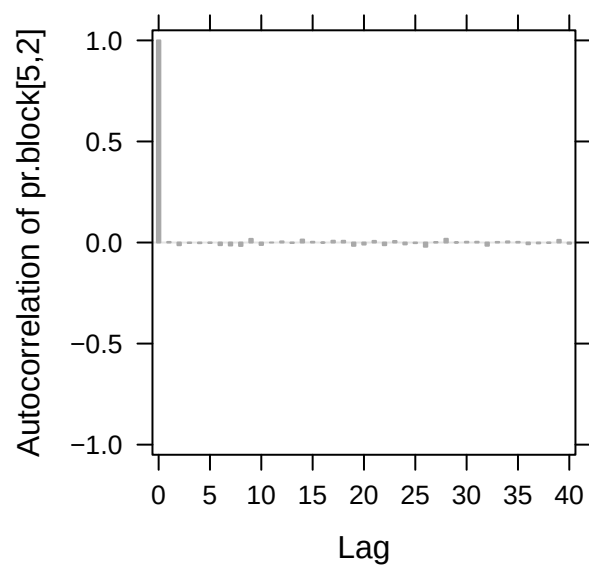
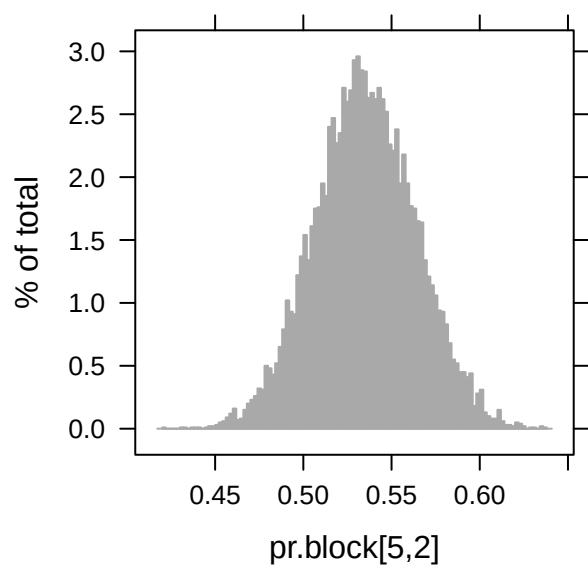
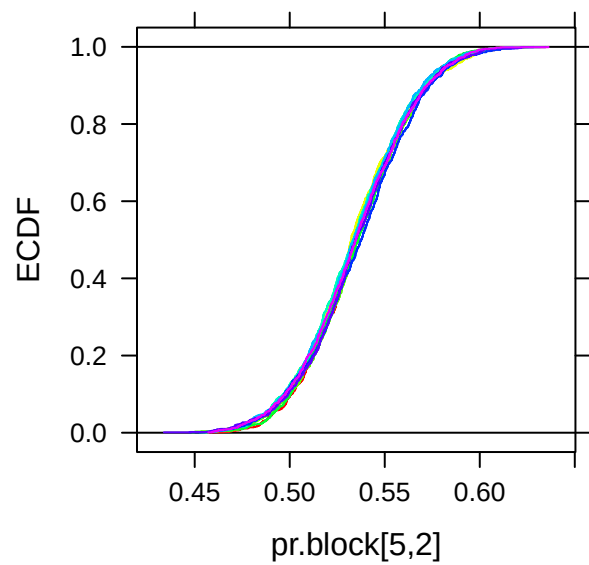
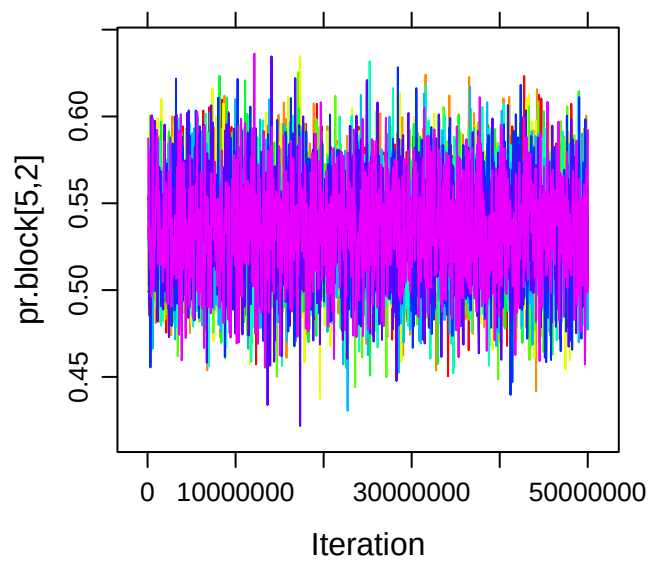


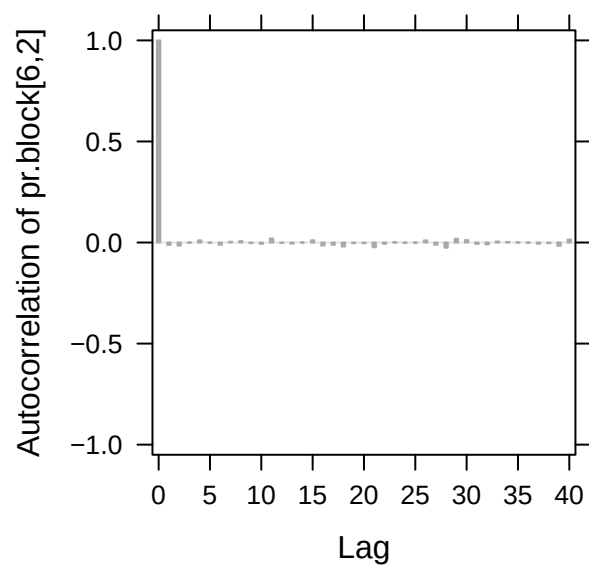
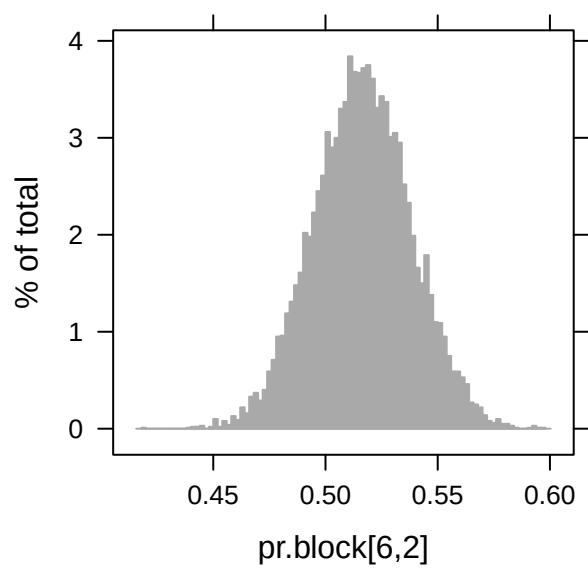
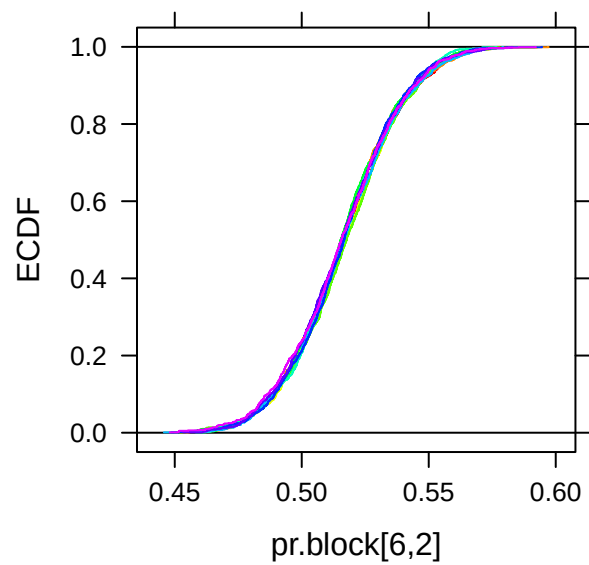
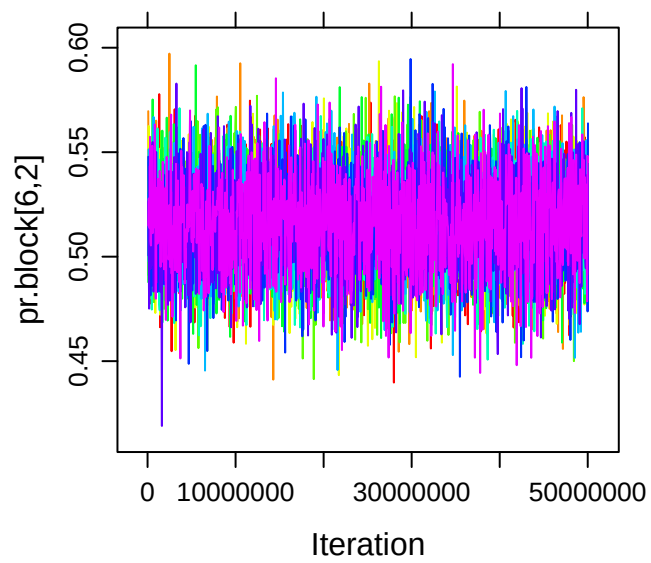


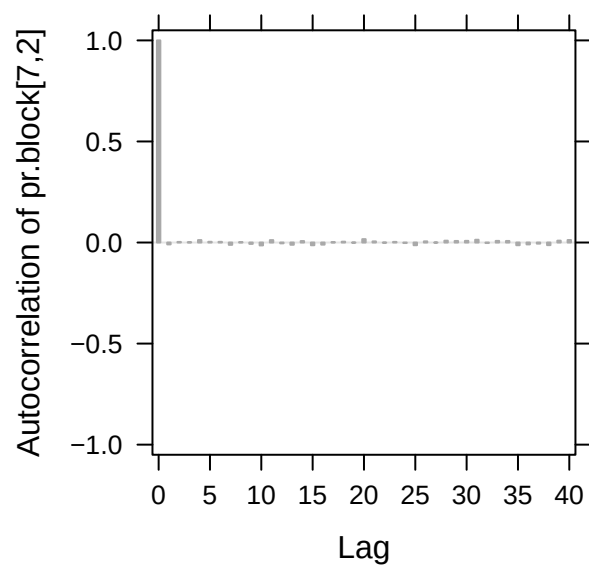
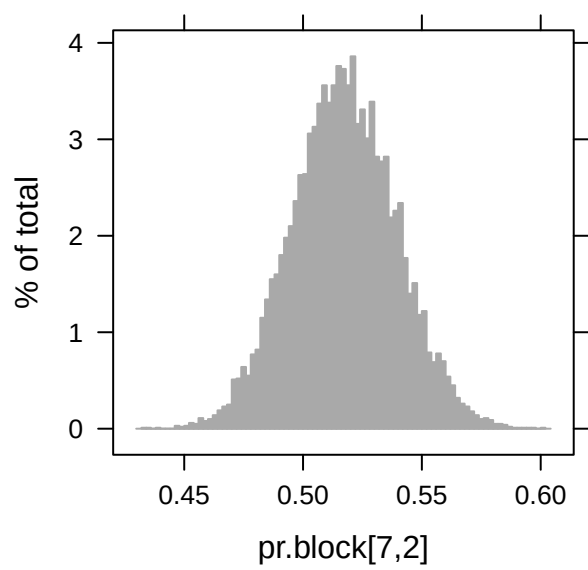
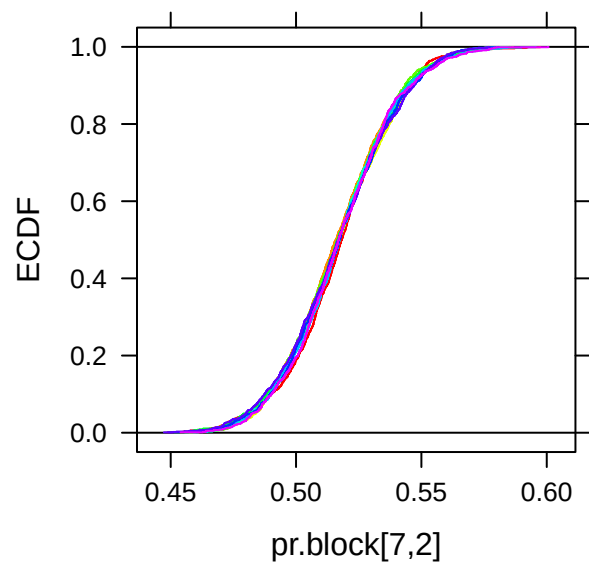
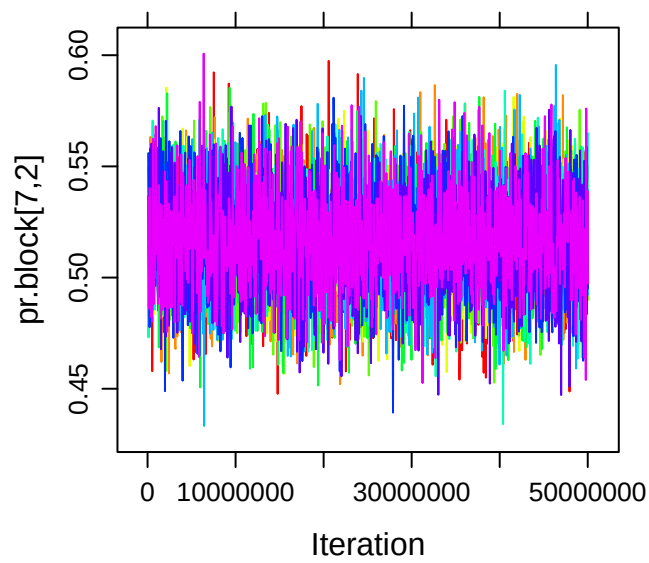


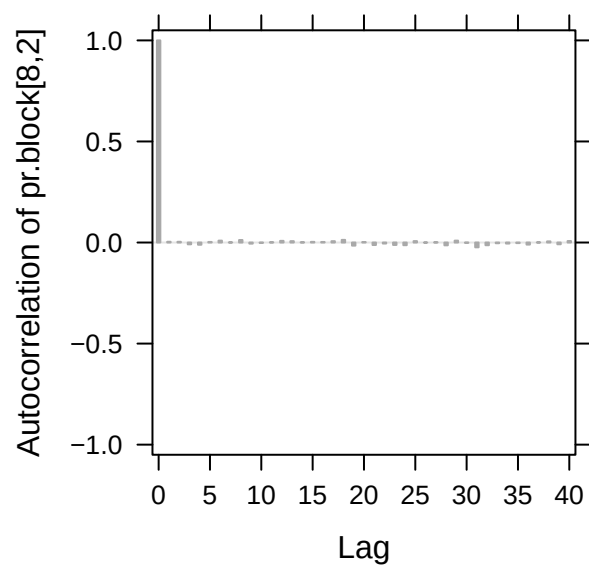
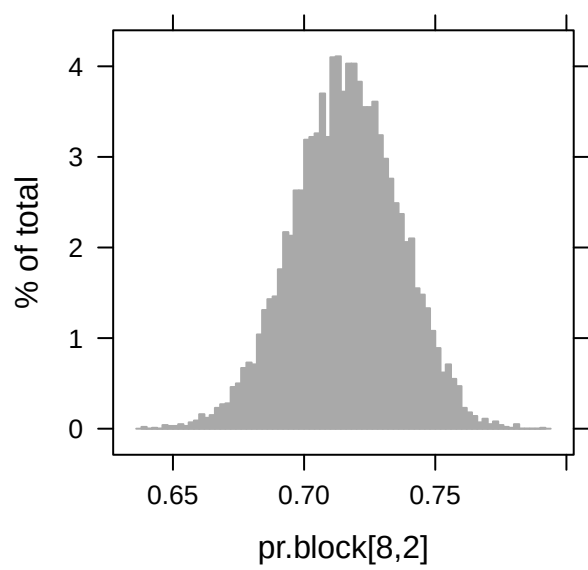
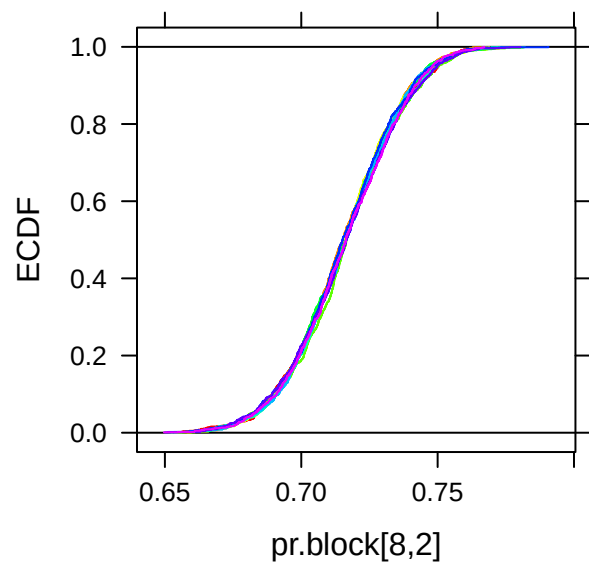
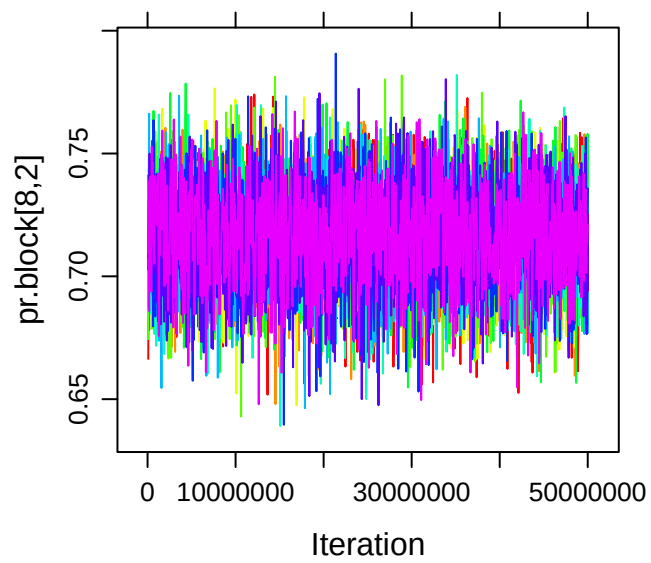














Model IV

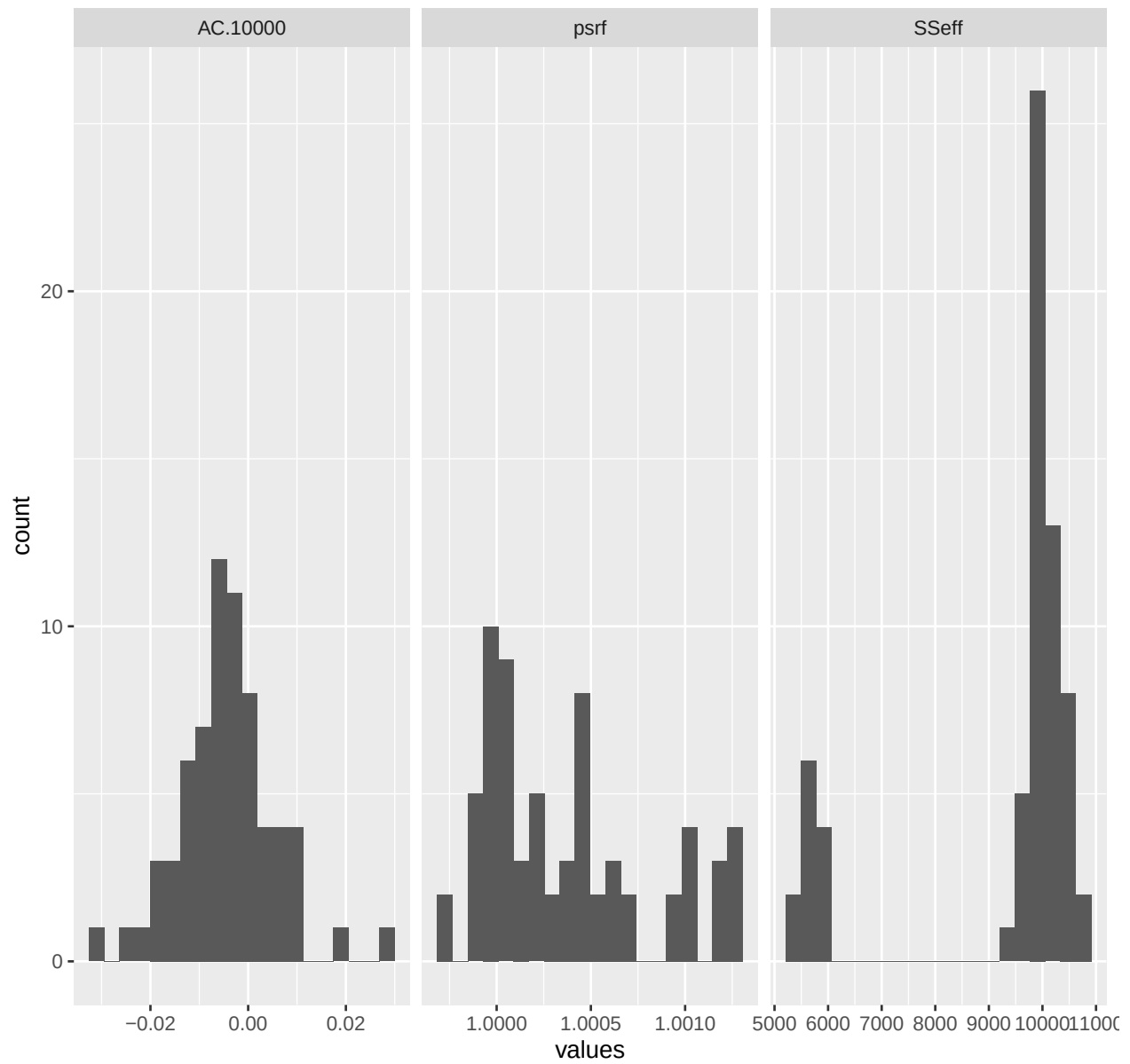
Number of chains: 10

Burnin: 100100

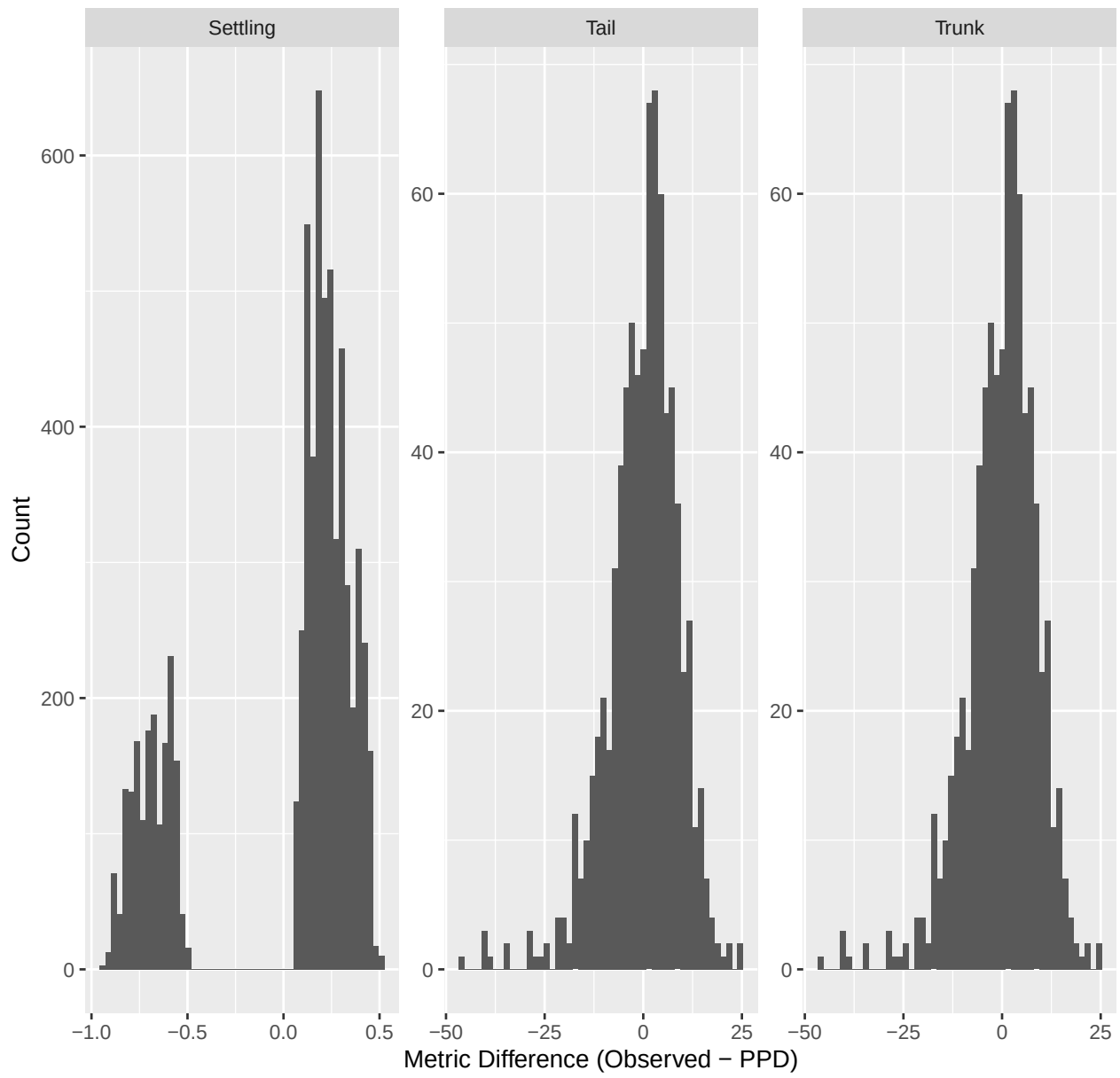
Samples per chain: 1000

Total samples: 10000

A) Posterior distribution diagnostics



B) Posterior predictive check distribution



C) By-parameter posterior summaries

Generating summary statistics and plots (these will NOT be saved for reuse)...

Calculating summary statistics...

Calculating the Gelman-Rubin statistic for 67 variables....

Note: Unable to calculate the multivariate psrf

